

# Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Independent research; computational work assisted by Instinct

## Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

## 1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

## 2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

### 3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

#### 3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG\_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

#### 3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]:  $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$ , with per-position mismatch penalties  $m_i$  from the reference distribution. Binding free-energy changes are  $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$  at  $T = 300 \text{ K}$  ( $RT = 0.596 \text{ kcal/mol}$ ). Flux balance analysis solves  $\max_v c^\top v$  subject to  $Sv = 0$ ,  $l \leq v \leq u$ , using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is  $0.8739 \text{ h}^{-1}$  [4]. Agreement is quantified with Pearson  $r$ , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

#### 3.3 Derivations

**(F1) Position weight matrix log-odds.** For aligned sites of length  $L$  with counts  $n_{i,b}$  at position  $i$  and base  $b$ , the pseudocount estimate is  $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$ . The log-odds weight against background  $q_b$  is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$  is the log-likelihood ratio of site model to background under positional independence:  $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$ .

**(F2) Normalised score.** With  $S_{\min} = \sum_i \min_b w_{i,b}$  and  $S_{\max} = \sum_i \max_b w_{i,b}$ ,  $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$ , because every term of  $S$  lies between its column minimum and maximum.

**(F3) Variant delta.** For reference and alternate windows  $x^r, x^a$  that differ at position  $j$ ,  $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$ : all other terms cancel, so a single-nucleotide delta depends only on column  $j$ .

**(F4) Tract blend.** The acceptor score is  $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$  with  $\lambda = 0.25$ , where  $T$  is the normalised pooled polypyrimidine-tract score. Linearity gives  $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$ .

**(F5) AUC as a Mann-Whitney statistic.** For scores  $\{s_i^+\}_{i=1}^m$  and  $\{s_j^-\}_{j=1}^n$ ,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[ \mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where  $R_+$  is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because  $R_+ - m(m+1)/2$  counts, for each positive, the negatives ranked below it. Equivalently,  $\text{AUC} = P(s^+ > s^-)$ .

**(F6) Within-position AUC.** To remove the prior carried by the intronic offset  $k$ , we report  $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$  over offsets with at least 10 positives and 10 negatives, weighting by positive count  $m_k$ .

**(F7) CNN objective.** With logit  $z_\theta(x)$  and class weight  $\omega = n_-/n_+$  computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[ \omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient  $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$ . Setting  $\omega = n_-/n_+$  makes both classes contribute equal total weight.

**(F8) Convolutional feature.** For one-hot input  $X \in \{0, 1\}^{8 \times 81}$  (reference and alternate channels) and filter  $K \in \mathbb{R}^{8 \times w}$ ,  $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$ . Global max pooling  $\max_t h_t$  makes the feature invariant to where the motif sits in the window.

**(F9) Binding free energy.**  $\Delta G = RT \ln K_d$  gives  $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$ , with  $RT = 0.596$  kcal/mol at 300 K.

**(F10) FBA duality.** The primal  $\max c^\top v$  s.t.  $Sv = 0$ ,  $l \leq v \leq u$  has the dual  $\min u^\top \alpha - l^\top \beta$  s.t.  $S^\top \pi + \alpha - \beta = c$ ,  $\alpha, \beta \geq 0$ . HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

**(F11) Codon Adaptation Index.** With relative adaptiveness  $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$ ,  $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$ , the geometric mean of  $w$  over the  $L$  scorable codons.

**(F12) ACMG Bayesian posterior.** Following Tavtigian et al.,  $O_P = 350^{\text{pts}/8}$  and  $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$ . With  $\pi = 0.1$  and  $\text{pts} = 6$ :  $O_P = 350^{0.75} \approx 80.92$ , so  $\text{Post} = 8.092/8.992 \approx 0.8999$ , matching the module's 0.8999 and the published 0.900.

**(F13) Profile-HMM forward recursion.** For match, insert and delete states,  $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$ , and similarly for  $I$  and  $D$ . The log-odds score is  $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$ .

**(F14) Bootstrap interval.** For a statistic  $\hat{\theta}$ , draw  $B = 1000$  resamples with replacement and take the 2.5th and 97.5th percentiles of  $\{\hat{\theta}_b^*\}$ .

### 3.4 Derivations for the Second-Round Validations

**(F15) Maximum-entropy splice-site score.** MaxEntScan [7] models a site  $x$  (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals  $\{\hat{p}_C(x_C)\}$  over a set of position subsets  $C$ . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp \left( \sum_C \lambda_C(x_C) \right), \quad Z = \sum_{x'} \exp \left( \sum_C \lambda_C(x'_C) \right), \quad (4)$$

since maximising  $H(p) = -\sum_x p \log p$  with constraints  $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$  gives stationarity  $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$ . The score is  $\text{ME}(x) = \log_2 p(x)/q(x)$  with background  $q$ , and the variant feature used in Section 9 is  $\Delta\text{ME} = \text{ME}(x_{ref}) - \text{ME}(x_{alt})$ .

**(F16) Logistic model with MaxEnt features.** With feature vector

$$z = (S_{ref}, S_{alt}, \Delta S, d, \text{ME}_{ref}/10, \text{ME}_{alt}/10, \Delta\text{ME}/10, e_k),$$

where  $d = 1$  for donors and 0 for acceptors,  $e_k$  one-hot encodes the offset,  $P(y = 1 | z) = \sigma(\beta^\top z + \beta_0)$ . The log-likelihood  $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$  has gradient  $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$  and Hessian  $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$ , which is negative semidefinite, so  $\ell$  is concave and the fit has a single optimum.

**(F17) Fisher  $z$  test for a correlation.** For sample correlation  $r$  from  $n$  pairs,  $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$ . By the delta method  $\text{Var}(r) \approx (1 - \rho^2)^2/n$  and  $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$ , so  $\text{Var}(z) \approx 1/n$ ; the refined value  $1/(n - 3)$  gives

$$Z = \text{artanh}(r) \sqrt{n - 3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (5)$$

We apply it to Spearman  $\rho$  (rank correlation), which is an approximation.

**(F18) Benjamini-Hochberg adjustment.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$ . Rejecting all  $q_{(i)} \leq \alpha$  controls the false discovery rate at  $\alpha m_0/m \leq \alpha$  for independent or positively dependent tests, because  $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$ .

**(F19) tRNA adaptation index.** For codon  $c$  with tRNA gene copy numbers  $t_j$  of the anticodons  $j$  that decode it and wobble efficiencies  $(1 - s_{jc})$ ,  $W_c = \sum_j (1 - s_{jc}) t_j$ ,  $w_c = W_c / \max_{c'} W_{c'}$  (zeros replaced by the geometric mean of non-zero  $w$ ), and for a gene of  $L$  codons  $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$  [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with  $w$  from tRNA supply instead of a reference gene set.

**(F20) Effective number of codons.** For an amino acid with  $k$  synonymous codons observed  $n$  times with frequencies  $p_i$ ,  $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$  is the unbiased homozygosity;  $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$  averages  $\hat{F}$  within degeneracy classes. Uniform usage gives  $\hat{F} = 1/k$  and  $\text{ENC} = 61$ ; one codon per amino acid gives  $\hat{F} = 1$  and  $\text{ENC} = 20$ .

**(F21) Shrake-Rupley surface area.** Atom  $a$  with radius  $r_a$  is expanded to  $R_a = r_a + r_p$  ( $r_p = 1.4 \text{ \AA}$ ). With  $N$  quasi-uniform points  $u_k$  on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[ \|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (6)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as  $N^{-1/2}$  for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

**(F22) Nearest-neighbour melting temperature.** For a duplex with  $\Delta H = \Delta H_{init} + \sum_i \Delta H_{i,i+1}$  and  $\Delta S$  likewise, at equilibrium  $\Delta G = \Delta H - T\Delta S = -RT \ln K$ . At  $T_m$  half the limiting strand is bound, so  $K = 1/C_{eff}$  with  $C_{eff} = C_1 - C_2/2$  for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{eff}}, \quad \Delta S_{[Na^+]} = \Delta S + 0.368 (N - 1) \ln [Na^+]. \quad (7)$$

**(F23) Negative-binomial GLM for counts.** DESeq2 models  $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$  with  $\text{Var} = \mu + \alpha\mu^2$ ,  $\mu_{ij} = s_j q_{ij}$  and  $\log_2 q_{ij} = x_j^\top \beta_i$ . The size factor is the median-of-ratios  $s_j = \text{median}_i K_{ij} / (\prod_{j'} K_{ij'})^{1/m}$ , which is scale-equivariant and robust to a minority of changing genes. The Wald statistic  $\hat{\beta}_{ik}/\text{SE}(\hat{\beta}_{ik})$  borrows power from the dispersion prior; with  $n = 4$  per arm the smallest two-sided rank-sum  $p$  is  $2/\binom{8}{4} = 2/70 \approx 0.029$ , which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

**(F24) Robinson-Foulds distance.** For unrooted trees  $T_1, T_2$  on the same leaves with non-trivial split sets  $\Sigma_1, \Sigma_2$ ,  $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$ . In the RF00005 case the observed  $d_{RF} = 2$  (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

### 3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

## 4 Results

### 4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG\_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within  $5 \times 10^{-7}$  (maximum absolute difference  $5.0 \times 10^{-7}$ ). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman  $\rho = 0.9995$ .

## 4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was  $0.873922 \text{ h}^{-1}$ , identical to an independent SciPy/HiGHS solve ( $0.873922$ ) and to the published  $0.8739 \text{ h}^{-1}$ .

## 4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form  $O_P = 350^{\text{pts}/8}$  to 0 across  $-16 \dots 16$  points; 6 points at prior 0.10 gives posterior 0.900.

## 4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG\_008021.1, NG\_007489.1, NG\_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

## 4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region  $+3 \dots +20$  variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

## 4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

# 5 Benchmark Comparison

Each row compares a sugarcoder module with an independent published reference; every number is generated from the named committed file.

| Module         | Reference             | Metric            | Result (file)                                        |
|----------------|-----------------------|-------------------|------------------------------------------------------|
| pgx_guidelines | CPIC diplotype tables | agreement         | 16836/16836 CYP2D6<br>(sweep_pgx_cplic.json)         |
| bio.codon CAI  | Biopython; PaxDb      | Pearson; Spearman | 0.99969; 0.5797<br>(sweep_codon_cai.json)            |
| rna_nussinov   | ViennaRNA 2.7         | bp F1 (tRNA)      | 0.328 vs 0.714<br>(sweep_rna.json)                   |
| profile_hmm    | Pfam PF00042          | auc_log_odds_bits | 0.952 global, 0.9792 local<br>(profile_hmm_cv*.json) |
| splice CNN     | SpliceAI 1.3.1        | AUROC n=757       | 0.929 vs 0.9775 (headto-head_interim.json)           |

## 6 Failure Modes and Negative Results

**openclinvar, first run: 500/500 abstentions.** ClinVar names such as NM\_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

**deepsplICE, first run: acceptor AUC 0.547.** This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

**profile\_hmm: residual gap to HMMER.** The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcode-ai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile\_hmm\_cv\_local.json); Pfam’s model still reaches 1.000.

**mutdock vs SKEMPI: no honest adapter.** mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol,  $|\Delta V|$  has Pearson  $r = 0.053$  and hydrophathy loss  $r = -0.129$  ( $n = 150$ ).

**openclinvar accuracy is not a hard test.** The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

## 7 Module Sweep: Additional Real-Data Results

### 7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep\_rna.json, sweep\_rna\_vienna\_methods.json).

| Family  | mean len | Nussinov | MFE   | centroid | MEA( $\gamma=1$ ) | MEA( $\gamma=4$ ) |
|---------|----------|----------|-------|----------|-------------------|-------------------|
| RF00005 | 73.0     | 0.328    | 0.714 | 0.716    | 0.703             | 0.704             |
| RF00001 | 117.9    | 0.185    | 0.531 | 0.494    | 0.505             | 0.504             |
| RF00010 | 354.5    | 0.189    | 0.545 | 0.594    | 0.592             | 0.572             |

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold_energy` now delegates to ViennaRNA.

## 7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF\_000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE\_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with  $\log_{10}$  abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.



Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

## 7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (\*9, \*41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



Figure 2. CPIC agreement before and after the fix (sweep\_pgx\_cplic.json).

## 8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on  $n=757$  variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference  $[-0.0087, 0.0081]$ ). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

| CNN threshold | TP   | FP  | sensitivity | FPR     | PPV (prior 3%) |
|---------------|------|-----|-------------|---------|----------------|
| 0.5           | 1794 | 453 | 0.77        | 0.0656  | 0.2663         |
| 0.8           | 1298 | 146 | 0.5571      | 0.02114 | 0.449          |
| 0.9           | 996  | 68  | 0.4275      | 0.00985 | 0.5731         |
| 0.95          | 729  | 33  | 0.3129      | 0.00478 | 0.6694         |
| 0.98          | 435  | 11  | 0.1867      | 0.00159 | 0.7838         |



Figure 3. Splice CNN operating points (cnn\_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd\_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

## 9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

| Threshold | TP   | FP  | Sensitivity | FPR     |
|-----------|------|-----|-------------|---------|
| 0.5       | 1899 | 259 | 0.815       | 0.03751 |
| 0.8       | 1503 | 100 | 0.6451      | 0.01448 |
| 0.9       | 1230 | 56  | 0.5279      | 0.00811 |
| 0.95      | 950  | 28  | 0.4077      | 0.00406 |
| 0.98      | 692  | 15  | 0.297       | 0.00217 |



Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

## 10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank  $p = 3.3e - 07$ , sign test  $p = 1.6e - 06$ ). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at  $p < 0.01$ ).



Figure 2: Spearman  $\rho$  (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                                  | Assembly         | $n$  | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|---------------------------------------------------------------------------|------------------|------|------------|--------------|---------------|----------|
| 10090  | <i>Mus musculus</i>                                                       | GCF_000001635.27 | 8089 | 99         | 0.0704       | 0.0775        | 4.4e-12  |
| 10116  | <i>Rattus norvegicus</i>                                                  | GCF_036323735.1  | 890  | 124        | 0.142        | 0.1405        | 3.1e-05  |
| 121845 | <i>Diaphorina citri</i>                                                   | GCF_000475195.1  | 592  | 1          | 0.1458       | —             | —        |
| 1295   | <i>Staphylococcus schleiferi</i>                                          | GCF_025558325.1  | 442  | 66         | 0.4583       | 0.5403        | 2e-36    |
| 1314   | <i>Streptococcus pyogenes</i>                                             | GCF_900475035.1  | 306  | 66         | 0.4954       | 0.6701        | 9.8e-45  |
| 185431 | <i>Trypanosoma brucei brucei</i> TREU927                                  | GCF_000002445.2  | 6302 | 0          | 0.3167       | —             | —        |
| 192222 | <i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819 | GCF_000009085.1  | 725  | 66         | 0.1208       | 0.2291        | 5.3e-10  |
| 196627 | <i>Corynebacterium glutamicum</i> ATCC 13032                              | GCF_000011325.1  | 323  | 72         | 0.5268       | 0.6307        | 7e-40    |
| 198214 | <i>Shigella flexneri</i> 2a str. 301                                      | GCF_000006925.2  | 1887 | 72         | 0.5511       | 0.6514        | 8.1e-249 |
| 208964 | <i>Pseudomonas aeruginosa</i> PAO1                                        | GCF_000006765.1  | 4796 | 72         | 0.2459       | 0.4149        | 2.5e-204 |
| 224308 | <i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168                  | GCF_000009045.1  | 3147 | 64         | 0.3295       | 0.4156        | 3.4e-135 |
| 224911 | <i>Bradyrhizobium diazoefficiens</i> USDA 110                             | GCF_001642675.1  | 494  | 72         | 0.4049       | 0.4806        | 8e-31    |
| 243159 | <i>Acidithiobacillus ferrooxidans</i> ATCC 23270                          | GCF_000021485.1  | 419  | 68         | 0.3378       | 0.4055        | 2.8e-18  |
| 243230 | <i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539                | GCF_020546685.1  | 319  | 66         | 0.4784       | 0.5698        | 2.4e-30  |

Table 2: Per-taxon results (all taxa with a matched assembly).

| Taxid  | Organism                                                         | Assembly         | $n$   | $n_{ribo}$ | $\rho_{gen}$ | $\rho_{ribo}$ | $q$      |
|--------|------------------------------------------------------------------|------------------|-------|------------|--------------|---------------|----------|
| 246200 | Ruegeria pomeroyi DSS-3                                          | GCF_000011965.2  | 395   | 70         | 0.2714       | 0.46          | 1.2e-22  |
| 2903   | Emiliana huxleyi CCMP1516                                        | GCF_000372725.1  | 484   | 4          | 0.4636       | –             | –        |
| 343509 | Sodalis glossinidius str. 'morsitans'                            | GCF_000010085.1  | 748   | 72         | 0.2849       | 0.4703        | 1.2e-43  |
| 36329  | Plasmodium falciparum 3D7                                        | GCF_000002765.6  | 4946  | 62         | 0.056        | 0.0253        | 0.078    |
| 403677 | Phytophthora infestans T30-4                                     | GCF_000142945.1  | 708   | 0          | 0.5278       | –             | –        |
| 4081   | Solanum lycopersicum                                             | GCF_036512215.1  | 426   | 4          | -0.1336      | –             | –        |
| 4097   | Nicotiana tabacum                                                | GCF_000715075.1  | 9304  | 0          | -0.0837      | –             | –        |
| 418985 | Tropilaelaps mercedesae                                          | GCA_002081605.1  | 2213  | 0          | 0.2207       | –             | –        |
| 44689  | Dictyostelium discoideum AX4                                     | GCF_000004695.1  | 3183  | 10         | -0.3039      | –             | –        |
| 4932   | Saccharomyces cerevisiae S288C                                   | GCF_000146045.2  | 5587  | 226        | 0.4099       | 0.5924        | 0        |
| 511145 | Escherichia coli str. K-12 substr. MG1655                        | GCF_000005845.2  | 3486  | 72         | 0.4961       | 0.5809        | 0        |
| 54291  | Klebsiella ornithinolytica                                       | GCF_030505655.1  | 1038  | 72         | 0.2053       | 0.5268        | 1.2e-78  |
| 6239   | Caenorhabditis elegans                                           | GCF_000002985.6  | 11571 | 0          | 0.0748       | –             | –        |
| 7227   | Drosophila melanogaster                                          | GCF_000001215.4  | 2251  | 150        | 0.0189       | 0.0813        | 0.00012  |
| 7460   | Apis mellifera                                                   | GCF_003254395.2  | 879   | 4          | -0.0468      | –             | –        |
| 8030   | Salmo salar                                                      | GCF_965601325.2  | 1779  | 0          | 0.1332       | –             | –        |
| 83332  | Mycobacterium tuberculosis H37Rv                                 | GCF_000195955.2  | 3258  | 76         | 0.2304       | 0.2373        | 6.6e-43  |
| 9031   | Gallus gallus                                                    | GCF_016699485.2  | 1891  | 68         | 0.2134       | 0.2274        | 1.6e-23  |
| 93061  | Staphylococcus aureus subsp. aureus NCTC 8325                    | GCF_000013425.1  | 1699  | 58         | 0.5221       | 0.6046        | 3.6e-182 |
| 9544   | Macaca mulatta                                                   | GCF_049350105.2  | 3732  | 84         | 0.0736       | 0.0808        | 1e-06    |
| 9606   | Homo sapiens                                                     | GCF_000001405.40 | 12343 | 86         | -0.0083      | 0.0106        | 0.24     |
| 9615   | Canis lupus familiaris                                           | GCF_011100685.1  | 329   | 81         | 0.1639       | 0.1673        | 0.0025   |
| 9823   | Sus scrofa                                                       | GCF_054392235.1  | 3010  | 84         | 0.0576       | 0.071         | 0.00011  |
| 9913   | Bos taurus                                                       | GCF_002263795.3  | 2694  | 81         | 0.0963       | 0.1177        | 1.2e-09  |
| 99287  | Salmonella enterica subsp. enterica serovar Typhimurium str. LT2 | GCF_000006945.2  | 2472  | 72         | 0.3119       | 0.4743        | 4.4e-144 |
| 9986   | Oryctolagus cuniculus                                            | GCF_964237555.1  | 1993  | 82         | 0.0999       | 0.1012        | 7.4e-06  |

$q$ : Fisher-z two-sided test of  $\rho_{ribo}$ , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep\_codon\_bh\_fdr.json).

## 10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman  $\rho$  with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman  $\rho$ ).

| Organism                                   | $n$  | $CAI_{gen}$ | $CAI_{ribo}$ | tAI    | –ENC   |
|--------------------------------------------|------|-------------|--------------|--------|--------|
| Escherichia coli str. K-12 substr. MG1655  | 3450 | 0.4961      | 0.5809       | 0.5277 | 0.3836 |
| Bacillus subtilis subsp. subtilis str. 168 | 3115 | 0.3295      | 0.4156       | 0.4146 | 0.1735 |
| Saccharomyces cerevisiae S288C             | 5474 | 0.4099      | 0.5924       | 0.6604 | 0.4471 |

## 11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD  $< 2$  Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.



Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

| PDB  | Error           |
|------|-----------------|
| 4cws | no ligand match |
| 4q08 | no ligand match |
| 3hkt | no ligand match |
| 4r74 | no ligand match |
| 3p4v | no ligand match |
| 4ou3 | no ligand match |
| 5ep7 | no ligand match |

Table 5: Per-complex docking results.

| PDB  | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 1cnw | 7.72 | 26    | -4.19       | -5.37     | -7.02     | 7.26              | 6.03              |
| 1o3l | 6.54 | 24    | -7.09       | -8.63     | -9.54     | 3.19              | 1.09              |
| 3po6 | 7.06 | 19    | -4.54       | -6.13     | -7.14     | 3.81              | 0.86              |
| 4db7 | 6.56 | 18    | -6.63       | -7.18     | -7.41     | 0.52              | 0.52              |
| 5sz5 | 7.7  | 17    | -7.33       | -7.33     | -8.14     | 0.39              | 0.39              |
| 4mo8 | 8.0  | 16    | -4.75       | -5.50     | -6.25     | 2.29              | 1.52              |
| 4tim | 2.16 | 11    | -3.41       | -4.13     | -5.59     | 1.66              | 1.66              |
| 1ejn | 5.62 | 25    | -8.20       | -8.43     | -7.93     | 2.51              | 2.15              |
| 5d3l | 6.06 | 30    | -6.22       | -6.32     | -6.70     | 8.34              | 1.64              |
| 3b3c | 3.98 | 10    | -5.07       | -5.76     | -6.48     | 1.42              | 1.42              |
| 3r4n | 6.89 | 23    | -6.79       | -7.37     | -7.66     | 0.81              | 0.81              |
| 2p4j | 8.96 | 49    | -10.13      | -10.73    | -11.07    | 1.58              | 1.58              |
| 5sz4 | 9.7  | 16    | -7.04       | -7.33     | -7.65     | 1.79              | 1.79              |
| 4xit | 7.27 | 32    | -12.19      | -12.80    | -13.06    | 0.96              | 0.96              |
| 5mjn | 8.54 | 18    | -6.42       | -6.93     | -7.76     | 5.97              | 1.41              |
| 2xjg | 7.82 | 23    | -9.04       | -9.53     | -9.72     | 0.25              | 0.25              |
| 5ix0 | 6.28 | 25    | -11.86      | -12.14    | -12.27    | 0.27              | 0.27              |
| 5y8y | 7.09 | 25    | -6.08       | -6.38     | -8.16     | 4.75              | 0.84              |
| 6g38 | 6.62 | 19    | -7.36       | -7.70     | -8.12     | 0.61              | 0.61              |
| 4o04 | 6.16 | 26    | -11.03      | -12.14    | -12.36    | 1.34              | 1.34              |
| 5ewk | 4.85 | 22    | -7.91       | -7.96     | -8.57     | 0.68              | 0.68              |

Table 5: Per-complex docking results.

| PDB  | pK   | heavy | $E_{cryst}$ | $E_{min}$ | $E_{top}$ | RMSD <sub>1</sub> | RMSD <sub>9</sub> |
|------|------|-------|-------------|-----------|-----------|-------------------|-------------------|
| 3d8w | 8.15 | 19    | -6.96       | -7.48     | -8.20     | 1.43              | 1.43              |
| 4rfc | 8.24 | 23    | -5.25       | -6.44     | -8.37     | 7.44              | 5.67              |
| 3r17 | 6.41 | 20    | -6.73       | -7.38     | -7.74     | 3.57              | 2.30              |
| 4knj | 7.95 | 21    | -6.40       | -7.32     | -8.23     | 4.81              | 1.87              |
| 6g6t | 7.7  | 19    | -7.42       | -7.76     | -8.31     | 2.10              | 0.61              |
| 4lxz | 6.74 | 19    | -6.37       | -6.47     | -7.29     | 6.21              | 1.95              |
| 5o87 | 6.61 | 12    | -5.46       | -6.58     | -6.70     | 0.44              | 0.44              |
| 4j93 | 4.7  | 21    | -7.50       | -7.81     | -8.02     | 0.44              | 0.44              |
| 6hrq | 7.09 | 23    | -6.44       | -6.94     | -7.94     | 6.23              | 2.21              |
| 4nh7 | 8.4  | 33    | -11.46      | -12.05    | -12.52    | 1.31              | 1.31              |
| 2x7t | 6.63 | 30    | -6.30       | -7.34     | -7.81     | 6.16              | 0.77              |
| 4fl1 | 3.28 | 31    | -5.80       | -5.98     | -7.02     | 9.18              | 2.47              |
| 3c84 | 7.85 | 16    | -6.59       | -6.98     | -7.23     | 0.39              | 0.39              |
| 4ytc | 7.89 | 26    | -9.65       | -10.13    | -10.15    | 0.49              | 0.49              |
| 2zdn | 6.25 | 27    | -7.86       | -8.84     | -9.05     | 1.02              | 1.02              |
| 5f60 | 7.37 | 37    | -8.55       | -8.90     | -9.99     | 4.69              | 1.63              |
| 6hqy | 6.44 | 22    | -5.86       | -6.97     | -7.28     | 2.03              | 1.48              |
| 5l8c | 7.0  | 35    | -8.93       | -9.64     | -11.31    | 9.53              | 0.86              |
| 1p1o | 5.76 | 13    | -7.26       | -7.37     | -7.79     | 0.60              | 0.60              |
| 5htz | 8.62 | 26    | -9.05       | -9.25     | -9.40     | 0.19              | 0.19              |
| 4oag | 6.3  | 27    | -5.33       | -7.22     | -7.84     | 9.16              | 3.85              |
| 2j4i | 9.0  | 36    | -8.00       | -8.56     | -8.54     | 1.38              | 1.38              |
| 5n24 | 8.92 | 18    | -7.04       | -7.44     | -7.80     | 1.68              | 1.40              |
| 1y3x | 5.14 | 33    | -9.50       | -9.83     | -10.15    | 0.70              | 0.70              |
| 1f5l | 5.28 | 15    | -6.27       | -7.20     | -7.32     | 2.87              | 1.51              |
| 5aml | 9.15 | 21    | -6.79       | -6.99     | -7.74     | 5.22              | 1.14              |
| 6u6w | 5.33 | 21    | -7.71       | -7.69     | -8.04     | 5.13              | 0.33              |
| 5mpn | 7.07 | 23    | -6.86       | -7.18     | -8.21     | 7.46              | 1.25              |
| 4jfm | 5.48 | 35    | -6.54       | -7.29     | -7.74     | 0.97              | 0.97              |
| 5ewy | 4.3  | 22    | -7.30       | -7.87     | -8.78     | 3.50              | 2.90              |
| 4zme | 4.3  | 19    | -6.04       | -6.27     | -6.60     | 10.33             | 0.18              |
| 4yxi | 6.5  | 11    | -5.93       | -6.43     | -6.62     | 0.39              | 0.39              |
| 2erz | 5.66 | 21    | -7.23       | -8.31     | -8.65     | 2.99              | 0.93              |
| 6oe0 | 8.92 | 27    | -7.95       | -8.86     | -9.35     | 0.86              | 0.86              |
| 4b2i | 3.0  | 9     | -4.33       | -5.04     | -5.25     | 1.97              | 0.61              |
| 4f5y | 5.43 | 46    | -10.70      | -11.84    | -10.97    | 3.66              | 3.66              |
| 6eq5 | 4.74 | 10    | -6.31       | -6.71     | -7.17     | 2.09              | 0.51              |
| 5g5v | 6.7  | 36    | -10.54      | -10.73    | -11.46    | 2.75              | 1.52              |
| 5cau | 8.8  | 34    | -9.64       | -9.84     | -9.93     | 0.65              | 0.65              |
| 1gi1 | 5.47 | 19    | -6.60       | -8.25     | -8.38     | 0.77              | 0.77              |
| 3hkq | 5.34 | 15    | -4.38       | -5.42     | -6.47     | 3.67              | 3.04              |
| 2xab | 9.27 | 22    | -9.46       | -9.81     | -9.94     | 0.21              | 0.21              |
| 3hll | 8.24 | 30    | -9.62       | -10.25    | -10.25    | 0.36              | 0.36              |
| 5f62 | 8.17 | 38    | -8.66       | -9.01     | -9.80     | 4.22              | 1.46              |
| 3ibi | 8.59 | 13    | -4.21       | -4.73     | -4.97     | 5.02              | 3.28              |
| 3m35 | 7.22 | 37    | -8.52       | -9.07     | -9.52     | 0.90              | 0.90              |
| 1g54 | 8.82 | 25    | -7.03       | -7.63     | -8.00     | 8.95              | 8.89              |
| 4r76 | 7.85 | 26    | -7.58       | -8.26     | -8.82     | 1.62              | 1.62              |
| 1zgi | 5.34 | 32    | -9.23       | -9.64     | -9.98     | 0.76              | 0.76              |
| 5cap | 7.28 | 29    | -8.09       | -8.54     | -8.89     | 1.44              | 1.44              |
| 1klj | 7.55 | 37    | -8.36       | -8.62     | -9.29     | 1.77              | 1.77              |
| 6rfn | 6.7  | 35    | -7.10       | -8.72     | -9.63     | 6.69              | 1.10              |

## 12 Structure and Primer Utilities Against Reference Tools

**SASA.** Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

**Primers.** On 5404 20-mers from NM\_000546.6, NM\_007294.4, NM\_000059.4, NM\_004333.6, NM\_005228.5, the melting temperature equals Biopython's DNA\_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin  $\Delta G$ , a negative result; the module now calls primer3 when installed.

### 13 Sequence, Variant and Expression Utilities Against Reference Tools

**Phylogenetics.** DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj\_tree/upgma\_tree, treecompare.symmetric\_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

**ORFs.** On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min\_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

**VCF.** 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

**Differential expression.** GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with  $\geq 10$  counts in  $\geq 4$  samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at  $\text{padj} < 0.05$  (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

| Gene     | log2FC | padj PyDESeq2 | padj Welch | padj Wilcoxon |
|----------|--------|---------------|------------|---------------|
| FKBP5    | 4.03   | 2.1e-26       | 0.06       | 0.15          |
| TSC22D3  | 3.206  | 2.1e-19       | 0.16       | 0.15          |
| PER1     | 3.221  | 2.3e-99       | 0.05       | 0.15          |
| DUSP1    | 2.955  | 1.3e-140      | 0.07       | 0.15          |
| KLF15    | 4.411  | 5.0e-114      | 0.17       | 0.15          |
| ZBTB16   | 7.432  | 1.1e-58       | 0.12       | 0.15          |
| CRISPLD2 | 2.628  | 1.7e-61       | 0.13       | 0.15          |

## 14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design  $\sim \text{cell} + \text{dex}$  gave 4,451 genes at adjusted  $p < 0.05$  and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of  $1.8\text{e-}09$  (benchmarks/sweep\_de\_pydeseq2.json).

**(F25) Negative-binomial GLM.** For gene  $g$  and sample  $j$ ,  $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$  with  $\mu_{gj} = s_j q_{gj}$  and  $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$ , so  $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$ .

**(F26) Median-of-ratios size factor.**  $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$ .

**(F27) Wald statistic.**  $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$ , compared with  $N(0, 1)$ ; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

### 14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted  $p < 0.05$ ,  $\log_2 \text{FC} > 1$ ) and tested them against the 16,802 tested genes with g:Profiler (version e114\_eg62\_p19\_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

**(F28) Hypergeometric tail.** With background size  $M$ , term size  $K$ , query size  $n$  and overlap  $k$ ,  $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$ , and fold enrichment  $= (k/n) / (K/M)$ .

| Source | Term                             | overlap | size | $p_{\text{adj}}$ |
|--------|----------------------------------|---------|------|------------------|
| GO:BP  | multicellular organismal process | 228     | 4764 | 2.8e-07          |
| GO:BP  | signaling                        | 209     | 4354 | 4e-06            |
| GO:BP  | cell communication               | 209     | 4365 | 5.2e-06          |
| GO:BP  | cell adhesion                    | 73      | 1031 | 1.5e-05          |
| GO:BP  | system process                   | 80      | 1189 | 2.1e-05          |
| GO:BP  | response to stimulus             | 261     | 5987 | 8.6e-05          |
| GO:BP  | signal transduction              | 192     | 4028 | 8.9e-05          |
| GO:BP  | anatomical structure development | 196     | 4191 | 0.00027          |
| GO:BP  | cell migration                   | 74      | 1134 | 0.00041          |
| KEGG   | Cytoskeleton in muscle cells     | 19      | 172  | 0.00042          |
| GO:BP  | developmental process            | 209     | 4580 | 0.00047          |
| GO:BP  | cell differentiation             | 152     | 3041 | 0.00054          |

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep\_dex\_enrichment.json).

**Negative result.** The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.

| Term                                         | overlap | size | fold (approx.) | $p_{g.SCS}$ |
|----------------------------------------------|---------|------|----------------|-------------|
| cellular response to glucocorticoid stimulus | 9       | 43   | 7.16           | 0.43        |
| response to corticosteroid                   | 10      | 91   | 3.76           | 1           |
| response to glucocorticoid                   | 10      | 81   | 4.22           | 1           |
| cellular response to corticosteroid stimulus | 9       | 48   | 6.42           | 1           |

Table 8: Glucocorticoid terms: large fold change, not significant.

## 14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached  $FDR < 0.05$ ; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

| Pathway                                            | found | total | $p$     | FDR  |
|----------------------------------------------------|-------|-------|---------|------|
| Anchoring fibril formation                         | 5     | 15    | 0.00037 | 0.11 |
| FOXO-mediated transcription                        | 13    | 110   | 0.00056 | 0.11 |
| Laminin interactions                               | 7     | 31    | 0.00028 | 0.11 |
| Metallothioneins bind metals                       | 5     | 16    | 0.00049 | 0.11 |
| Non-integrin membrane-ECM interactions             | 12    | 83    | 0.00015 | 0.11 |
| Collagen chain trimerization                       | 8     | 44    | 0.00044 | 0.11 |
| FOXO-mediated transcription of oxidative stress, m | 8     | 49    | 0.00087 | 0.13 |
| Adipogenesis                                       | 15    | 147   | 0.00096 | 0.13 |
| Degradation of the extracellular matrix            | 15    | 148   | 0.001   | 0.13 |
| Response to metal ions                             | 5     | 21    | 0.0016  | 0.17 |

Table 9: Top Reactome pathways (benchmarks/sweep\_dex\_reactome.json).

**(F29) Benjamini-Hochberg.** For ordered  $p_{(1)} \leq \dots \leq p_{(m)}$ ,  $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$ .

## 14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported  $p$  below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

**(F30) Edge enrichment ratio.**  $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$ .

## 14.4 Network hubs

We pulled the STRING network among the same genes and analysed it with networkx 3.4.2 on STRING API/network (v12): 201 connected nodes, 402 edges, a largest component of 183 nodes. The highest-degree hubs are FOXO1 (21), CEBPD (15), FOXO3 (14), LEP (13), COL4A1 (13), GPX3 (13). Of ten canonical glucocorticoid-receptor targets fixed before the run, 8 are in the network, with mean degree 6.25 against 4.00 overall; a degree permutation test (10,000 draws) gives  $p = 0.049$ . With eight genes this is weak evidence and we do not claim it as a finding.

**(F34) Betweenness centrality.**  $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$ , with  $\sigma_{st}$  the number of shortest  $s$ - $t$  paths and  $\sigma_{st}(v)$  those through  $v$ .

**(F35) Permutation  $p$ -value.**  $\hat{p} = (1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$ , which is never zero and is valid for any fixed statistic  $T$ .

## 14.5 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed  $FDR < 0.05$  and 5 passed  $FDR < 0.25$  (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

**(F32) Enrichment score.** With ranked scores  $r_j$  and set  $S$ ,  $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$ ,  $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$ , and  $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$  (signed).

**(F33) Normalised ES.**  $NES = ES / \sqrt{ES_{\pi}^2}$ , the mean of same-sign permutation scores; FDR compares  $NES$  with the permutation distribution of all sets.

| Hallmark set                    | NES   | nominal $p$ | FDR   |
|---------------------------------|-------|-------------|-------|
| Adipogenesis                    | 1.70  | 0           | 0.038 |
| UV Response Up                  | 1.55  | 0.00567     | 0.107 |
| Fatty Acid Metabolism           | 1.55  | 0.00559     | 0.131 |
| TNF-alpha Signaling via NF-kB   | 1.50  | 0.00273     | 0.141 |
| Reactive Oxygen Species Pathway | 1.57  | 0.0249      | 0.156 |
| Androgen Response               | 1.39  | 0.0557      | 0.256 |
| Xenobiotic Metabolism           | 1.42  | 0.0287      | 0.266 |
| Hypoxia                         | 1.39  | 0.0294      | 0.275 |
| UV Response Dn                  | 1.36  | 0.0591      | 0.280 |
| mTORC1 Signaling                | -1.31 | 0.0361      | 0.283 |

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep\_dex\_gsea\_hallmark.json).

## 14.6 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at  $FDR < 0.05$ . Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches  $FDR 0.026$  under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

## 15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty

| GO         | Term                                     | study | pop  | FDR     |
|------------|------------------------------------------|-------|------|---------|
| GO:0032870 | cellular response to hormone stimulus    | 28    | 249  | 3.4e-05 |
| GO:0009725 | response to hormone                      | 38    | 421  | 3.4e-05 |
| GO:0007155 | cell adhesion                            | 39    | 513  | 0.00098 |
| GO:0032502 | developmental process                    | 132   | 2766 | 0.0011  |
| GO:0007165 | signal transduction                      | 130   | 2703 | 0.0011  |
| GO:0030334 | regulation of cell migration             | 47    | 689  | 0.0011  |
| GO:2000145 | regulation of cell motility              | 48    | 719  | 0.0012  |
| GO:0044057 | regulation of system process             | 28    | 322  | 0.0015  |
| GO:0071495 | cellular response to endogenous stimulus | 33    | 419  | 0.0022  |
| GO:0040012 | regulation of locomotion                 | 48    | 740  | 0.0022  |

Table 11: goatools GO:BP (benchmarks/sweep\_dex\_goatools.json).

12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes gap.open = -11 is the cost of the first gap residue, i.e. parasail open 11.

**(F31) Affine gap cost.** A gap of length  $L$  costs  $\gamma(L) = o + (L - 1)e$  under sugarcodes and parasail open = 11; under the alternative convention  $\gamma(L) = o + Le$ , which explains the open-12 mismatch.



Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

## 16 Sequence properties, restriction digests and file formats

This section checks four bio/ libraries against Biopython 1.88 ProtParam + Restriction and gffutils. Three pass exactly. One, the restriction-site finder, had a real bug that we fixed.

### 16.1 Protein physicochemical properties

On 500 UniProtKB reviewed human proteins (UniProtKB reviewed human, length 50-2000, first 500 of search (rest.uniprot.org, 2026-09-25)), every property matches ProtParam (Table 12).

| property     | $n$ | within tol.     | max abs diff | Pearson      |
|--------------|-----|-----------------|--------------|--------------|
| mw           | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| gravy        | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| aromaticity  | 500 | 500 (rel 1e-06) | 2.8e-17      | 1.0000000000 |
| instability  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| pI           | 500 | 500 (abs 0.01)  | 5.9e-05      | 0.9999999999 |
| ext_reduced  | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |
| ext_oxidized | 500 | 500 (rel 1e-06) | 0            | 1.0000000000 |

Table 12: bio.proteinprops vs Biopython ProtParam (benchmarks/sweep\_proteinprops\_restriction.json).

**(F36) Average molecular weight.**  $M = \sum_a n_a m_a - (L - 1) m_{\text{H}_2\text{O}}$ , with  $m_a$  the average residue mass (free amino acid) and one water lost per peptide bond.

**(F37) Net charge.**  $Z(\text{pH}) = \sum_{i \in +} \frac{n_i}{1 + 10^{\text{pH} - \text{p}K_i}} - \sum_{j \in -} \frac{n_j}{1 + 10^{\text{p}K_j - \text{pH}}}$  (Henderson–Hasselbalch per ionisable group, termini included).

**(F38) Isoelectric point.** pI is the root of  $Z(\text{pH}) = 0$ ;  $Z$  is strictly decreasing in pH, so bisection on  $[0, 14]$  converges, and the residual  $6 \times 10^{-5}$  difference is the bisection tolerance.

**(F39) Instability index.**  $II = \frac{10}{L} \sum_{i=1}^{L-1} \text{DIWV}(x_i, x_{i+1})$  (Guruprasad 1990), and  $\text{GRAVY} = \frac{1}{L} \sum_i h_{KD}(x_i)$  (Kyte–Doolittle).

**(F40) Extinction coefficient at 280 nm.**  $\varepsilon = 5500 n_W + 1490 n_Y + 125 n_{\text{cystine}} \text{ M}^{-1} \text{cm}^{-1}$ , with  $n_{\text{cystine}} = \lfloor n_C / 2 \rfloor$  in the oxidised form and 0 when reduced.

## 16.2 Restriction sites: a bug in degenerate recognition sequences

We digested pBR322 (J01749.1, circular, 4,361 bp) and phage lambda (NC\_001416.1, linear, 48,502 bp) with all 610 enzymes and compared cut positions with Biopython Restriction. Before the fix (sugarcode-ai 699fee), 533 and 529 enzymes agreed. Two causes explained the disagreements. First, the reverse-strand pattern for a non-palindromic-looking site was built with a plain DNA reverse complement that leaves IUPAC codes unchanged. AccI (GT<sup>^</sup>MKAC) is in fact palindromic under IUPAC complementation ( $M \leftrightarrow K$ ), but the plain reverse complement GTKMAC was searched as a second pattern, which matches GTGAAC and GTTCAC, neither an AccI site. On pBR322 this gave 5 AccI sites where Biopython finds 2. Second, on linear DNA the finder reported Type IIS cuts that fall outside the molecule. After the fix (sugarcode-ai fe2185d), 608 and 608 enzymes agree. The remaining two, LpnPI, MspJI, are modification-dependent enzymes that do not cut unmethylated DNA; we leave them open.

**(F41) IUPAC complement.** The map  $c$ :  $A \leftrightarrow T$ ,  $C \leftrightarrow G$ ,  $R \leftrightarrow Y$ ,  $K \leftrightarrow M$ ,  $B \leftrightarrow V$ ,  $D \leftrightarrow H$ , with S, W, N fixed, is the unique involution on the 15 codes that complements every base in the set a code denotes. A site  $s$  is palindromic iff  $s = \overline{c(s)}$  (reversed), which for AccI holds only under  $c$ , not under the four-letter map.

### 16.3 Stockholm alignments

bio.stockholm parsed 15 Rfam seed alignments (4,471 sequences). 15/15 match Biopython on sequence identifiers and order, aligned sequences and SS\_cons, and survive a write–parse round trip unchanged.

| file        | AC      | seqs | width | all checks |
|-------------|---------|------|-------|------------|
| RF00001.sto | RF00001 | 712  | 230   | yes        |
| RF00005.sto | RF00005 | 954  | 118   | yes        |
| RF00010.sto | RF00010 | 458  | 996   | yes        |
| RF00004.sto | RF00004 | 208  | 278   | yes        |
| RF00015.sto | RF00015 | 177  | 261   | yes        |
| RF00017.sto | RF00017 | 91   | 412   | yes        |
| RF00023.sto | RF00023 | 477  | 954   | yes        |
| RF00050.sto | RF00050 | 146  | 221   | yes        |
| RF00059.sto | RF00059 | 115  | 211   | yes        |
| RF00162.sto | RF00162 | 457  | 239   | yes        |
| RF00167.sto | RF00167 | 133  | 113   | yes        |
| RF00168.sto | RF00168 | 47   | 274   | yes        |
| RF00174.sto | RF00174 | 434  | 461   | yes        |
| RF00234.sto | RF00234 | 18   | 236   | yes        |
| RF00504.sto | RF00504 | 44   | 215   | yes        |

Table 13: bio.stockholm vs Biopython AlignIO (benchmarks/sweep\_stockholm\_biopython.json).

**(F42) Pairwise identity.**  $PID(x, y) = \#\{i : x_i = y_i, x_i, y_i \notin \{-, .\}\} / \#\{i : x_i, y_i \notin \{-, .\}\}$ , case-insensitive; our implementation equals an independent count on every file.

### 16.4 GFF3 annotation

On the RefSeq E. coli K-12 annotation (9,523 records), bio.gff and gffutils 0.14 (sqlite, in-memory) agree on every record (9,523/9,523) and on the children of 300/300 genes, and write–parse is lossless. A first comparison showed two child mismatches; both were CDS features split over two lines with a shared ID, which gffutils renames X\_1. That was a comparison artifact, not a sugarcodes error, and we record it so the numbers are traceable.

| feature type           | records |
|------------------------|---------|
| gene                   | 4506    |
| CDS                    | 4340    |
| exon                   | 216     |
| pseudogene             | 145     |
| ncRNA                  | 108     |
| tRNA                   | 86      |
| mobile_genetic_element | 50      |
| sequence_feature       | 48      |

Table 14: Feature types in GCF\_000005845.2 (benchmarks/sweep\_gff\_gffutils.json).

## 16.5 Newick trees

bio.newick parsed 6 Open Tree of Life synthetic subtrees with 3,655 leaves in total, including quoted labels with spaces and parentheses and empty leaves left by the height limit. Leaf sets equal Biopython Bio.Phylo and DendroPy, internal-node counts and internal labels equal Bio.Phylo, and write–parse is lossless in 6/6 trees. These trees have no branch lengths, so this checks topology and labels only.

| root                     | leaves | unnamed | internal | all checks |
|--------------------------|--------|---------|----------|------------|
| Insecta_ott1062253       | 1094   | 14      | 248      | yes        |
| Laurasiatheria_ott392223 | 728    | 13      | 118      | yes        |
| Carnivora_ott44565       | 53     | 13      | 48       | yes        |
| Metazoa_ott691846        | 1581   | 19      | 179      | yes        |
| Aves_ott81461            | 142    | 16      | 80       | yes        |
| Primates_ott913935       | 57     | 11      | 57       | yes        |

Table 15: bio.newick vs Bio.Phylo/DendroPy (benchmarks/sweep\_newick\_biophylo.json).

**(F43) Tree size identity.** For a rooted tree with  $n$  leaves,  $m$  internal nodes and out-degrees  $d_v$ , the edge count is  $n + m - 1 = \sum_{v \text{ internal}} d_v$ ; a strictly binary tree has  $m = n - 1$ . The Carnivora subtree has 53 leaves and 48 internal nodes, so it is far from binary, which exercises the parser on polytomies.

## 16.6 SAM records and a second bug

bio.sam parsed 10,943 real reads (1000 Genomes phase 3 NA12878 chrom20 low-coverage BAM, 20:1000000-1200000). Name, flag, reference, position, mapping quality, CIGAR operations, mate fields, sequence, qualities and typed tags equal pysam for all 10,943 reads. The derived reference end did not: 10,905/10,943 before the fix. The 38 failures were unmapped mates (flag 0x4) that bwa places at the mapped mate’s position and gives a CIGAR. The SAM specification says those fields are not an alignment, and htslib reports no end; sugarcodes computed one. After sugarcodes-ai 87cf78e all 10,943 agree.

**(F44) Reference span of a CIGAR.**  $\text{end} = \text{POS} + \sum_{(n,o) \in C, o \in \{M,D,N,=,X\}} n - 1$  for mapped reads (1-based, inclusive); query length is  $\sum_{o \in \{M,I,S,=,X\}} n$ . For flag &4  $\neq 0$  the end is undefined.

## 16.7 FASTQ reads

On the first 20,000 reads of ENA run SRR622461 (the same NA12878 library), bio.fastq equals Biopython on identifiers (20,000), sequences (20,000) and per-base Phred scores (20,000), detects offset 33, and round-trips losslessly. GC was 0.4252 against 0.4252 (Biopython, ambiguous bases removed) and 0.422 (ambiguous bases kept in the denominator). The difference is a convention: sugarcodes divides by A+C+G+T. We document it rather than change it.

**(F45) Phred score.**  $Q = -10 \log_{10} P_{err}$ , encoded as ASCII  $Q + 33$ ; the mean read quality reported is the arithmetic mean of  $Q$ , not  $-10 \log_{10}$  of the mean error probability, which would be lower for reads with a few bad bases.

## 16.8 PDB and mmCIF structures: a third bug

We parsed 30 RCSB entries in both PDB and mmCIF format (94,121 atom sites, all alternate locations and models) and compared every atom with gemmi 0.7.5. The PDB-format parser was identical in 30/30 entries. The mmCIF parser failed on all 30 real files (error: loop\_ row count 3 is not a multiple of 4 columns (all 30 files)). Two causes: POSIX shell-style splitting broke atom names containing a prime ( $O5'$ ,  $C1'$ ), and loops were closed early when a row contained an unquoted item name as a value, which RCSB files do in the revision-history tables. A CIF 1.1 tokenizer (a quote closes only when followed by whitespace) and closing loops only at row boundaries fixed both (sugarcode-ai fc1fd1e); mmCIF is now identical to gemmi in 30/30 entries. Unit tests with toy CIF strings had passed throughout. Only real files exposed the bug.

**(F46) Loop row invariant.** A loop with  $c$  columns and  $t$  value tokens is well formed iff  $t \equiv 0 \pmod{c}$ ; closing the loop at the first reserved-looking token is only safe when the running count  $t \bmod c = 0$ , which is the rule we now apply.

## 16.9 Pileup

Using the same reads, bio.pileup and pysam 0.24.1 pileup (stepper nofilter, flag\_filter=4, no quality filters, overlaps kept) agree on base counts at 197,437/197,437 positions and on deletion counts at 197,437/197,437. sugarcode also writes deletions as ‘\*’ in the base counts, following samtools mpileup; the comparison maps these to the deletion count.

**(F49) Column depth.**  $D(p) = \sum_r \mathbb{I}[p \in [\text{POS}_r, \text{end}_r] \wedge \text{op}_r(p) \in \{M, =, X, D\}]$ , split into base-covering reads ( $M, =, X$ ) and deletions ( $D$ ); reference skips ( $N$ ) and clipped bases do not count.

## 16.10 Transcription-factor motifs

We read 10 JASPAR 2024 profiles and scanned BRCA1 (10 kb) and phage lambda (48.5 kb) on both strands, comparing with Biopython 1.88 Bio.motifs (jaspar reader, pseudocount 0.5, uniform background, log2). Log-odds matrices agree to  $2 \times 10^{-15}$ , per-window scores to  $4 \times 10^{-6}$  (Biopython stores scores in single precision), and the hit sets at 80% of the score range are identical for every motif and sequence.

| motif             | length | hits BRCA1 | hits lambda |
|-------------------|--------|------------|-------------|
| MA0003.4 TFAP2A   | 14     | 27         | 61          |
| MA0079.3 SP1      | 11     | 235        | 599         |
| MA0080.4 SPI1     | 14     | 16         | 61          |
| MA0099.3 FOS::JUN | 10     | 77         | 286         |
| MA0105.4 NFkB1    | 13     | 29         | 52          |
| MA0106.3 TP53     | 18     | 8          | 21          |
| MA0108.2 TBP      | 15     | 154        | 530         |
| MA0112.3 ESR1     | 17     | 11         | 18          |
| MA0113.3 NR3C1    | 17     | 2          | 36          |
| MA0139.1 CTCF     | 19     | 13         | 40          |

Table 16: Motif hits at 80% of range, identical to Biopython (benchmarks/sweep\_motif\_biopython.json).

**(F50) Log-odds score and relative threshold.**  $S(w) = \sum_{i=1}^L \log_2 \frac{(c_{i,w_i} + \alpha)/(N_i + 4\alpha)}{q_{w_i}}$  with pseudocount  $\alpha = 0.5$  and background  $q = 1/4$ ; a window is a hit when  $S(w) \geq S_{\min} + f(S_{\max} - S_{\min})$ ,  $f = 0.8$ , where  $S_{\max} = \sum_i \max_b$  and  $S_{\min} = \sum_i \min_b$  of the log-odds matrix.

## 16.11 BED intervals: a fourth bug

We fetched two hg38 tracks from the UCSC REST API: CpG islands on chr20 (847 intervals) and RepeatMasker on chr20:0–5 Mb (11,699 intervals), wrote them as BED and compared bio.bed with bioframe 0.8.0 (merge, overlap). Parsing and 600 random overlap queries agreed exactly (300/300 and 300/300). merge\_intervals did not: its docstring promised bedtools-style merging, yet it refused to join book-ended intervals. RepeatMasker has 3,112 book-ended pairs, so sugarcodes returned 11,364 merged intervals where bedtools/bioframe return 8,251. CpG islands have no book-ended pairs and hid the bug (847 = 847). We added a bedtools-compatible min\_dist argument (default 0; -1 for overlap-only) and a matching CLI flag; after the fix both tracks match bioframe exactly (8,251 = 8,251).

**(F51) Merge rule.** For intervals sorted by start with running end  $e^*$ , interval  $[s, e)$  joins the current block iff  $s \leq e^* + d$ ;  $d = 0$  joins book-ended intervals ( $s = e^*$ ) and  $d = -1$  requires at least one shared base ( $s < e^*$ ).

## 16.12 GenBank flat files: a fifth bug

We fetched 25 GenBank records one by one from NCBI nuccore (phage, viral, mitochondrial, bacterial, RefSeqGene and 13 human RefSeq mRNAs) and compared bio.genbank with Biopython 1.88 SeqIO genbank on 1,867 features. Sequences and feature counts already agreed, but only 1,829/1,867 feature locations and 5/302 CDS /translation qualifiers matched. The parser appended every continuation line of a multi-line qualifier to the feature location instead of to the qualifier: /translation kept its first line only, and when a continuation began with digits the location itself was corrupted (E. coli thrA CDS: 337..2799 became 337..27991). Partial locations ( $\text{;1..;200}$ ) and single bases produced no spans at all. We rewrote qualifier continuation (INSDC joining: no spaces for /translation, one space otherwise, doubled quotes unescaped), added a location parser for join/order, complement, partial ends, between-base sites and remote parts, and kept all IUPAC letters in ORIGIN so coordinates cannot shift. After the fix: spans 1,867/1,867, strands 1,867/1,867, /translation 302/302, CDS extraction 302/302. The fix exposed a downstream case: KCNQ1's record NG\_016178.2 carries only a partial CDS, complement( $\text{;42918..;43038}$ ). The old parser read no spans and bio.splice reported 'no CDS spans'; with spans read, the junction map came back 'ok' but empty. We made it fail loud with status 'partial CDS in record'. The splice-region discovery pipeline reads hg38 directly and does not use this parser, so its results are unchanged.

**(F52) Qualifier continuation.** For a qualifier whose value spans lines  $\ell_1, \dots, \ell_m$ ,  $v = \ell_1 \oplus_{\sigma} \ell_2 \oplus_{\sigma} \dots \oplus_{\sigma} \ell_m$  with separator  $\sigma = \emptyset$  for /translation and  $\sigma = \text{one space}$  otherwise; a line starting with / opens a new qualifier only when the running count of quote characters in  $v$  is even.

## 16.13 Core sequence operations

On the same 25 GenBank records, bio.sequence.translate reproduced the /translation qualifier for 285/285 standard-code CDSs (17 CDSs with other genetic codes, codon\_start $\neq$ 1, transl\_except or ribosomal slippage were set aside) and matched Bio.Seq.translate on all 285. Reverse complements agreed on 25/25 whole

records, IUPAC motif search on 192/192 (motif, record) pairs, Wallace  $T_m$  on 500/500 lambda oligos (8–13 nt), and ORF calls ( $\geq 100$  aa, both strands) on 22/22 records (239 = 239 ORFs) against an independent Biopython frame scan. GC fraction agreed to  $10^{-12}$  on 24/25; the exception is the human mitochondrial genome, whose single N sugarcode leaves out of the denominator. Protein masses on 500 UniProt proteins differ from ProteinAnalysis by at most  $5.0\text{e-}05$  (relative), from two-decimal residue masses. Reading the code for this comparison turned up a latent bug the data could not: `clean_dna` silently dropped IUPAC ambiguity letters, so `ATGRCTTAA` translated as `ML` (frame shifted) instead of `MX`. No CDS in this set carries an ambiguity code, so the real-data numbers are identical before and after; ambiguity letters now become N (sugarcode 143d1c4), with a regression test. We report it as a code-review finding, not a data finding.

**(F53) Frame preservation.** Cleaning must be a letter-wise map  $c : \Sigma \rightarrow \{A, C, G, T, N\} \cup \{\epsilon\}$  with  $c(x) = \epsilon$  only for non-letters; then codon  $k$  of the cleaned sequence is  $c(s_{3k})c(s_{3k+1})c(s_{3k+2})$  and every downstream codon keeps its frame.

## 16.14 k-mer sets and sketches: exact agreement, a biased estimator

We fetched six coronavirus genomes (SARS-CoV-2, bat RaTG13, pangolin GX-P4L, SARS-CoV Tor2, bat ZC45, MERS) and compared `bio.kmer` with `sourmash` 4.8.14 at  $k = 21$ . With `sourmash` keeping every hash (scaled=1), distinct canonical k-mer counts agreed for 6/6 genomes, and Jaccard and containment agreed exactly on all 15 pairs (maximum difference 0.0). The sketch estimator is a negative result. `bio.kmer`’s minimizer sketch (window  $w = 100$ , about 600 k-mers per genome) under-estimated Jaccard: for SARS-CoV-2 vs RaTG13 the exact value is 0.321, the minimizer estimate 0.236 and `sourmash` `FracMinHash` (scaled=100, about 300 hashes) 0.312. Across the 15 pairs the mean absolute error was 0.0121 for minimizers and 0.0044 for `FracMinHash`, with half the sketch size. Jaccard over minimizer sets is known to be biased because sequence changes shift which k-mer wins each window; `FracMinHash` keeps a k-mer if and only if its hash falls below a fixed threshold, so the kept subset is a uniform sample of the union. The module’s docstring already labels the values as estimates; we record the bias and do not recommend minimizer Jaccard for distance estimation.

**(F54) FracMinHash estimator.** With hash  $h$  and threshold  $H/s$ ,  $\text{FMH}_s(A) = \{a \in A : h(a) < H/s\}$  and  $\hat{f} = |\text{FMH}_s(A) \cap \text{FMH}_s(B)|/|\text{FMH}_s(A \cup B)|$ , which is (nearly) unbiased because inclusion depends on the k-mer alone; the Mash distance is  $D = -\frac{1}{k} \ln \frac{2J}{1+J}$ .

## 16.15 Population-genetics statistics on 1000 Genomes

We pulled 1000 Genomes phase 3 genotype counts for 22 well-studied SNPs (APOE, LCT, SLC24A5, HERC2, ALDH2, ABCC11, ADH1B, EDAR, F5, HFE, MTHFR, COMT and others) from the Ensembl GRCh37 REST service, one record per rsID, giving 110 (SNP, super-population) genotype tables. `bio.gstats.hwe_exact` matched an independent full enumeration of the exact HWE distribution on 110/110 tables (maximum relative error  $1.1\text{e-}12$ ); 15 tables have  $p < 0.05$ , as expected for a set that includes SNPs with strong population structure inside super-populations. Treating EUR as ‘cases’ and EAS as ‘controls’ for 20 SNPs, allelic chi-square (with and without Yates), genotypic chi-square and Woolf odds ratios matched `scipy` and `statsmodels` to  $\leq 1\text{e-}13$  relative error, and Benjamini-Hochberg and Bonferroni adjustments matched `statsmodels` exactly (maximum difference  $2\text{e-}19$ ). No bug was found in this module.

| SNP        | alleles | EUR (AA/AB/BB) | EAS (AA/AB/BB) | allelic $\chi^2$ | OR       |
|------------|---------|----------------|----------------|------------------|----------|
| rs1426654  | A/G     | 500/3/0        | 0/12/492       | 1954.6           | 2.77e+04 |
| rs16891982 | C/G     | 4/54/445       | 498/6/0        | 1756.6           | 0.000393 |
| rs3827760  | A/G     | 492/11/0       | 12/104/388     | 1516.9           | 622      |
| rs1229984  | T/C     | 1/27/475       | 244/215/45     | 972.8            | 0.0129   |
| rs12913832 | A/G     | 90/186/227     | 502/2/0        | 932.6            | 0.00114  |
| rs17822931 | C/T     | 378/113/12     | 36/150/318     | 840.0            | 22.5     |
| rs4988235  | G/A     | 154/187/162    | 504/0/0        | 686.1            | 0.00048  |
| rs671      | G/N     | 503/0/0        | 349/135/20     | 191.3            | 424      |
| rs53576    | A/G     | 55/243/205     | 212/231/61     | 179.9            | 0.291    |
| rs2231142  | G/T     | 414/83/6       | 251/213/40     | 124.7            | 3.93     |
| rs1799945  | C/G     | 348/137/18     | 478/23/3       | 114.4            | 0.143    |
| rs4680     | G/A     | 133/237/133    | 264/198/42     | 102.7            | 0.388    |
| rs1042713  | G/A     | 183/252/68     | 96/263/145     | 53.7             | 1.94     |
| rs1800562  | G/A     | 461/41/1       | 504/0/0        | 44.0             | 0.011    |
| rs429358   | T/C     | 358/134/11     | 421/79/4       | 22.4             | 0.515    |
| rs1801133  | G/A     | 204/231/68     | 259/192/53     | 10.9             | 0.731    |
| rs7412     | C/T     | 441/61/1       | 407/93/4       | 9.5              | 1.67     |
| rs10427255 | C/T     | 131/236/136    | 94/257/153     | 5.8              | 1.24     |
| rs1815739  | T/C     | 90/257/156     | 98/251/155     | 0.2              | 0.964    |
| rs762551   | C/A     | 58/206/239     | 55/220/229     | 0.1              | 0.967    |

Table 17: EUR vs EAS allele-frequency contrasts from 1000 Genomes phase 3, computed by bio.gstats and matching scipy/statsmodels (benchmarks/sweep\_gstats\_scipy.json).

**(F55) Exact HWE distribution.** For  $n$  individuals with  $n_A$  copies of allele A and  $n_B = 2n - n_A$ , the probability of  $h$  heterozygotes is  $P(h) = \frac{n!}{n_{AA}!h!n_{BB}!} \frac{2^h n_A!n_B!}{(2n)!}$  with  $n_{AA} = (n_A - h)/2$ ,  $n_{BB} = (n_B - h)/2$ ; the two-sided p-value is  $\sum_{h:P(h) \leq P(h_{obs})} P(h)$ .

## 16.16 UniProt lookup: a wrong protein and a cache that never hit

bio.uniprot.search maps a gene symbol to its UniProt entry, and several modules (binding sites, mutation docking) build on it. We checked it against HGNC’s curated symbol-to-UniProt mapping for 60 human genes, fetching each HGNC record separately. Before the fix 59/60 matched. The miss was CDKN2A: the query gene:CDKN2A also matches CDKN2A-AS1, and taking the top hit returned Q9UH64 (the antisense-locus protein) instead of P42771 (p16-INK4a). We switched to gene\_exact and prefer the entry whose primary gene name equals the symbol; after the fix 60/60 match and all 60 returned sequences have the length of the entry fetched by accession. Reading the connector turned up a second bug. The on-disk cache named each file by abs(hash(url)); Python salts string hashes per process, so a new process never found its own cache and offline=True always failed. The same pattern was in the ChEMBL, gnomAD and structure connectors. All four now use SHA-256, and a new process running offline reads TP53 and CDKN2A from cache (P04637, P42771).

**(F56) Stable cache key.** A cache key must be a pure function of the request,  $k(u) = \text{SHA256}(u)$ ; a per-process salted hash  $k_s(u)$  gives  $\Pr[k_{s_1}(u) = k_{s_2}(u)] \approx 2^{-64}$  for independent salts, so the hit rate across processes is effectively zero.

## 16.17 NCBI Gene lookup: Huntington resolved to the serotonin transporter

`bio.entrez.gene_id` resolves a symbol to an NCBI Gene ID and feeds the gene-analysis module. Against HGNC’s curated Gene IDs for the same 60 symbols it scored 58/60. HTT returned 6532, which is SLC6A4 (the serotonin transporter, alias HTT), not huntingtin (3064); TTN returned 7276, which is TTR (transthyretin, alias TTN), not titin (7273). The query field [sym] also matches aliases and the search took the top hit. The function now fetches up to ten hits and returns the one whose official symbol equals the query; after the fix 60/60 match (sugarcode 93d5b12). For a user asking about Huntington disease or titin cardiomyopathy the old answer would have been about the wrong gene.

**(F57) Symbol resolution rule.** Given candidate IDs  $u_1, \dots, u_r$  from an alias-aware search, return  $u^* = u_{\min\{i: \text{sym}(u_i)=q\}}$  if that set is non-empty, else  $u_1$ ; an alias match is used only when no official-symbol match exists.

## 16.18 Reactome pathway mapping

`bio.reactome.pathways_for_uniprot` asks Reactome’s live ContentService which lowest-level pathways contain a protein. As an independent check we used Reactome’s own bulk file UniProt2Reactome.txt for the 61 HGNC-curated UniProt accessions of our 60 genes. A first comparison against canonical-accession rows matched only 59/61: KRAS (P01116) has 9 canonical rows but the service returned 71 pathways, and LMNA (P02545) 2 vs 8. The service pools the isoform rows (P01116-1, P01116-2); pooling them in the file gives 61/61 identical sets (775 = 775 protein-pathway pairs). No bug, but a caveat for users: KRAS4A and KRAS4B pathways come back together.

## 16.19 ChEMBL activities: “best first” from an arbitrary page

`bio.chembl.activities_for_target` promises measured IC50/Ki/Kd values ‘best first’. It fetched the first 50 rows in ChEMBL’s default order and sorted those, so ‘best’ meant best of an arbitrary page. On 12 cancer and CNS targets (between 1,731 and 21,238 activities each) the old top-10 shared at most one molecule with the true top-10, and the reported best was far from the real one; for ABL1 it was 1,200 nM when ChEMBL holds exact measurements at 0.015 nM. The function now asks ChEMBL to sort all activities that carry a pChEMBL value by pChEMBL, descending (sugarcode c6db085). The best value now equals the best pChEMBL-valid measurement on 12/12 targets; this check is partly circular because the reference uses the same ordering, so the evidence for the bug is the before-fix column. Restricting to pChEMBL also drops data-entry artefacts that sort first by raw value: 0.0 nM (JAK2, IDH1),  $5 \times 10^{-9}$  nM (EGFR) and  $-0.63$  nM (DRD2).

**(F58) pChEMBL.**  $\text{pChEMBL} = -\log_{10}(A \cdot 10^{-9})$  for an activity  $A$  in nM with an exact relation; ranking by pChEMBL descending equals ranking by  $A$  ascending on valid records, and  $A \leq 0$  has no pChEMBL.

## 16.20 gnomAD constraint: a version-2 cutoff on version-4 data

`bio.gnomad.gene_constraint` reads pLI and LOEUF from gnomAD r4 and flags a gene as strongly loss-of-function constrained when  $\text{LOEUF} < 0.35$ . That cutoff comes from gnomAD v2. gnomAD’s own help page says v4 o/e and LOEUF values run higher than v2, that v2 thresholds do not transfer, and suggests  $\text{LOEUF} < 0.45$  (the most constrained 15% of 17,063 MANE Select transcripts) for Mendelian interpretation. Under

| target | activities | best before (nM) | best after (nM) | pChEMBL |
|--------|------------|------------------|-----------------|---------|
| EGFR   | 21,238     | 0.45             | 0.01            | 11.00   |
| ABL1   | 6,290      | 1200             | 0.015           | 10.82   |
| BRAF   | 9,702      | 0.3              | 0.02            | 10.70   |
| JAK2   | 18,707     | 1                | 0.0106          | 10.97   |
| FLT3   | 6,684      | 0.8              | 0.01            | 11.00   |
| ERBB2  | 3,534      | 5                | 0.04            | 10.40   |
| PIK3CA | 9,583      | 1                | 0.016           | 10.80   |
| IDH1   | 9,132      | 70               | 0.04            | 10.40   |
| ALK    | 4,300      | 0.4              | 0.07            | 10.15   |
| KIT    | 6,127      | 4                | 0.02            | 10.70   |
| CDK4   | 1,731      | 10               | 0.026           | 10.59   |
| DRD2   | 13,215     | 0.031            | 0.0149          | 10.83   |

Table 18: Best reported activity per target before and after the fix (benchmarks/sweep\_chembl\_order.json).

the old cutoff TP53 (LOEUF 0.418, pLI > 0.999) was called ‘not strongly constrained’. On our 60 genes the old rule flagged 13 and the v4 rule flags 19; the newly flagged genes are APC, FMR1, LMNA, NF1, SCN5A, TP53. Agreement with the independent pLI > 0.9 call rose from 51/60 to 57/60 (sugarcode 0a265e7). The sweep also hit gnomAD’s API rate limit (HTTP 400, ‘try again in a few minutes’); the connector used to retry three times within about 1.5 s, which only burns quota, and now fails fast with a distinct rate-limit error so callers can wait or fall back to the cache.

**(F59) LOEUF.** With observed pLoF count  $O$  and neutral expectation  $E$ ,  $\text{LOEUF} = q_{0.95}(\lambda \mid O)/E$  where the posterior of the true count is  $\lambda \mid O \sim \text{Gamma}(O + 1, 1)$  under a flat prior and Poisson likelihood; a gene is flagged when  $\text{LOEUF} < \tau_v$  with version-specific  $\tau_{v4} = 0.45$ .

## 16.21 Splice-junction maps against ClinVar canonical-site variants

bio.splice.junction\_map builds donor and acceptor windows, keyed by cDNA position, from each gene’s RefSeqGene record, taking the CDS with the most exons as canonical. ClinVar names canonical splice-site variants on the MANE Select transcript (c.N+1, c.N+2, c.N-1, c.N-2), which gives an independent test of both the numbering and the sequence: the named position must be a junction in the map, and the base at the named intronic offset must equal ClinVar’s reference allele. We queried 12 genes; CDH1 is not in the module’s vendored RefSeqGene list and gets an explicit ‘no RefSeqGene junction map’ status rather than a guess (a coverage limit, noted for follow-up). Across the other 11 (TP53, BRCA1, BRCA2, MLH1, MSH2, APC, PTEN, ATM, PALB2, CFTR, NF1) ClinVar returned 161 such SNVs. The map had the junction for 159 and the right reference base for 159; the two misses are named on other transcripts (KCNK12, a neighbouring gene returned by the MSH2 search, and a non-MANE APC isoform). The ‘most exons’ rule picked the MANE transcript for every gene here, NF1 included. A first run of our own checker scored 96 because it assumed 15 intronic bases in the acceptor window; the windows hold 14 intronic bases plus the first exonic base, consistently across all three map builders. No module bug.

**(F60) Window offsets.** For a donor at exon end  $b$  the window is  $x_{b-2} \dots x_{b+6}$ , so intronic position  $+k$  is index  $2 + k$ ; for an acceptor at exon start  $c$  the window is  $x_{c-14} \dots x_c$ , so intronic position  $-k$  is index  $14 - k$  (0-based); on the minus strand both windows are reverse-complemented before indexing.

## 16.22 CFD off-target scores

modules.cfd\_offtarget scores guide/off-target pairs with the Cutting Frequency Determination model. We re-implemented the scoring loop of the authors' published calculator on their released matrices (mismatch\_score.pkl, pam\_scores.pkl) and scored the 600 BRCA1 guide/off-target pairs against all 16 PAM dinucleotides: 9,600/9,600 scores agree, maximum difference 2.2e-16. No bug.

**(F61) CFD score.**  $\text{CFD}(g, o, p) = \pi(p_2 p_3) \prod_{i: g_i \neq o_i} m(\text{rg}_i: d\overline{o}_i, i)$ , where  $\overline{o}_i$  is the complement of the off-target base,  $m$  the position-specific mismatch activity and  $\pi$  the PAM activity.

## 16.23 Short tandem repeats: a flank homopolymer hid compound repeats

modules.str\_scope.find\_strs finds tandem repeats with 1–6 bp units. We ran it and pytrf 1.5.0 (the finder behind Krait and TRF-style pipelines) with pytrf's default minimum repeat counts (12/7/5/4/4/4 for mono- to hexanucleotides) on all 25 GenBank records in our dataset set, from phiX174 to a 193,689-bp RefSeqGene region. Before the fix sugarcode found 208 of pytrf's 211 repeats and nothing pytrf missed. The three losses had one cause: the scan moved left to right and, after taking a short homopolymer, jumped to the end of that run, past the start of an overlapping longer repeat. In HSV-1 (NC\_001806.2) a seven-base G run swallowed the start of (GGGGTT)<sub>4</sub>; in NG\_005905.2 T and A runs did the same to (TTTTG)<sub>4</sub> and (AAAT)<sub>4</sub>. The finder now takes every maximal periodic run at every unit length and resolves overlaps longest first, trying the other phases of a run before dropping it (sugarcode 727344a). After the fix it finds 211/211, with 0 extra calls; 209 have identical coordinates and the other 2 are the same span reported in the disease-motif phase (CGG rather than GCG in PTEN's 5' UTR), which the module does on purpose. The same module labels repeat genotypes. The old rule used only the distance from a normal upper bound (normal if  $\leq 2$  above it, intermediate up to 10 above, else 'pathogenic-range candidate'). Compared with the GeneReviews allele tables it mislabels HTT CAG counts 27, 28, 36 (27 and 28 are intermediate, 36 is reduced-penetrance pathogenic) and 154 FMR1 CGG counts between 5 and 300, from 47 to 200: every premutation allele (55–200) was called a pathogenic expansion, which in counselling terms confuses a carrier with an affected full-mutation allele. expansion\_call now takes an optional locus and uses the published bands for HTT and FMR1 (0 disagreements on HTT 5–120); without a locus the old behaviour is unchanged.

**(F62) Maximal periodic runs.** For unit length  $u$ , position  $j$  is periodic if  $x_j = x_{j+u}$ ; a maximal run of periodic positions  $[a, b]$  defines the span  $[a, b + u]$  holding  $\lfloor (b + u - a - k)/u \rfloor$  whole units from phase  $k \in \{0, \dots, u - 1\}$ . Candidates over all  $u$  are kept greedily by length (ties: smaller  $u$ , then leftmost) subject to disjoint whole-unit intervals, which is the rule pytrf-style finders apply implicitly by reporting the longest repeat at each locus.

## 16.24 Bayesian ACMG scoring against ClinGen expert panels

modules.acmg\_bayesian scores ACMG/AMP criteria with the Tavtigian et al. (2018) Bayesian model in its point form (supporting 1, moderate 2, strong 4, very strong 8; pathogenic  $\geq 10$ , likely pathogenic 6–9, likely benign  $-1$  to  $-6$ , benign  $\leq -7$ ). The ClinGen Evidence Repository publishes, for every variant a Variant Curation Expert Panel (VCEP) has classified, the criteria it applied and its final call. We used 7,929 current classifications from 33 panels across 117 genes (the server closed the bulk download after about 18 MB, so only complete rows were used). ClinGen writes modified strengths as suffixes: PM2.Supporting, PVS1.Strong, PM3.Very Strong, BS1.Stand Alone. The module accepted a strength only as a separate field,

so in 5,699 rows at least one code was rejected, and it reproduced 4,583/7,929 panel calls. The rejections were listed in the output rather than hidden, but anyone pasting ClinGen notation got a wrong class. The parser now reads the suffix, and a stand-alone benign strength acts like BA1 (sugarcode cdf160b). The module now reproduces 7,515/7,929 calls (94.8%), with no rejected codes. As a baseline we coded the original ACMG/AMP 2015 combining rules (Richards et al., Table 5), counting each criterion at its applied strength and calling conflicting evidence uncertain. They reproduce 7,267/7,929 (91.7%); the point model is closer to what the panels actually do. The largest remaining disagreement is 229 variants the panels call uncertain and the model calls likely benign; 177 of them net to exactly  $-1$  point, typically a lone BP4. The 2015 rules need two benign supporting criteria for likely benign, and the panels follow that; the 2018 posterior bands do not. We report this as a known difference and did not change the bands. BA1 with pathogenic criteria also met is rare and panels treat it differently (21 benign, 3 pathogenic, 3 uncertain significance, 4 likely benign), so the module keeps BA1 as an override but flags the conflict.

**(F63) Point form of the Bayesian model.** With prior  $\pi = 0.1$  and  $O_{PVSt} = 350$ , a criterion of strength  $k$  points has odds  $350^{k/8}$ ; summed points  $S$  give posterior  $P(S) = \frac{\frac{\pi}{1-\pi} 350^{S/8}}{1 + \frac{\pi}{1-\pi} 350^{S/8}}$ , so  $P(10) = 0.994$ ,  $P(6) = 0.900$ ,  $P(-1) = 0.051$  and  $P(-7) = 0.0007$ .

## 16.25 ClinVar interpretation: exon-spanning ranges and review stars

modules.openclinvar infers a variant’s molecular consequence from its HGVS name when the name alone decides it, and abstains otherwise. We pulled 720 ClinVar records (60 per gene for BRCA1, BRCA2, TP53, MLH1, MSH2, CFTR, APC, PTEN, LDLR, MYH7, SCN5A and RYR1) and scored the module’s call as correct when ClinVar lists that consequence for the record. Of 682 records with a ClinVar consequence the module got 646 right, abstained on 32 and was wrong on 4. All four were ranges judged by their first coordinate only: c.453+625\_545+920delinsT in MLH1 starts in an intron but removes the exon between two introns, c.5468-64\_5480dup in BRCA1 runs from the intron into the exon, and c.-19\_\*21del in TP53 deletes the whole coding sequence; the module called them intronic or UTR, the lowest-weight classes in its evidence model. Ranges are now read at both ends: ends in different exon/intron segments mean a boundary is crossed (splice disruption) and a range from the 5’ UTR into the CDS removes the start codon (sugarcode 7067655). After the fix: 651 right, 30 abstentions, 1 wrong; the remaining one (MSH2 c.2459-2\_2472del) is annotated by ClinVar on an older transcript, and c.2459-2 is an acceptor-site base, so we kept the call. clinvar\_stars maps a review status to ClinVar’s 0–4 gold stars, which weight ClinVar evidence in the module’s trace. Against the 9 statuses in ClinVar’s review-status table it was wrong once: ‘criteria provided, multiple submitters’ (the status used for somatic clinical-impact and oncogenicity classifications) got one star instead of two, because the rule required the germline suffix ‘no conflicts’. Now 9/9.

**(F64) Segment rule for c. ranges.** Write each end as  $(a, s, k)$ : anchor  $a$  (with region 5’ UTR, CDS or 3’ UTR), intron side  $s \in \{+, -, \emptyset\}$  and offset  $k$ . Two ends lie in the same segment iff both are exonic, or both are intronic with  $(a_1, s_1) = (a_2, s_2)$  or  $s_1 = +, s_2 = -, a_2 = a_1 + 1$ . A range whose ends lie in different segments crosses at least one splice site.

## 16.26 CRISPR on-target scores against measured guide activity

modules.crispr\_opt has four on-target scorers: Rule Set 2 (the Azimuth gradient-boosted model, with or without the peptide-position features), Rule Set 1 (Doench 2014 logistic weights), and a hand-built heuristic

used only when a guide sits too close to the sequence edge for a 30-mer context. We scored all 5,310 guides in the Doench 2016 FC+RES set (17 genes; file from the Azimuth repository) and compared with the measured, within-gene rank-normalised activity. Spearman correlations: Rule Set 2 with position 0.7106, sequence-only 0.6693, Rule Set 1 0.3948, heuristic 0.1125; the repository’s own prediction column gives 0.7148. Median per-gene values are 0.6965, 0.6599, 0.468 and 0.1357. Both published models were trained on these guides, so their numbers are in-sample and flattering; the heuristic was not fitted and is close to useless ( $\rho \approx 0.11$ ). Two consequences for users. `design_guides` ranks with Rule Set 1 even though Rule Set 2 ships in the same module; we have not switched it, because the case for switching needs a held-out screen and we have not run one. And the edge fallback should be read as ‘no reliable score’, which the module already labels in its `on_target_model` field (`heuristic_edge_fallback`). No code bug was found.

**(F65) Spearman correlation.** For scores  $s$  and measurements  $y$  with ranks  $R(s_i), R(y_i)$ ,  $\rho = \text{corr}(R(s), R(y))$ , which for untied data equals  $1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}$  with  $d_i = R(s_i) - R(y_i)$ ; the per-gene median guards against between-gene differences in dynamic range.

## 16.27 Replicate planning: a two-entry z table

`modules.gene_analysis.power_estimate` tells a user how many replicates per group an experiment needs. It added 0.84 to 1.96 when the target power was at most 0.8 and 1.28 otherwise, which is right only at 0.8 and 0.9. Against `statsmodels`’ `NormalIndPower` solver on 41 combinations of power (0.5–0.99), effect and variance, the old rule matched 7 times and over-asked by up to  $2.03\times$  (power 0.5 needs 8 replicates for a one-SD effect, it said 16); at 0.95 and 0.99 it under-asked. The function now uses exact normal quantiles and takes `alpha` as a parameter (`sugarcode 4defcdb`); it matches `statsmodels` in 41/41 cases. This is a formula check, not a real-data sweep, and the rest of `gene_analysis` is not yet validated.

**(F66) Two-sample size.** For a two-sided test at level  $\alpha$  with power  $1 - \beta$ , difference  $\delta$  and per-observation variance  $\sigma^2$ ,  $n = \lceil 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2 / \delta^2 \rceil$  per group.

## 16.28 QSAR descriptors against RDKit on approved drugs

`modules.qsar_bench` computes its own 2D descriptors from a small SMILES parser, so a QSAR model can be built without a chemistry toolkit. We compared them with RDKit 2026.03.6 on the 993 ChEMBL phase-4 molecules RDKit reads. Before the fix the parser rejected 110 of them: every SMILES with a / or \ stereo bond (88 molecules, including tamoxifen and diethylstilbestrol) and every sodium, potassium or silver salt (22). On the 883 it did read, molecular weight matched in 843/883 (charged bracket atoms such as the [O-] of a nitrate got an implicit hydrogen), rotatable bonds in 96/883 (the rule subtracted the ring rank from all non-terminal single bonds, so ring bonds counted as rotors: nicotine got 4 instead of 1) and fraction  $\text{sp}^3$  in 296/883 (carbonyl and alkene carbons counted as  $\text{sp}^3$ ). All four are fixed (`sugarcode bf8665b`): the parser reads directional bonds and common counter-ions, bracket atoms carry only their written hydrogens, a bond is rotatable only if it is single, acyclic, between non-terminal atoms and not next to a triple bond, and  $\text{sp}^3$  carbons are non-aromatic carbons with only single bonds. Now all 993 molecules parse and molecular weight, heavy atoms, heteroatoms, formal charge, ring rank, rotatable bonds and fraction  $\text{sp}^3$  agree on 993/993 each; donors agree on 992/993 (the miss is a dihydrate whose water RDKit does not count). Rotatable bonds match RDKit’s NonStrict definition; RDKit’s default Strict count, which

also drops amide C–N bonds, agrees on 655/993. The acceptor count stays a labelled heuristic (264/993 against RDKit’s Lipinski acceptors) and should not be read as Lipinski HBA.

**(F67) Ring bond test.** A bond  $(i, j)$  lies on a ring iff  $j$  is reachable from  $i$  in  $G \setminus \{(i, j)\}$ ; the number of independent rings is the cyclomatic number  $r = |E| - |V| + c$  for  $c$  connected components, which counts rings but not ring bonds. The module’s Morgan fingerprint is meant to follow ECFP. Its starting atom label used only hydrogens written inside brackets, so CH<sub>3</sub>, CH<sub>2</sub> and CH carbons, and ring and chain atoms, began with the same identifier. On 5,000 random pairs of 300 drugs its Tanimoto similarities ranked with RDKit’s radius-2, 2048-bit Morgan at Spearman 0.7815, and the nearest neighbour agreed for 202/300 molecules. The label now uses RDKit’s default invariants (element, heavy degree, total hydrogens, charge, ring membership; sugarcode 9a33eb4): Spearman 0.9005, nearest neighbour 236/300, mean absolute Tanimoto difference 0.0146. The bits are not identical to RDKit’s (different hashing, no removal of duplicate environments), so fingerprints from the two should not be mixed in one model.

**(F68) Tanimoto on bit vectors.** For fingerprints  $a, b \in \{0, 1\}^L$ ,  $T(a, b) = \frac{|a \wedge b|}{|a| + |b| - |a \wedge b|}$ ; two fingerprint schemes are compared by the rank correlation of  $T$  over sampled pairs and by agreement of each molecule’s arg-max neighbour. Split by expert panel (Table 19), the point model reproduces more calls than the 2015 rules for 21 of 33 panels, fewer for 11 and the same for 1. The rules do better for these panels: Myeloid Malignancy; Phenylketonuria; Platelet Disorders; Rett and Angelman-like Disorders; Familial Hypercholesterolemia; Mitochondrial Diseases; FBN1; Hereditary Breast, Ovarian and Pancreatic Cancer; Coagulation Factor Deficiency; Thrombosis; VHL. We did not check why against each panel’s written specification; for these panels a user should read the module’s class next to the rule-based one.

## 16.29 Elastic network modes: a doubled Hessian diagonal

modules.evofold\_4d.anm\_modes computes anisotropic network model (ANM) modes from C $\alpha$  coordinates; its fluctuation profile is offered as a B-factor proxy and feeds hinge detection. We built the ANM Hessian for the first chain of all 30 PDB structures in our set with ProDy 2.4.1 (cutoff 10 Å,  $\gamma = 1$ ; ProDy’s own eigen-solver fails with the installed SciPy, so both Hessians were diagonalised with numpy). Before the fix no structure matched: the module looped over ordered residue pairs and subtracted each contact block from both diagonal blocks on both visits, so every diagonal block was doubled, rows of the Hessian no longer summed to zero and the six rigid-body zero modes disappeared. The ‘lowest modes’ it reported were therefore not the collective motions of the protein; the fluctuation profile correlated with ProDy’s at  $r$  as low as 0.19, and with crystallographic B-factors at median 0.28. Each contact is now added once (sugarcode feced35). All 30/30 structures now give ProDy’s six lowest non-zero frequencies to the module’s four reported decimals, with fluctuation profiles correlated at  $r \geq 1.0000$ ; the median B-factor correlation is 0.38, the same as ProDy’s (0.38). 2 chains have more than six zero modes because of chain breaks; the module skips all zero modes, which is the right behaviour.

**(F69) ANM Hessian.** For contacts  $r_{ij} < r_c$  with unit vector  $\hat{e}_{ij}$ ,  $H_{ij} = -\gamma \hat{e}_{ij} \hat{e}_{ij}^\top$  for  $i \neq j$  and  $H_{ii} = -\sum_{j \neq i} H_{ij}$ , so each block row sums to zero and  $H$  has six zero eigenvalues (three translations, three rotations) for a connected network; mean-square fluctuations are  $\langle \Delta r_i^2 \rangle \propto \sum_k \|u_k^{(i)}\|^2 / \lambda_k$  over non-zero modes.

| expert panel                                                  | variants | point model | 2015 rules |
|---------------------------------------------------------------|----------|-------------|------------|
| Myeloid Malignancy VCEP                                       | 810      | 740         | 776        |
| Monogenic Diabetes VCEP                                       | 801      | 780         | 723        |
| Phenylketonuria VCEP                                          | 770      | 748         | 765        |
| Platelet Disorders VCEP                                       | 582      | 560         | 562        |
| Severe Combined Immunodeficiency Disease VCEP                 | 375      | 368         | 328        |
| Rett and Angelman-like Disorders VCEP                         | 367      | 346         | 353        |
| Familial Hypercholesterolemia VCEP                            | 343      | 328         | 335        |
| Malignant Hyperthermia Susceptibility VCEP                    | 335      | 324         | 311        |
| CDH1 VCEP                                                     | 331      | 325         | 278        |
| RASopathy VCEP                                                | 329      | 319         | 288        |
| Lysosomal Diseases VCEP                                       | 308      | 296         | 273        |
| Cerebral Creatine Deficiency Syndromes VCEP                   | 266      | 256         | 249        |
| Glaucoma VCEP                                                 | 265      | 240         | 228        |
| Mitochondrial Diseases VCEP                                   | 225      | 181         | 204        |
| ACADVL VCEP                                                   | 225      | 220         | 149        |
| Hearing Loss VCEP                                             | 201      | 184         | 172        |
| PTEN VCEP                                                     | 198      | 186         | 183        |
| Cardiomyopathy VCEP                                           | 171      | 168         | 165        |
| DICER1 and miRNA-Processing Gene VCEP                         | 125      | 117         | 116        |
| Leber Congenital Amaurosis/early onset Retinal Dystrophy VCEP | 125      | 124         | 116        |
| TP53 VCEP                                                     | 124      | 114         | 103        |
| FBN1 VCEP                                                     | 104      | 93          | 101        |
| Hereditary Breast, Ovarian and Pancreatic Cancer VCEP         | 93       | 86          | 87         |
| Coagulation Factor Deficiency VCEP                            | 69       | 52          | 56         |
| Thrombosis VCEP                                               | 62       | 53          | 59         |
| InSiGHT Hereditary Colorectal Cancer/Polyposis VCEP           | 59       | 58          | 53         |
| von Willebrand Disease VCEP                                   | 57       | 57          | 55         |
| Brain Malformations VCEP                                      | 43       | 39          | 34         |
| Pulmonary Hypertension VCEP                                   | 42       | 38          | 31         |
| ENIGMA BRCA1 and BRCA2 VCEP                                   | 40       | 34          | 33         |
| Hereditary Hemorrhagic Telangiectasia VCEP                    | 32       | 32          | 30         |
| VHL VCEP                                                      | 31       | 28          | 30         |
| Epilepsy Sodium Channel VCEP                                  | 21       | 21          | 21         |

Table 19: Panel calls reproduced by the Bayesian point model and by the ACMG/AMP 2015 combining rules, per ClinGen expert panel (benchmarks/sweep-acmg\_erepo\_per\_panel.json).

### 16.30 Solvent-accessible surface area against FreeSASA

modules.structural\_biophysics.shrake\_rupley has its own Shrake–Rupley routine for per-atom and per-residue solvent-accessible surface area (SASA, 960 points per atom), separate from the bio.structures SASA checked earlier. We parsed 14 PDB structures from our set (1622–2058 protein heavy atoms) with the module’s own parser and gave the same atoms and radii to FreeSASA’s Lee–Richards algorithm (100 slices, probe 1.4 Å), a different numerical method. Total SASA agreed within 0.17% for every structure, and per-residue areas correlated at  $r \geq 0.99994$  with a mean absolute difference of about 0.3 Å<sup>2</sup> per residue. No bug. The module’s Vina-style score is checked in the next subsection; its pose refinement is not yet validated.

**(F70) Shrake–Rupley estimate.** Atom  $i$  with expanded radius  $R_i = r_i + r_p$  carries  $N$  near-uniform sphere points  $x_{ik}$ ; its area is  $A_i = 4\pi R_i^2 \frac{1}{N} \sum_k \mathbf{1}[\|x_{ik} - c_j\| \geq R_j \forall j \neq i]$ , with sampling error  $O(N^{-1/2})$  per atom that largely cancels in residue and molecule totals.

### 16.31 Vina-form docking score against AutoDock Vina

The same module’s vina\_score claims the AutoDock Vina 1.1 functional form on a fixed pose. Reading Vina’s source showed three departures. It measured surface distances with Bondi radii (C 1.70, N 1.55, O 1.52 Å) where Vina uses its X-Score radii (C 1.9, N 1.8, O 1.7, S 2.0 Å); it counted sulfur as hydrophobic, which Vina does not; and it added  $w_{rot}N_{rot}$  to the energy, whereas Vina divides the intermolecular energy by  $1 + w_{rot}N_{rot}$ . To test the pair terms numerically with no atom-typing ambiguity we scored stacked all-carbon rings, where every atom is a Vina hydrophobic carbon, with AutoDock Vina 1.2.7 in score-only mode. Before the fix the module gave -1.5646, -0.8557 and -0.3662 kcal/mol at 3.6, 4.0 and 4.6 Å separation where Vina gives -2.31, -1.65 and -0.634. After the fix (sugarcode 28b8590) it matches Vina within 0.0046 kcal/mol at all 8 separations from 3.4 to 6.0 Å; the residual is Vina’s grid interpolation and three-decimal output. The rotor division is checked against the source only. Element-level typing still cannot separate polar from apolar carbon or place donors without hydrogens, so on real complexes the score remains an approximation, as the module says.

**(F71) Vina score.** With surface distance  $s_{ij} = r_{ij} - R_i^{XS} - R_j^{XS}$ ,  $e = \sum_{ij} [w_1 e^{-(s/0.5)^2} + w_2 e^{-((s-3)/2)^2} + w_3 s^2 \mathbf{1}[s < 0] + w_4 h(s) + w_5 b(s)]$  over pairs within 8 Å, and  $\Delta G = e / (1 + w_{rot}N_{rot})$  with  $w = (-0.0356, -0.0052, 0.840, -0.03, 0.0585)$ ;  $h$  is 1 for  $s < 0.5$  falling to 0 at 1.5,  $b$  is 1 for  $s < -0.7$  falling to 0 at 0.

### 16.32 Library-evaluation graph keys against RDKit canonical SMILES

molecule\_eval deduplicates generated libraries with an order-invariant graph digest (four rounds of Weisfeiler–Lehman relabelling over element, aromaticity and charge, then a sorted edge list) rather than canonical SMILES. A digest like this must give the same key for every SMILES spelling of a molecule and different keys for different molecules. We tested both on 993 ChEMBL phase-4 drugs, comparing against RDKit 2026.03.6 canonical SMILES with stereochemistry removed, and wrote three randomised SMILES per drug. Before the fix 3 drugs changed key under re-spelling (acetazolamide ChEMBL19 among them). All three were written by ChEMBL with a directional bond marker between two aromatic ring atoms, as in /N=c1/sc. The shared SMILES parser read every / or \ as an explicit single bond, which broke the aromatic ring; RDKit keeps such a bond aromatic. After the fix (sugarcode 15a8ba7), where the marker only sets direction and bond order follows the implicit rule, there are 0 invariance failures, 0 keys shared by molecules RDKit calls different, and 0 RDKit molecules split across keys; both give 980 unique structures.

The directional-bond change left the QSAR descriptor and Morgan benchmarks unchanged when re-run. Internal diversity (mean pairwise Tanimoto distance, radius-2 2048-bit fingerprints, 300 drugs, all pairs) is 0.9055 with the module’s fingerprint and 0.8994 with RDKit’s. Four WL rounds can in principle merge graphs that differ only beyond radius four; no such case occurred in this set.

**(F72) Internal diversity.** For a library of  $n$  unique molecules with fingerprints  $f_i$ ,  $D = \binom{n}{2}^{-1} \sum_{i < j} (1 - T(f_i, f_j))$  with  $T(a, b) = |a \wedge b| / (|a| + |b| - |a \wedge b|)$ ; the module samples at most  $10^4$  pairs and bootstraps a 95% interval.

### 16.33 Nussinov folding against an independent implementation, exhaustive oracle and ViennaRNA

riboswitch uses an exact Nussinov dynamic program (maximum number of nested base pairs, minimum loop 3) to score designed switches. We checked the recursion three ways. On 100 short sequences (length up to 16) we enumerated every non-crossing matching by brute force; the module matched the exhaustive optimum on all 100 (the oracle itself needed two fixes during the check - reused positions and a stale pruning bound - which is why we keep it in the repository). On 145 real Rfam seed sequences from four riboswitch families (TPP RF00059, FMN RF00050, glycine RF00504, cobalamin RF00174), an independently written Nussinov recurrence (pair- $i$ -with- $k$  decomposition instead of the module’s bifurcation form) agreed on the optimal pair count for all 145 (0 mismatches). Against ViennaRNA 2.7.2 minimum-free-energy folds the median base-pair F1 is 0.194 (family means 0.159 FMN to 0.278 glycine), with the module placing  $1.289 \times$  as many pairs: maximising pair count is not the same as minimising free energy, and we do not claim MFE agreement. The module’s estimated\_energy\_kcal\_mol field (-3 per GC pair, -2 per AU/GU) is a heuristic ranking score, not a thermodynamic energy; designs needing energies should go through ViennaRNA, as in the rna\_nussinov module check.

**(F73) Nussinov recursion.**  $N_{ij} = \max\{N_{i+1,j}, N_{i,j-1}, \max_{k > i+m, (s_i, s_k) \in \mathcal{P}} [1 + N_{i+1,k-1} + N_{k+1,j}]\}$  with pair set  $\mathcal{P} = \{AU, UA, GC, CG, GU, UG\}$  and minimum loop  $m = 3$ ; the module evaluates the equivalent bifurcation form  $N_{ij} = \max\{N_{i+1,j}, N_{i,j-1}, [j - i > m](s_i, s_j) + N_{i+1,j-1}, \max_{i < k < j} (N_{ik} + N_{k+1,j})\}$ .

### 16.34 Prime-editing guide design against published, validated pegRNAs

prime\_design builds pegRNAs: a 20-nt spacer, a 3’ extension of reverse-transcription template (RTT) plus primer-binding site (PBS). Three departures from the published anatomy were found by replaying two experimentally validated pegRNAs through the module. (i) The PBS was reverse-complemented out of the RT-template region, so PBS and RTT overlapped; the PBS must anneal to the 3’ flap of the nicked strand, i.e. it is the reverse complement of the protospacer’s PAM-proximal end. (ii) For plus-strand PAMs the RTT started at the edit, not at the nick. (iii) When the RT template was truncated to the chosen length, the module kept the nick-distal end; reverse transcriptase reads from the PBS, so the RTT’s 3’ end must pair with the nick base. After the fixes (sugarcode 947b55e) the module reproduces both published extensions exactly: HEK3 +1 CTT insertion (Anzalone et al. 2019; Addgene 132778; also validated by pegFinder, Chow et al. 2021) RTT TCTGCCATCAAAGC + PBS GTGCTCAGTCTG, and RNF2 +5 G>T RTT AACGAACACATCAGG + PBS TAATGACTAAGATG. Before the fix the ‘PBS’ came out as TCTGCCATCAAAA - RTT sequence, not flap. End to end, from the HEK3 genomic context (Anzalone substrate AVA024) with the +1 CTT edit, the top design uses the published spacer and a PBS from the correct flap (10 nt auto-chosen by the Wallace- $T_m$  rule vs the published 12 nt; same annealing tract), and its RTT contains the published template at the nick. A crash on

insertion edits with empty ref was fixed along the way. The PBS/RTT boundary convention (spacer positions 17/18, one nt PAM-proximal of the canonical Cas9 nick) was adopted to match every published validated pegRNA we could locate, including Anzalone’s own in-vitro pegRNAs.

**(F74) pegRNA extension.** Spacer  $g = g_1 \dots g_{20}$  with PAM at positions 21-23; PBS =  $\overleftarrow{g_{17-L} \dots g_{16}}$  (length  $L \in [10, 16]$ , chosen to bring Wallace  $T_m = 2N_{AT} + 4N_{GC}$  near 32 °C); RTT =  $\overleftarrow{e_{n+1} \dots e_{n+\ell}}$  for edited strand  $e$ , nick at  $n$ ,  $\ell \in [10, 20]$ ; extension = RTT • PBS appended 3’ of the scaffold.

### 16.35 Hill occupancy, burst noise and Wright-Fisher drift against theory

Two modules were checked at the formula level. `promoter_lib`’s thermodynamic occupancy agrees with the Hill form  $1/(1 + (K_d/c)^h)$  to 2.2e-16 over a grid of Hill coefficients,  $K_d$  and concentrations (half-occupancy at  $c = K_d$  holds exactly; the  $h = 1$  case also matches a logistic/expit evaluation to 1.1e-16), and its expression-noise summary matches Poisson-burst theory ( $\text{Var} = \mu(1 + B)$ ,  $\text{Fano} = 1 + B$ ) on all 4 tested settings. `stability_ai`’s seeded Wright-Fisher simulation of construct loss (per-generation plasmid-loss and mutation terms, then viability selection  $f' = a(1 - s)/(1 - as)$  with  $a = f(1 - L - m)$ , binomial resampling) tracks an independently written deterministic recursion within 0.00204 at population  $2 \times 10^6$  across plasmid and genomic constructs, is reproducible under a fixed seed, and gives the expected extremes (0% extinction for a low-burden plasmid, 100% for a high-burden toxic one). No bug in either; both are theory checks, so the parameters themselves (loss rates, selection coefficients) remain literature-free defaults and are not validated against experimental stability data - a limitation we state rather than a defect.

**(F75) Wright-Fisher with loss and selection.**  $a_g = f_g(1 - L - m)$ ,  $f_{g+1} \sim \text{Bin}(N, a_g(1 - s)/(1 - a_g s))/N$ , with plasmid-loss rate  $L = (2 \times 10^{-3})(1 + 4b)$  for burden  $b$ , mutation  $m = 10^{-4}k$  for size  $k$  kb ( $\times 2$  if toxic), selection  $s = 0.08e$  for expression load  $e$ .

### 16.36 FASTA parsing against Biopython on Rfam seed files

`bio.fasta` was compared with Biopython 1.88 SeqIO on three real Rfam seed-alignment FASTA files (TPP RF00059: 16245 records, FMN RF00050: 6377, glycine RF00504: 9474) - 32096 records in total. Identifiers, full descriptions and sequences agree exactly on every record (0 mismatches), the streaming reader agrees with the in-memory parser on all three files, and a write-read round trip over 200 records is lossless. One deliberate difference: Biopython rejects a file that starts with blank lines (‘comments at the beginning of the file’), while `bio.fasta` skips them; we treat this as leniency, not a defect, and record it here.

**(F76) FASTA round trip.** For records  $r_i = (h_i, s_i)$ , write-read gives

$\text{mathrm{parse}}($

$\text{mathrm{write}}(r)) = r$  with sequence lines wrapped at width  $w$ :

$\text{mathrm{write}}(r) =$

$> !h_i$

,

|

,

$s_i[kw : (k + 1)w]_{k=0}^{\lceil \text{ceil}[s_i]/w \rceil - 1}$ ; identity holds for  $w = 50$  on 200 RF00059 records.

### 16.37 CRISPRater guide-efficacy scores against the author-computed reference scores

crisprater vendors the published ten-feature linear model of Labuhn et al. 2018 (NAR 46:1375) with its intercept, coefficients and class thresholds. We scored all 3141 sgRNAs from the paper's Supplementary Table 3 (author-computed CRISPRater scores, PMC5814880) and compared. As vendored, 3139 of 3141 guides mismatched. The cause was the GC feature: the module summed GC over spacer positions 4-14, but the published model uses GC4-13 (Figure 4C of the paper; the narrower window also avoids double-counting position 14, which is its own negative feature). After the fix (sugarcode 55748fa) the module matches the authors' scores on 3138 of 3141 guides (99.90%) within  $5 \times 10^{-4}$ . The three residual outliers each differ from the published model by about one feature weight (exactly one weight in two cases), consistent with isolated anomalies in the supplementary table itself, which also contains two duplicate sequences; we document them rather than fit to them. Class thresholds (low  $<0.56$ , high  $>0.74$ ) were confirmed against the paper text.

**(F77) CRISPRater score.**  $s = 0.6505037 + 0.14177$

,  
 $\mathrm{GC}_{413} + 0.06966$   
,  $[b_{20}=G] + 0.04216$   
,  $[b_3 \text{ in } A, T] + 0.03303$   
,  $[b_{12} \text{ in } A, G] + 0.02355$   
,  $[b_6=G] - 0.04746$   
,  $[b_4 \text{ in } A, T] - 0.04878$   
,  $[b_{18} \text{ in } A, G] - 0.06982$   
,  $[b_5 \text{ in } A, C] - 0.07088$   
,  $[b_{14}=G] - 0.08161$   
,  $[b_{15}=A]$ , with  
 $\mathrm{GC}_{413}$  the GC fraction of spacer positions 4-13.

### 16.38 Platform performance analytics against independent recomputation

omega\_stats aggregates recorded metric observations into dashboards, release comparisons and 51 diagnostics. Every computed quantity was recomputed independently with numpy/statistics on synthetic series (4 modules, accuracy and latency, 30 observations each): per-series mean/standard-deviation/min/max match to the module's rounding (0 mismatches over 8 series), subnetwork mean-of-means matches, least-squares drift slopes match numpypolyfit exactly (0 mismatches), pass rates use the correct direction per kind (latency  $\leq$  target, others  $\geq$ ; 0 mismatches), bootstrap intervals are seed-deterministic and contain the mean, release-comparison effect sizes match the pooled-variance Cohen form (0 mismatches), and the diagnostics block has exactly 51 entries. The release gate behaved correctly on a planted mixed case: a 5% candidate improvement in accuracy with a 5% latency regression gates fail. No bug.

**(F78) Standardized release effect.** For baseline  $a$  and candidate  $b$ ,  $d = (\bar{b} - \bar{a}) / \sqrt{(s_a^2 + s_b^2)/2}$  with population variances; relative improvement is  $(\bar{b} - \bar{a})/|\bar{a}|$ , sign-flipped for lower-is-better kinds, and the gate fails when any series drops more than the tolerance (5%).

### 16.39 16S diversity statistics against scipy and explicit Gower centering

microbiome\_exp computes per-sample alpha diversity (Shannon, Gini–Simpson, richness, Pielou evenness), a Bray–Curtis beta-diversity matrix on relative abundances, classical PCoA ordination, and permutation-based differential abundance. On a 12-sample synthetic cohort every alpha quantity matches scipystatsentropy and direct forms (0 mismatches across both entry points), the Bray–Curtis matrix matches scipyspatialdistancebraycurtis to 0.0, and a planted control-enriched taxon is recovered with the correct sign and a significant permutation  $p$  (1.968 log2 fold,  $p = 0.034$ ; unplanted cohort median  $p = 0.51$ ). The PCoA had a real bug: the Gower centering subtracted means of the unsquared distances, shifting the leading eigenvalue by 0.80. After the fix (sugarcode 2de74ce) the first two eigenvalues match an explicit  $-\frac{1}{2}JD^2J$  computation to 8.3e-16. Disease associations in the module are heuristic flags over a small built-in table, and the module itself reports them as hypotheses, not diagnoses; we did not validate the association table against cohort literature.

**(F79) Gower-centered PCoA.** With Bray–Curtis matrix  $D$ ,  $G = -\frac{1}{2}JD^{\circ 2}J$  for  $J = I - n^{-1}\mathbf{1}\mathbf{1}^\top$ ; coordinates  $U_k \text{diag}(\sqrt{\lambda_k})$  from the top eigenpairs of  $G$ . Centering must use means of  $D^{\circ 2}$ , the elementwise square - using means of  $D$  does not double-center the squared matrix.

### 16.40 m6A site prediction against the SRAMP benchmark and single-nucleotide miCLIP sites

rna\_decoder scores candidate m6A sites by a DRACH motif search weighted by region priors (highest near the stop codon and in the 3'UTR) and a local AU-richness exposure term. We validated it on two independent real datasets. First, motif biology: of 9536 single-nucleotide miCLIP CIMS sites in HEK293 cells (GSE63753, Linder et al. 2015), 77.3% match the DRACH consensus the module searches for, and the five most common site 5-mers (GGACT, GGACA, AGACT, GAACT, TGACT) are all DRACH. Second, discrimination: the SRAMP mature-messenger benchmark (Zhou et al. 2016) labels 1210 transcript positions on the 40 transcripts we fetched live from Ensembl (positions are 1-based adenosine coordinates: 135 of 258 positives land on A, vs 36 under a 0-based reading). Because the benchmark predates the current Ensembl build, 708 labels no longer sit at a DRACH-centred A and were excluded, leaving 99 positives and 403 negatives. The check found a real defect: with the module's original defaults the whole sequence was treated as coding, so the final 100 nt of any transcript inherited the near-stop prior; scored that way the module achieved AUC 0.5496, below its own exposure feature alone (0.562). The default now auto-detects the coding region from the first ATG (AUC 0.5886), and with true Ensembl CDS coordinates the module reaches AUC 0.6702 on 88 positives and 361 negatives over 38 CDS-aligned transcripts. The regional prior direction is validated by the data: 49 of 88 positives (55.7%) fall in near-stop CDS or 3'UTR against 130 of 361 negatives (36.0%), the established m6A topology. This remains a heuristic scorer, far below trained predictors, and the m5C/psi/m1A motif maps are placeholders the module itself flags; we report the AUCs, not a discovery.

**(F80) m6A site score.**  $s = 0.55r + 0.45(1 - \text{GC}_{31})$  with region prior  $r \in \{0.35 \text{ (5'UTR)}, 0.55 \text{ (CDS)}, 0.85 \text{ (< 100nt from stop)}, 0.75 \text{ (3'UTR)}\}$  over a 31-nt flank; sites are DRACH occurrences, modification at the cen-

tral A. Discrimination is measured by Mann–Whitney  $AUC = (\sum_{i \in \text{pos}} R_i - P(P+1)/2) / (PN)$  with midranks  $R_i$ .

### 16.41 Live evidence mining against PubMed E-utilities and ClinicalTrials.gov

evidence\_mining queries two public APIs - NCBI E-utilities (esearch then efetch) and the ClinicalTrials.gov v2 studies endpoint - and normalises both into one evidence table. We replayed its traffic against independent live calls made seconds apart with the same query ('ivacaftor cystic fibrosis', limit 10 per source). The module's PubMed identifiers match the independent esearch idlist exactly and in order (10/10), its trial identifiers match the independent v2 call exactly and in order (10/10), the record\_count field equals the emitted table length, all PubMed URLs are correctly formed, all PMIDs are numeric, all trial IDs are well-formed NCT identifiers, and 20 of 20 records carry a non-empty title. Failure semantics are as documented: strict mode is fail-closed (an offline cache miss raises), non-strict mode returns both source errors in-band with zero records rather than silently discarding a failed source, and offline replay from the content-addressed cache reproduces the live record order exactly. No bug. This validates transport, parsing and bookkeeping; it does not validate the mined effect-direction/sample-size text heuristics, which remain unverified extraction guesses.

**(F81) Evidence-table merge.** Records  $r = (s, i, t, u, d)$  with source  $s \in \{\text{PubMed}, \text{CTG}\}$ , identifier  $i$ , title  $t$ , URL  $u$ , date  $d$ ;  $\text{mine}(q) = \{r\}_{s,i}$  with  $\text{record\_count} = |\text{mine}(q)|$ , strict mode  $\Rightarrow$  any source error aborts, non-strict  $\Rightarrow$  errors  $\subseteq$  result. Verified:  $i_{\text{PubMed}}^{\text{module}} = i_{\text{PubMed}}^{\text{research}}$  and  $i_{\text{CTG}}^{\text{module}} = i_{\text{CTG}}^{\text{v2}}$  elementwise on 10+10 live records.

### 16.42 Rare-disease signatures against the HPO annotation corpus

rarenet.ai ranks six rare diseases from symptom sets, variants and omics over a curated table of symptom signatures, causal genes and prevalences. We audited the table against the Human Phenotype Ontology gene-to-phenotype corpus (333,984 gene–phenotype–disease rows, current release). The audit found one stale curation: Gaucher disease was keyed to the retired gene symbol GBA, which appears nowhere in the current corpus - any gene-keyed lookup would silently miss it; the current symbol GBA1 restores the mapping. After the fix, all 6/6 gene–disease pairs are confirmed (CFTR–OMIM:219700, HTT–143100, FBN1–154700, PAH–261600, GBA1–230800, DMD–310200), and of the 24 curated symptoms 12 match the disease's annotated HPO term names directly while 21 match once clinical synonyms are applied (lens dislocation = Ectopia lentis, poor growth = Failure to thrive, calf hypertrophy = Calf muscle hypertrophy). The three residue are annotation absences in this HPO release, not curation errors: no adult-onset row for HTT/OMIM:143100, no musty-odor row for PAH/OMIM:261600, no fatigue row for GBA1/OMIM:230800. Behaviourally, the module ranks the correct disease first for all 6/6 complete signatures. Limitations recorded: the symptom vocabulary is lay-oriented, the prevalence fields are unverified literature defaults, and six diseases is a demo-scale table.

**(F82) Signature score.**  $c = \min\{0.99, \min(1, 0.25|S \cap S_d| + 0.1 \sum_{s \in S \cap S_d} f(s)^{-1}) + 0.4[g_d \in V] + 0.15[g_d \in O]\}$  for input symptoms  $S$ , disease signature  $S_d$ , causal gene  $g_d$ , variant genes  $V$ , omics-disrupted genes  $O$ , and symptom-frequency weight  $f$ ; candidates rank by  $c$ . Table audit:  $g_d$  must appear in the HPO corpus under the OMIM identifier of the disease.

### 16.43 Tumour-promoter selectivity table against Human Protein Atlas pathology data

oncocircuit designs tumour-selective kill circuits over a five-entry promoter table with cancer associations and numeric tumour-selectivity values. We audited the table against the Human Protein Atlas v23 pathology resource (403,241 gene–cancer patient-sample rows). PSA/KLK3 is confirmed outright: detected in all ten prostate-cancer samples and in none of the 89 samples across the other 19 cancers (detection and High-expression shares in the claimed cancer both 1.0). AFP holds at High-expression level (liver is the maximum-High cancer; High share in claimed tissue 0.5), although low-level AFP detection is pan-cancer. HER2/ERBB2 is directionally right with breast the maximum-High cancer (0.364 High rate) and gastric much weaker (0.083) than the table implies. Two entries do not survive as written: Survivin/BIRC5 is near-ubiquitous across tumours (breast 1.0, lung 0.917, colorectal 0.917 detection; its High expression concentrates in testis), so it works as a proliferation-associated second input but its 0.80 selectivity figure is unsupported; and hTERT/TERT protein is essentially undetectable by immunohistochemistry in every cancer, so the pan-cancer hTERT claim - which rests on promoter-reactivation and promoter-mutation biology, not protein abundance - can be neither confirmed nor refuted with this evidence class. The circuit behaviour itself is internally consistent: the dual-input AND design expresses the payload at 9.5168 in tumour against 0.0502 in normal and 0.1463 in single-positive tissue, while the single-input comparator leaks (single-positive equals tumour output). We kept the table's selectivity numbers as uncalibrated literature-free defaults and recorded the evidence class each claim actually needs.

**(F83) Detection specificity.** For gene  $g$  and claimed cancer set  $C$ ,  $\text{detshare} = \sum_{c \in C} \text{det}_{g,c} / \sum_c \text{det}_{g,c}$  with  $\text{det} = n_{\text{High}} + n_{\text{Med}} + n_{\text{Low}}$  per HPA patient counts; PSA/KLK3 attains 1.0. Dual-input gate: payload  $\propto \prod_i a_i$  vs single-input  $\propto a_1$ ; selectivity requires output in single-positive tissue  $\ll$  tumour.

### 16.44 Educational DNA-to-code utilities against Ensembl proteins and consequence rules

`dna_to_code` maps molecular-biology concepts to code analogies and exposes small executable helpers. We ran the helpers on 38 real mouse Ensembl coding sequences. Translation reproduced the Ensembl protein exactly for 38 of 38 transcripts, and the codon table matched NCBI translation table 1 on all 64 codons. The mutation helper had a real gap: it only reported silent and nonsense flags, so start-codon and stop-codon substitutions were silently lumped with missense changes. On 220 real substitutions the old flags agreed with Sequence Ontology consequence terms on 158; every start-lost and stop-lost change was misreported. The fixed helper returns a consequence term and agrees on 220 of 220. A second error sat in the teaching snippet for transcription, which swapped T for U on the template strand; on real cDNAs that gives 0 of 38 correct mRNAs, against 38 of 38 for the corrected reverse-complement form. Negative note: the Ensembl VEP service returned server errors throughout the run, so the reference classifier is an independent Biopython codon-level implementation of the same rules, not VEP itself.

**(F84) Codon-level consequence.** For substitution at CDS position  $p$  (0-based), codon index  $k = \lfloor p/3 \rfloor$ , reference codon  $r$  and mutant codon  $m$  with  $a = T(r)$ ,  $b = T(m)$  under table  $T$ : start\_lost if  $k = 0 \wedge r = \text{ATG} \wedge m \neq \text{ATG}$ ; stop\_lost if  $a = * \wedge b \neq *$ ; stop\_retained if  $a = b = *$ ; stop\_gained if  $b = *$ ; synonymous if  $a = b$ ; else missense. Template identity:  $\text{mRNA} = \text{rev}(t)$  with  $A \rightarrow U$ , i.e. the coding strand with  $T \rightarrow U$ .

## 16.45 Proteochemometric drug–target baseline against live ChEMBL and UniProt

dti\_bench trains a target-conditioned ridge model over ligand fingerprints, protein composition and explicit interaction terms, and ingests exact-relation pChEMBL pairs from the live ChEMBL API. The audit found that the ingestion client could never succeed on the live API: it read component sequences inline from the target resource, which embeds no sequences, so every call raised before any data was retrieved. The fixed client fetches the component resource and fails closed on multi-component targets. On six kinases it retrieved 1,466 exact-relation records; all six target sequences match UniProt canonical exactly, and the median aggregation reproduces 481 of 481 independently recomputed molecule medians. On the pooled real set a warm-random split gives RMSE 0.953 against 1.403 for a naive-mean baseline ( $R^2$  0.539, seed-deterministic split, all 293 test records inside the applicability domain). A chronological split degrades to  $R^2$  0.134, and a cold-target split holding out ABL1 and EGFR collapses to  $R^2$  -0.883 - but the module’s applicability gate flags all 405 of those predictions as outside-domain, which is the intended honest behaviour for a composition baseline. Ligand similarity rank-agrees with the RDKit Morgan generator (Spearman 0.864 over 1,770 pairs). Caveat: the year-granularity time split lets the boundary year straddle train and test.

**(F85) Cold-start applicability gate.** Ridge on standardised features  $z$ :  $\hat{y} = b + \sum_j \beta_j z_j$ ,  $\beta = (Z^\top Z + \alpha I)^{-1} Z^\top y$ . Applicability: inside iff  $\max_{s \in T} \text{Tan}(f(x), f(s)) \geq 0.35$  and  $\max_{q \in Q} \text{pid}(p, q) \geq 0.30$  with positional identity  $\text{pid} = n^{-1} \sum_i [p_i = q_i]$ . Cold-target holdout:  $R^2 = -0.883$  with 405/405 flagged outside.

## 16.46 Mutation docking against Platinum experimental affinity changes

mutdock predicts how a point mutation in a binding pocket changes small-molecule binding. We validated it against the Platinum database of experimentally measured affinity changes (1,008 single-point records). For 36 entries we rebuilt the true binding pocket from the fetched wild-type PDB structure (residues within 6 Å of the ligand) and compared against  $\Delta\Delta G = RT \ln(K_{mt}/K_{wt})$ . Direction is right in 72.2% of cases, but the rank correlation with experiment is only 0.036. The cause is visible in the predictions: the docking rescore is composition-based, so a same-class substitution leaves it unchanged and the output collapses onto the class-penalty table. Auditing that table on 776 Platinum entries shows every covered class pair has the correct sign, since binding-site mutations hurt binding on average, but the magnitudes are miscalibrated and five of sixteen pairs use an undocumented default. We kept the table and documented the gap rather than refit it to the validation data. Formula checks pass: the reported fold-change equals  $e^{\Delta\Delta G/0.593}$  exactly.

**(F86) Mutation  $\Delta\Delta G$  heuristic.**  $\Delta\Delta G = 0.6 (G_{mt}^{dock} - G_{wt}^{dock}) + 0.4 P(c_{wt} \rightarrow c_{mt})$  with class-penalty table  $P$ ; fold-change =  $e^{\Delta\Delta G/RT}$ ,  $RT = 0.593$  kcal/mol at 298 K. Platinum audit: sign correct for 11/11 covered pairs; Spearman  $\rho = 0.036$ .

## 16.47 Biomaterial degradation model against its own reported half-lives

bio\_material designs production and predicts degradation for four biomaterials. The audit found an internal contradiction: the degradation curve decayed with half-life  $1/k$  while the reported half-life was  $\ln 2/k$  (for PHA in soft tissue, 42.0 d versus the reported 28.9 d). The fixed module uses the first-order form  $M_0 e^{-kt}$ ; all 16 material-by-site cases now agree with their reported half-lives and track  $e^{-kt}$  to within 0.15 percentage points. The remaining physics helpers match independent recomputation: rubber elasticity  $G = \nu RT$ ,  $E = 3G(1 - \nu)^2$ ; Herschel–Bulkley stress; the analytic cosh profile of steady reaction–diffusion (agreement  $3.5 \times 10^{-18}$ ); and a deterministic polymerization integration. Table audit: PHA modulus and tensile match the

Goodfellow PHB datasheet exactly; PLA tensile sits inside the datasheet range while its modulus is 0.3 GPa below it; spider silk matches classic dragline values; bacterial cellulose is unverified. Degradation rates and site factors remain heuristic.

**(F87) First-order degradation.**  $M(t) = M_0 e^{-kt}$ ,  $t_{1/2} = \ln 2/k$ ; effective modulus of a porous network  $E = 3\nu RT(1-p)^2$ . Pre-fix curve used  $2^{-kt}$  ( $t_{1/2} = 1/k$ ) against a reported  $\ln 2/k$ .

## 16.48 Cell-free expression optimizer grid alignment

cell\_free\_opt grid-searches a Gaussian response surface for cell-free protein synthesis yields. The audit found that the default grid never evaluated the documented optimal template concentration, capping the reported optimum at 1.879 g/L against the model maximum 2.0 g/L. After the fix the search returns the exact optimum for grid steps 1-4 and agrees with an independent brute-force argmax. Cost, batch normalisation, sensitivity elasticities and Pareto filtering all match independent recomputation. Caveat: the Euler-integrated kinetic trace overshoots the analytic integral (final 2.034 vs 1.9 g/L at 6 h), a declared discretisation.

**(F88) CFPS response surface.**  $y = 2 \prod_i e^{-((x_i - o_i)/w_i)^2}$  g/L over reagents  $x_i$  with optima  $o_i$  and widths  $w_i$ ; kinetic trace  $r(t) = v_{max} e^{-t/tau}$ , cumulative  $v_{max} tau(1 - e^{-t/tau})$ ; grid must contain  $o_i$  for all  $i$ .

## 16.49 CAR-T construct curation against approved products

car\_t\_designer recommends CAR architectures from curated antigen, scFv and costimulatory tables. Its CD19 recommendation (FMC63, CD8 $\alpha$  hinge, 4-1BB, CD3 $\zeta$ ) is exactly the Kymriah architecture, and its BCMA recommendation (C11D5.3, CD8 $\alpha$ , 4-1BB) matches the Abecma FDA label; the alternative CD28 CD19 product is acknowledged in the rationale. The costimulatory choice rule, safety-switch inclusion, toxicity formula and all helper functions (selectivity ratios, logic gates, exhaustion and killing curves, trial persistence) reproduce independent recomputation. Predicted response for CD19 is 0.6, in a plausible range for the declared class-level heuristic; the module states it is not fitted to per-antigen clinical data.

**(F89) CAR heuristic scores.** CRS risk =  $\min(0.3 + 0.2 e_c, 0.9)$ , off-tumour =  $0.15 |N|$  for costim expansion  $e_c$  and normal tissues  $N$ ; exhaustion  $E(d) = 1 - e^{-rd/14}$ ; kill fraction =  $f(1 - e^{-hoR})$  at E:T ratio  $R$  with antigen-positive fraction  $f$ ; persistence  $p(0.85)^m$ .

## 16.50 Biofoundry scheduler against workflow dependencies

biofactory\_1\_a generates robotic protocols and schedules them. The audit found that the scheduler ignored workflow order: steps started whenever their resource was free, so the golden-gate thermocycle ran at  $t = 0$  in parallel with its own input dispenses and the reported makespan was 120 min against the true 274 min. After the fix every step starts at the later of resource-free and predecessor-end; all three workflows now give ordered schedules whose makespan equals the sequential sum (golden-gate 274, gibson 126, pcr-screen 173 min). The Bayesian condition update, PID trace, reagent half-life compensation, Zhang  $Z'$  factor and DAG rerouting all match independent recomputation exactly. Protocol steps sit within standard NEB Golden Gate and Gibson method bounds.

**(F90) Assay and update identities.** Zhang  $Z' = 1 - 3(\sigma_s + \sigma_b)/|\mu_s - \mu_b|$ ; conjugate update  $\pi^{-1} = \pi_0^{-1} + n/u$ ,  $\mu = (\mu_0/\pi_0 + \sum o_i/u)\pi$  for noise variance  $u$ ; reagent activity  $a = \ell 2^{-t/t_{1/2}}$  with compensation  $1/a$ ; list schedule  $s_i = \max(r_{free}, e_{i-1})$ .

## 16.51 Metabolic pathway designer against KEGG and fermentation stoichiometry

metabodesigner searches a curated reaction graph for glucose-to-product routes and scores them with an LP flux model. Every reaction was checked against its KEGG ENZYME record (name and substrate/product compound IDs) and every metabolite formula against KEGG COMPOUND. The audit found that the glycolysis graph had no triose-phosphate isomerase, so the aldolase product DHAP was a dead end and the LP returned 1.0 mol lactate and 1.0 mol ethanol per mol glucose against the textbook 2. The thiolase step also consumed one acetyl-CoA instead of two, and by cancellation the 3-hydroxybutyrate yield stayed at the correct 1.0. After adding EC 5.3.1.1, a byproduct-recycling closure, the 2:1 thiolase stoichiometry and the PDH  $\text{CO}_2$ , the yields are 2.0, 2.0 and 1.0 mol/mol; 11 of 17 reactions match KEGG exactly, 5 are correctly named lumped steps and one has a class-level EC; 15 of 16 checkable reactions are carbon balanced. The remaining imbalance is citrate synthase, which treats oxaloacetate as a cofactor. Free-energy and kcat defaults are placeholder priors and were not validated.

**(F91) Pathway yield by LP.**  $\max v_T$  s.t.  $S_{int}v = 0$ ,  $0 \leq v \leq 1$ ; molar yield  $Y = (Sv)_T / (-(Sv)_{glc})$ ; carbon balance  $\sum_s n_C(s) = \sum_p n_C(p)$  with acyl-CoA counted as  $n_C - 21$ ; glycolysis requires  $\text{DHAP} \rightleftharpoons \text{GAP}$  for  $Y_{lac} = 2$ .

## 16.52 Biosensor design against PubChem descriptors and binding literature

bio\_switch couples curated ligand-binding domains to reporters and models dose-response, noise and processing. Its descriptor table was checked against PubChem for all five small-molecule analytes: molecular weights, TPSA and donor counts match, acceptor counts differ only by definition (Lipinski N+O counting versus PubChem’s refined rules), and logP values differ by method (clogP versus XLogP3). The audit found that the Lipinski violation count was cast to a boolean, so lactose (two violations: 8 donors, 11 acceptors) was classed moderate instead of low under the module’s own rule, and the returned field read True instead of 2. A second curation error put the TetR-tetracycline  $K_d$  at 10 nM against the measured  $K_a = 3 \pm 2 \times 10^9 \text{ M}^{-1}$  (0.2-1 nM, Takahashi 1991, PMID 1812784); both are fixed. The fluoride riboswitch  $K_d$  of 60  $\mu\text{M}$  matches the crcB literature exactly; the theophylline aptamer value (0.3  $\mu\text{M}$ ) sits within the 0.1-0.32  $\mu\text{M}$  literature range. Hill response, tunable-sensor c10/c90 (grid-quantized, 7.94 versus continuous 9.0 for  $n = 2$ ), gate logic, temporal filtering, feedback curves and degradation kinetics all reproduce independent recomputation, and the Gillespie simulation converges to the deterministic occupancy (0.1664 versus 0.1667 bound fraction) with seed reproducibility and a correct copies-per-fL conversion. Documented caveats: the reaction-diffusion profile uses a semi-infinite slab ( $\exp(-x/\lambda)$ ) where a finite cell gives flatter cosh profiles, off-target cross-reactivity is an order-of-magnitude proxy, and the remaining curated  $K_d$  values are plausible curations, not individually verified.

**(F92) Sensor dose-response and noise.** Hill  $R(c) = V_{max}c^n / (K_d^n + c^n)$ , threshold ratio  $c_{90}/c_{10} = 81^{1/n}$ ; occupancy  $\bar{b} = L / (L + K_d)$  from birth-death propensities  $k_{on}L(N - b)$ ,  $k_{off}b$ ; copies per fL =  $6.022 \times 10^2 C_{\mu\text{M}}$ ; intracellular gradient  $c(x) = c_0 e^{-x/\lambda}$ ,  $\lambda = \sqrt{D/k}$ .

### 16.53 Dark-genome regulatory decoder against JASPAR and UCSC CpG islands

dark\_genome decodes the regulatory landscape of non-coding sequence: TF motif hits, enhancer clusters, CpG islands and lncRNA candidates, plus biophysical surrogates. Its eight TF consensus motifs were checked against JASPAR position weight matrices: five match the matrix argmax sequence exactly (AP-1, NF- $\kappa$ B, TATA-box, E-box, CCAAT), while p53, GC-box and HNF4 follow the canonical literature consensus (El-Deiry RRRCWWGYYY, matching the p21/WAF1 site GAACATGTCC; GC-box GGGCGG; HNF4 DR1) that differ from the assay-specific JASPAR versions fetched. The audit found that motif\_binding\_energy credited any base at a degenerate motif position, so R matched C/T: on poly-C with motif RRR the module reported match 1.0 against the true 0.0, and the p53, TATA-box and HNF4 minimum energies on a real promoter were all inflated. After the fix to set-based IUPAC matching all eight energies equal independent recomputation. Against the UCSC cpGIslandExt track on hg38 the module detects both real promoter islands tested (MLH1, CDKN2A) but calls 3 false positives in a 13 kb desert window because it omits UCSC's post-merge length filter; the tradeoff is documented rather than refit. All 64 motif hits reported for the 10 kb MLH1 locus pass an independent IUPAC regex check. The weighted coordinate field, damped graph propagation, nucleosome occupancy, enhancer-gene causality, logic gates, temporal trajectories and the precision-weighted perturbation update all match independent recomputation exactly.

**(F93) Regulatory field and propagation.** Potential  $\phi = 0.3g + 0.3a + 0.25h + 0.1(1 - m) + 0.05 \min(1, 20\rho)$  for GC  $g$ , ATAC  $a$ , H3K27ac  $h$ , methylation  $m$  and motif density  $\rho$ ; contacts  $w_{ij} = (1 + |i - j|)^{-0.8}$ ; damped propagation  $s_{t+1} = (1 - \delta)s_0 + \delta A_{row}^{-1} A s_t$ ; CpG  $O/E = n_{CGN}/(n_{CG}) > 0.6$  with GC  $> 0.5$ ; energy  $E = -8\text{match} + 0.8\text{AT shape}$  with set-based IUPAC matching.

### 16.54 Cell-fate GRN engine against HGNC and reprogramming cocktails

cellfatenet simulates a curated 18-node lineage GRN with Hill dynamics, epigenetic gating and optimal-control reprogramming. Curation checks: 19 of 20 gene symbols are approved HGNC symbols (OCT4 is an alias of POU5F1); the cardiac driver edges cover the full Ieda GMT cocktail (GATA4, MEF2C, TBX5; PMID 20691899), while OSKM lacks KLF4/MYC and the neuronal BAM set lacks POU3F2 (coverage caveats). The audit found two defects. First, production was  $\sigma(2s)$ , a 0.5 basal floor that left every gene half-expressed with no regulators (quiet TNNT2 steady state 1.43); a thresholded  $\sigma(6(s - 0.5))$  brings the quiet state to 0.20 with single-driver activation preserved. Second, the optimal-control stack was self-defeating: controls could force structural markers, unspecified targets counted as zeros so driver activation was penalized - the pre-fix fibroblast-to-cardiomyocyte recipe repressed MEF2C and TBX5, the opposite of the GMT protocol - the finite-difference step sat below solver noise, and negative concentrations turned the  $x^2$  Hill term into a sign exploit. After restricting controls to transcription-factor nodes, scoring only specified targets, setting  $\varepsilon = 10^{-3}$  and clamping the Hill input, the cardiac search overexpresses MEF2C and reaches TNNT2 1.98 / MYH6 2.01 with stochastic success 1.0. Deterministic trajectories match an independent integrator to 0.0; the Langmuir chromatin occupancy, noisy-ensemble statistics and attractor stability ( $\max |\dot{x}| = 3.1 \times 10^{-8}$ ) all check out.

**(F94) Fate dynamics and control.**  $\dot{x} = g \odot \sigma(6(W\phi(x) - 0.5)) + u - \gamma x$ ,  $\phi(x) = x_+^2 / (0.25 + x_+^2)$ ,  $\gamma = 0.35$ ; occupancy  $= K_a c / (1 + K_a c)$  with  $K_a = e^{-\Delta G/RT}$ ; objective  $J = \sum_{i \in S} (x_i(T) - t_i)^2 + 0.08 \|u\|_2^2 + 0.04 \|u\|_1$  over specified targets  $S$  with controls restricted to TF nodes; success  $= e^{-\|x(T) - t\|_S}$ .

## 16.55 Literature knowledge graph against live PubMed, iCite and statsmodels

biogpt\_lit builds an evidence-weighted claim graph from the literature: a precision-biased relation extractor, citation/sample-size/replication edge weights, contradiction detection, temporal consensus, multi-hop latent paths, gap prioritization and a DerSimonian-Laird meta-analysis engine. The sweep ingested 8 live PubMed abstracts (query: SIRT1 resveratrol activation, all 2026) through the module's own entrez client and merged NIH iCite citation counts (only one PMID has an iCite record so far; the query returns very recent papers - documented). The audit found one defect: the extractor accepted any token before a relation word as the subject, producing the garbage triple (and, inhibits, Neuronal) from PMID:42746974 - one of four live-extracted claims. A function-word subject filter fixes it; post-fix all three extracted triples are valid ((formulations, activates, SIRT1), (SIRT1, activates, AMPK), (AMPK, inhibits, mTORC1)) and the negation/loss-of-function unit suite passes 7/7. The meta-analysis engine matches statsmodels combine\_effects (DL) to below  $10^{-9}$  on fixed effect 0.29756, random effect 0.32299 and  $\tau^2 = 0.08470$ . Evidence weights, consensus/confidence, contradiction detection, temporal agreement (0.692), multi-hop path sets, bottleneck path strength (0.515), gap leverage and the update\_validation flip all equal independent recomputation exactly.

**(F95) Evidence weighting and DerSimonian-Laird pooling.**  $w = 0.3 + 0.3 \min(\log_{10}(1 + c)/3, 1) + 0.25 \mathbb{I}_{\text{rep}} + 0.15 \min(n/100, 1)$ ; fixed effect  $\bar{\theta} = \sum w_i \theta_i / \sum w_i$  with  $w_i = 1/v_i$ ;  $Q = \sum w_i (\theta_i - \bar{\theta})^2$ ;  $\tau^2 = \max(0, (Q - (k - 1)) / (S_1 - S_2 / S_1))$ ; random weights  $1/(v_i + \tau^2)$ ; temporal agreement  $a_t = w_t^+ / (w_t^+ + w_t^-)$ ; path strength =  $\min_{e \in p} w_e$ .

## 16.56 Gene digital twin against NCBI Gene, UniProt, RefSeq and PubMed

gene\_analysis profiles a gene locus (composition, longest ORF, regulatory landscape, CRISPR targets, variant interpretation) and merges live NCBI Gene, UniProt, ClinVar and Reactome records. For TP53, BRCA1, CFTR, HBB and EGFR the live profile returned the correct NCBI Gene identifier and chromosome, the Swiss-Prot canonical accession and canonical protein length in 5 of 5 genes. On RefSeq mRNAs NM\_000518.5, NM\_000546.6 and NM\_000492.4 the longest ORF equals the annotated CDS (147, 393 and 1480 residues) and molecular weights agree with Biopython to within  $4.2 \times 10^{-5}$  relative. Publication counts match an independent PubMed query exactly and replicate numbers equal statsmodels NormalIndPower in 4 of 4 designs. Bug 44 was found on real transcripts: HGVS c. coordinates were indexed into the transcript without the 5' UTR offset, and full ClinVar names were parsed from the accession digits, so the sickle variant c.20A>T read codon TTC or ACT instead of GAG. After the fix, 6 of 6 clinical hotspots (HBB E7, Q40; TP53 R175, R248, R273; CFTR G551) map to the correct reference base and residue.

**(F96) HGVS coordinate mapping and sample size.** Locus position  $p = s_{CDS} + c$  for  $c > 0$  and  $p = s_{CDS} + c + 1$  for  $c < 0$ , with  $s_{CDS}$  the 0-based start of the longest forward ORF; codon start  $s_{CDS} + 3 \lfloor (p - 1 - s_{CDS}) / 3 \rfloor$ ; replicates  $n = \lceil 2\sigma^2(z_{1-\alpha/2} + z_{1-\beta})^2 / \delta^2 \rceil$ ; Boltzmann populations  $p_k = e^{-E_k/RT} / \sum_j e^{-E_j/RT}$  with  $RT = 0.593$  kcal/mol.

## 16.57 Liquid-biopsy stack against TCGA PanCancer and cfDNA fragment literature

liquid\_biopsy combines ctDNA calling, a cfDNA fragment-length model, Dirichlet-mixture haplotype EM, tumor architecture assembly and longitudinal tracking. Its tumor-type marker panels were scored against TCGA PanCancer Atlas mutation calls from cBioPortal for six cancers. TP53 was the most frequently mutated checked gene in lung adenocarcinoma (52.1% of 566 samples) and was mutated in 59.8% of 179

pancreatic samples, yet was missing from both panels; adding it raised panel coverage from 46.3% to 76.1% (lung) and 66.5% to 74.9% (pancreatic). Bug 45: haplotype candidates were built by filling unobserved loci with 0, so a haplotype observed only as partial fragments was never a candidate; with truth 111/000 the EM returned 000, 011 and 110. After the fix it returns 111 and 000 at 0.479 each. The fragment model's docstring claimed entropy rises with tumor fraction; the model itself shows entropy peaking near balanced mixtures (4.60 bits at 0.3, 4.55 at 0.6), whereas KL divergence and the 100-150 bp fraction rise monotonically. Anchors match Underhill et al. (134-144 bp versus 167 bp).

**(F97) Fragment mixture and haplotype EM.** Gamma component with mode  $m$  and spread  $s$ :  $k = 1 + (m/s)^2$ ,  $\theta = m/(k-1)$ ; entropy  $H = -\sum_b p_b \log_2 p_b$ ;  $D_{KL}(p||q_h) = \sum_b p_b \log(p_b/q_{h,b})$ ; fragment likelihood  $P(f|h) = (1-\epsilon)^{n_o-m} \epsilon^m$  over  $n_o$  observed loci with  $m$  mismatches; update  $\pi_j = (\sum_i r_{ij} + \alpha)/(N + \alpha K)$ ; panel coverage =  $|\bigcup_{g \in P} S_g|/N$ .

## 16.58 AAV capsid engineer against real VP1 sequences and scoring recomputation

vector\_opt designs capsid variants by trading receptor-affinity gain against NAb escape and conservation penalties over five variable regions. Its scoring was recomputed exactly from the stated formulas with the module's own random sequence for three seeds, and directed evolution round  $r$  equals a fresh engineering run seeded  $1+r$ . The region annotations were checked against the real capsid sequences of AAV2 (UniProt P03135, 735 residues) and AAV9 (Q6JC40, 736 residues): a global alignment gives 82.3% identity, and window identity confirms VR-VIII (0.33) and VR-IV (0.43) as the most variable regions. Two annotation limits were found and documented: receptor contact is serotype-dependent (AAV2's heparan-sulfate contact R585/R588 sits inside VR-VIII, PMID 9445046, while the table follows the AAV9 default), and the GH-loop conservation constant 0.35 reflects the loop tip, since the measured identity over the 590-620 span is 0.81.

**(F98) Capsid variant scoring.** Receptor gain  $g = 0.15 m c_r u_1$  with  $c_r \in \{1, 0.2\}$ ,  $u_1 \sim U(0.7, 1.3)$ ; NAb loss  $l = 0.2 m u_2 \mathbb{1}_{\text{epitope}} + 0.02 m u_2 (1 - \mathbb{1}_{\text{epitope}})$ ; integrity  $I = 1 - 0.08 m (1 - C_r)$ ; transduction  $T = \text{clip}(0.5 + g - (1 - I), 0, 1)$ ; ranking score  $0.5 T + 0.5 \min(l, 1)$ ; global identity  $604/734 = 0.823$ .

## 16.59 Neurosurgical planner against an admissible-path oracle

neuroplan\_ai plans tumor approaches: corridor scoring over an eloquent-region risk field, weighted A  $\times$  pathfinding, MRI segmentation, tractography clearance and a NeuroTwin resection simulator. The corridor algebra, NeuroTwin probabilities, segmentation threshold and centroid all recompute exactly. Two bugs were fixed. Bug 47: tract clearance was measured to streamline vertices only, so a fiber passing near the tumor between samples was reported 7.30 mm clear when the segment comes within 6.0 mm; clearance is now point-to-segment. Bug 46: the A docstring claimed an admissible heuristic, but the frontier uses a 50-fold inflated weight; against an admissible oracle with the identical cost, the returned path cost 640.6 versus the 340.1 optimum (88% gap) on an oblique entry. The docstring now states the greedy bound, and four dead duplicate definitions were removed. The normalized atlas places brainstem lowest, motor cortex highest and visual cortex most posterior, consistent with anatomy.

**(F99) Risk field, path cost and tract clearance.**  $R(p) = \sum_i w_i e^{-d_i^2/2\sigma^2}$ ,  $\sigma = 12$ ; step cost  $c = \ell (1 + \lambda \min(R, 1.25))$ ,  $\lambda = 8$ ; frontier  $g + 50h$ ; segment clearance  $\min_t \|p_0 + t(p_1 - p_0) - c\| - r$ ,  $t = \text{clip}(\frac{(c-p_0) \cdot (p_1-p_0)}{\|p_1-p_0\|^2}, 0, 1)$ ; deficit  $p = \min(0.95, w k f / (1 + \text{clr}/5))$ ; aggregate  $1 - \prod(1 - p_i)$ .

## 16.60 Structure predictor against 1UBQ, AlphaFold DB and the 1978 Chou-Fasman rules

alpha\_fold.ui implements Chou-Fasman prediction, idealized backbone building, pocket detection and refinement, and can analyze real structures. On real data: ubiquitin 1UBQ parses to 76 residues with X-ray resolution 1.8 Å and exactly recomputed B-factor statistics; the AlphaFold model of HBB (P68871) has mean pLDDT 97.18 with no low-confidence residues. Bug 48: the helix-extension rule admitted any residue with  $P_\alpha > 100$ , so on ubiquitin the prediction was 92% helix while 1UBQ has one helix and five strands. The 1978 breaker rule (stop when the mean propensity of the next tetrapeptide drops below 100) with per-residue  $P_\alpha/P_\beta$  conflict resolution predicts the 23-34 helix plus strand regions,  $Q_3 = 0.513$  against the literature assignment - the expected range for this method. The parameter table matches the canonical 1978 values for all 20 amino acids. Generated backbones round-trip through a standard parser with consecutive C $\alpha$ -C $\alpha$  distances of 3.799-3.801 Å, and restrained refinement lowers the energy from 100.96 to -8.99 with exact 3.8 Å bonds.

**(F100) Chou-Fasman nucleation, breaker and conflict rules.** Helix nucleus:  $\geq 4$  of 6 residues with  $P_\alpha \geq 100$ ; strand nucleus:  $\geq 3$  of 5 with  $P_\beta \geq 100$ ; extension stops when  $\bar{P}$  over the next tetrapeptide  $< 100$ ; overlaps resolved by  $\bar{P}_\alpha$  vs  $\bar{P}_\beta$  over the  $\pm 3$  window;  $Q_3 = \frac{1}{n} \sum_i \mathbb{I}[\hat{s}_i = s_i] = 0.513$  on ubiquitin; refinement energy  $E = E_{LJ} + E_{elec} + k_b \sum (r - 3.80)^2$ .

## 16.61 Fragment molecule generator against PubChem and live ChEMBL

chemgpt\_engine assembles molecules from a 12-fragment library and scores fragment-additive ADMET. Seven assembled molecules - phenol, aniline, benzoic acid, fluorobenzene, 2-aminopyridine, 4-methylimidazole and phenethylamine - each resolved to the correct real compound in PubChem (queried by SMILES), with molecular weights within 0.03 Da. The scoring formulas recompute exactly, and the Pareto front (20 of 24 candidates) matches a brute-force non-dominated filter over an independently regenerated candidate list. Novelty claims were checked against the live ChEMBL similarity service: phenol’s nearest neighbor is phenolate sodium at 100% similarity, while the top generated design has no neighbor at or above 80%, so its “novel scaffold” label is measured, not asserted. No defect was found in this module.

**(F101) Fragment-additive assembly and Pareto filtering.**  $MW = \sum_f MW_f - 2.016(k - 1)$  for  $k$  fragments;  $\log S = 0.5 - \log P - 0.01(MW - 200)/50$ ; dominance  $a \succeq b \iff \forall j o_j^a \geq o_j^b \wedge \exists j o_j^a > o_j^b$ ; Boltzmann weights  $w_i = e^{-E_i/0.593} / \sum_j e^{-E_j/0.593}$  sum to 1.

## 16.62 Circuit stability scorer against recomputation and real restriction sites

syn\_stab.ai estimates the evolutionary stability of synthetic circuits: expression burden, plasmid-loss and mutation survival, a Wright-Fisher-like population simulation and sequence-level risk. All formulas recompute exactly, and the population simulation matches the closed selection recursion in the deterministic limit. Bug 49: the sequence-risk palindrome counter compared each 6-mer with its plain reverse, so real inverted repeats - EcoRI GAATTC, HindIII AAGCTT, SnaBI TACGTA - all counted zero; the reverse-complement check counts them, while textual palindromes such as GATTAG correctly remain at zero. Strategy comparison ranks genomic integration above promoter weakening, consistent with the loss-rate model.

**(F102) Burden, retention and selection recursion.** Burden  $b = \min(1, 0.25 g e)$ ; plasmid retention  $(1 - \lambda)^t$  with  $\lambda = \lambda_0(1 + 4b)$ ; mutation survival  $e^{-\mu L t}$ ; update  $f \leftarrow f(1 - \mu)(1 - c)/(1 - fc)$  with diffusion

noise  $\sqrt{f(1-f)/N}$ ; inverted repeat:  $x = \overline{\text{comp}(x)}$ .

### 16.63 Phage designer against recomputation and adsorption kinetics

phage\_designer plans tail-fiber retargeting, chimeric endolysins, adsorption, escape risk and cocktails. The fiber and lysin formulas, escape probability  $1 - e^{-\mu s N}$  and host-range scoring all recompute exactly. Bug 50: adsorption\_kinetics accepted time\_min but never used it - bound phage was identical at  $t = 1$  and  $t = 60$  minutes. Adsorption is a first-order process (Schlesinger 1932), so the function now returns  $P_0(1 - e^{-k B t})$ ; post-fix, bound phage grows with time and matches the closed form to machine precision.

**(F103) Fiber adsorption score and first-order kinetics.** Adsorption score  $a = \min(0.99, 0.55 + 12/K_d)$ ; lysin potency  $\bar{\alpha}$  with MIC =  $\max(0.1, 8(1.1 - \bar{\alpha}))$   $\mu\text{g/mL}$ ; escape  $P = 1 - e^{-\mu s N}$ ; kinetics  $B(t) = P_0(1 - e^{-k C t})$ .

### 16.64 CRISPR phage forge against the real NDM-1 plasmid and a host-chromosome screen

phageforge designs CRISPR-armed phages against resistance genes: backbone selection, guide design, payload budget, kill curves, escape risk and cocktail coverage. On the real blaNDM-1 CDS (813 bp, from plasmid pKpANDM-1, FN396876.1, 4318 bp) the module selects the P1 backbone (15 kb capacity, lytic, the eligible maximum for the 4.5 kb payload) and all three guides verify as 20-mers inside the CDS. Bug 51 couples two defects: the off-target screen ran against the target gene itself while the report claimed a screen against a supplied phage genome - an input that does not exist - and the CFD scanner skipped every perfect-match site, making the worst off-target class invisible (a planted exact match returned zero hits pre-fix). Perfect matches are now reported at CFD 1.0, design\_phage accepts background\_genome, and the report states exactly what was screened; against a real 40 kb window of the K. pneumoniae MGH 78578 chromosome (NC\_009648.1) all three blaNDM-1 guides show zero CFD hits. Bug 52: the cocktail coverage fraction was always 1.0 whenever any host was covered (union denominator); an optional target panel gives a real fraction (0.75 with one uncovered target in the sweep case). Kill curve, escape risk, payload budget and microbiome-sparing recompute exactly; backbone capacities were checked against the NCBI-verified genome sizes (T7 39,937 bp; lambda 48,502; T4 168,903; P1 94,800; M13 6,407) and are payload headroom, plausible but not individually literature-sourced (documented).

**(F104) Phage infection, escape and screening formulas.** Infected fraction  $f = 1 - e^{-a \text{MOI}}$ ; surviving  $S = (1 - f)(1 - r) + r$ ; effective burst  $f B$ ; expected escapees  $E = N \mu^k$  for  $k$  independent targets with  $P_{\text{esc}} = 1 - e^{-E}$ ; guide off-target risk  $R = \sum_s \text{CFD}(g, s)$  over background sites within three mismatches, perfect matches included; cocktail coverage  $|\mathcal{T} \cap \mathcal{C}|/|\mathcal{T}|$ ; microbiome sparing  $0.6 + 0.3(1 - 0.5 e)$ .

### 16.65 Variant copilot against a live ClinVar panel and UniProt

bio\_copilot is a variant-interpretation workbench: a mutation-to-phenotype computation graph (variant annotation, structural perturbation, pathway effect, phenotype inference with inconsistency flags), CRISPR guide enumeration, FASTA/PDB structured output, a UniProt-grounded question router, Monte-Carlo uncertainty propagation and beta-binomial experiment feedback. The curated gene slice was checked against live UniProt: all three lengths match exactly (TP53 P04637 393 aa, BRCA1 P38398 1863 aa, KRAS P01116 189 aa) and the TP53 DNA-binding domain 102-292 equals the UniProt feature, while the BRCA1 RING and BRCT windows are wider local approximations (documented against the UniProt zinc finger 24-65 and

the BRCT domains 1642-1736 and 1756-1855). An eight-variant panel was scored against live ClinVar records. Bug 53: a premature stop set pathway flux to 1.0, but the risk formula  $0.15 + 0.5f$  capped at 0.65, so TP53 R213\* and BRCA1 Q563\* - both Pathogenic in ClinVar, one expert-panel reviewed - were returned as “uncertain significance”. The fix adds a per-gene loss-of-function flag with a PVS1-style floor  $r \geq 0.75$  (ACMG/AMP guideline, PMID 25741868) and deliberately withholds it from gain-of-function oncogenes: KRAS L168\* is Likely benign in ClinVar and other KRAS truncating records are VUS or conflicting, so the module stays in the uncertainty band and raises an explicit caveat flag. Exact-band agreement rose from 4/8 to 6/8; the two remaining calls are one band away with documented rationale (oncogene-nonsense conservatism, and a KRAS hotspot-coverage caveat - the curated hotspot list was not refit to the validation panel). The guide enumerator returns a set identical to crispr\_opt on the real 813 bp blaNDM-1 CDS (137 guides, both strands, same forward-coordinate convention), and to\_pdb output on the real 189-residue KRAS sequence parses cleanly (189 C $\alpha$  atoms, residue names matching, consecutive C $\alpha$ -C $\alpha$  3.8297 Å against the documented 3.8 Å ideal-helix geometry). Uncertainty propagation matches the analytic product-of-Gaussians mean and the beta-binomial update matches scipy exactly.

**(F105) Variant risk with a mechanism-aware nonsense floor.**  $r = 0.15 + 0.5f + 0.25\mathbb{I}_h$ ; flux  $f = \min(1, \frac{\Delta\Delta G}{3}c)$  with  $c = 1.5$  at hotspots and 0.8 otherwise, and  $f = 1$  for nonsense;  $\Delta\Delta G = 0.8|\Delta H| + 1.2\mathbb{I}_h + 0.3(1 - \mathbb{I}_h) + 0.5\mathbb{I}_d$  over hydrophobicity change  $\Delta H$ , hotspot  $h$  and domain  $d$  indicators; the floor  $r \leftarrow \max(r, 0.75)$  applies exactly when the variant is nonsense and the gene carries the loss-of-function flag (PVS1-style); bands  $r > 0.7$  likely pathogenic,  $r > 0.4$  uncertain, else likely tolerated.

**(F106) Product of independent Gaussians.**  $E[\prod_i X_i] = \prod_i \mu_i$  by independence; for  $(\mu, \sigma) = (0.8, 0.1)$  and  $(0.7, 0.05)$  the analytic mean is 0.56, matched by a  $2 \times 10^4$ -sample Monte-Carlo mean whose 95% interval contains it.

**(F107) Beta-binomial experimental feedback.** Prior mean  $\mu_0$  with strength  $s$ :  $a = \mu_0 s + n_s$ ,  $b = (1 - \mu_0)s + (n - n_s)$ ; posterior mean  $a/(a + b)$ , equal to the scipy beta mean ( $\mu_0 = 0.5$ ,  $s = 10$ , 9/10 successes  $\rightarrow 0.7$ ).

## 16.66 Cell-image analyser against analytic shapes and BBBC001 ground truth

bioimage.ai segments microscopy frames (flat-field correction, Otsu or adaptive threshold, distance-transform splitting), measures per-cell morphology and computes 50 screening diagnostics. Two defects were found on geometric ground truth. Bug 54: the perimeter was the boundary-pixel count, which measures a 45-degree edge at  $1/\sqrt{2}$  of its length - circularity scored 1.27 for a perfect disk and 1.69 for a rotated square. The Freeman chain convention (orthogonal steps  $\times 1$ , diagonal  $\times \sqrt{2}$ ) gives 0.92 for the disk - the standard region-props value, the residual being pixelation - and exact perimeters for cardinal shapes (76 for a 20-pixel square,  $56\sqrt{2}$  for the diamond). Bug 55: distance-maximum markers fired on mid-ramp pixelation plateaus, so one convex elongated cell was split into three fragments; regional maxima by morphological reconstruction leave a single marker (the same cell now measures aspect 3.74 against nominal 4), while a genuine two-peak touching pair still splits. The Otsu threshold matches an independent between-class implementation exactly, and on all six real BBBC001 images (Broad benchmark, human HT29 cells) the module counts sit 4-9% above the inter-observer mean and track the manual counts with Pearson  $r = 0.9983$  - inside the noise band of the two human counters, who disagree by up to 27% on one image.

**(F108) Otsu thresholding.** Class probabilities  $\omega(k) = \sum_{i \leq k} p_i$ , means  $\mu(k) = \sum_{i \leq k} i p_i$ ; pick  $k^* = \arg \max_k (\mu_T \omega(k) - \mu(k))^2 / (\omega(k)(1 - \omega(k)))$ .

**(F109) Freeman chain perimeter and circularity.**  $P = N_{\text{ortho}} + \sqrt{2} N_{\text{diag}}$  over the traced outer boundary chain; circularity  $C = 4\pi A / P^2$ ; a filled disk scores  $\approx 0.91$ , an  $n \times n$  square  $4\pi n^2 / (4(n-1))^2$ .

**(F110) Regional maxima by reconstruction.** Reconstruct  $d - 1$  under the mask  $d$  by iterated geodesic dilation  $r \leftarrow \min(\delta(r), d)$  to stability; maxima  $\{d > r\}$ ; markers label the maxima and each foreground pixel takes its nearest marker (Voronoi split).

## 16.67 Evolution sandbox against population-genetics theory

bioplayground simulates a Wright-Fisher construct locus (selection, symmetric mutation, multinomial drift, bottlenecks), directed evolution, Lotka-Volterra competition and diversity metrics. One defect was found. Bug 56: the reported generation count was ten times the number of trajectory snapshots - 310 for a 300-generation run, 100 for a construct lost at generation 92 - and the playground report computed diversity on the last ten-generation snapshot rather than the final state ( $h = 0.0198$  reported when the construct was already lost, true  $h = 0$ ). The fix counts generations directly and exposes the final frequency vector, which the report now uses. Validation is against analytic theory: the two-allele mutation equilibrium  $p_t = \frac{1}{2} + \frac{1}{2}(1 - 2\mu)^t$  holds (0.5001 vs 0.5012 at  $N = 4 \times 10^4$ ), a single-species Lotka-Volterra run tracks the closed-form logistic within  $8 \times 10^{-4}$  mid-course and exactly at carrying capacity, coexistence calls match competition theory in both regimes, and the Shannon/heterozygosity metrics equal scipy exactly. A design limitation is documented: the construct starts at frequency 1, so with the default mutation rate every run fixes at generation 1 - loss requires mutation strong enough to seed competing alleles.

**(F111) Selection-mutation recursion and equilibrium.**  $p' = p w_0 / \bar{w}$  with  $\bar{w} = \sum_i p_i w_i$ ; symmetric mutation  $m_i = s_i(1 - \mu) + \mu(1 - s_i)/(n - 1)$  conserves  $\sum_i m_i = 1$ ; two-allele equilibrium  $p_t = \frac{1}{2} + (p_0 - \frac{1}{2})(1 - 2\mu)^t$ .

**(F112) Logistic competition.**  $\dot{x}_i = g_i x_i (1 - \sum_j c_{ij} x_j)$ ; single-species solution  $x(t) = K / (1 + A e^{-gt})$ ,  $A = K/x_0 - 1$ ,  $K = 1/c$ ; coexistence requires interspecific competition below intraspecific for both species.

**(F113) Diversity metrics.** Heterozygosity  $h = 1 - \sum_i x_i^2$ ; Shannon  $H = -\sum_i x_i \ln x_i$ ; effective alleles  $e^H$ .

## 16.68 Bioprint multiphysics against constitutive-equation ground truth

bioprint\_pro calibrates bioinks (Cross-model rheology, Hagen-Poiseuille pressure, first-order crosslinking) and runs a multiphysics chain: extrusion, reaction-diffusion gelation, poro-viscoelastic mechanics and a maturation ODE, with 51 computed diagnostics and a seeded global optimizer. Two defects were found. Bug 57: the extrusion flow used the viscosity at a guessed shear ( $6.7 \text{ s}^{-1}$ ) while the resulting wall shear was  $33.9 \text{ s}^{-1}$ ; the shear-thinning ink is far thinner there, so the flow rate came out  $4.36 \times$  low ( $0.213$  vs  $0.928 \text{ mm}^3/\text{s}$ ). The fixed point of  $q = \pi r^4 \Delta p / (8 \eta (4q / \pi r^3) L)$  is now solved by bisection and matches an independent root finder. Bug 58: the stress model was labeled Oldroyd-B while returning a shear-thinning stress; an Oldroyd-B fluid has constant shear viscosity, so the label now reads Giesekus-family. The calibrator recomputes

exactly for all 15 windows (alginate at 8 mm/s:  $160 \text{ s}^{-1}$ ,  $6.512 \text{ Pa}\cdot\text{s}$ ,  $104.2 \text{ kPa}$ ), the gelation PDE matches  $1 - e^{-kt}$  at the surface, and the maturation ODE matches an independent RK4 integration to  $10^{-7}$ .

**(F114) Cross/Carreau rheology and pipe flow.** Cross:  $\eta = \eta_\infty + (\eta_0 - \eta_\infty)/(1 + (\lambda \dot{\gamma})^n)$ ; Carreau-Yasuda:  $\eta_\infty + (\eta_0 - \eta_\infty)(1 + (\lambda \dot{\gamma})^a)^{(n-1)/a}$ ; wall shear  $\dot{\gamma}_w = 4Q/\pi r^3$ ; self-consistent rate  $Q = \pi r^4 \Delta p / (8 \eta(\dot{\gamma}_w) L)$ .

**(F115) Gelation, relaxation and maturation.** Conversion  $x = 1 - e^{-kt}$ ,  $t_{95} = -\ln 0.05/k$ ; modulus  $E_0(1 + 9x)$ ; stiffness  $2 + 48 g^2 \text{ kPa}$ ; standard-solid strain  $\varepsilon(\tau) = \varepsilon_0(0.35 + 0.65e^{-\tau/\tau})$ ; Euler buckling  $E\pi^2/(12n^2)$ ; maturation  $\dot{c} = 0.18c(1 - c/5)o - 0.03c$  with oxygen  $\dot{o} = 0.25(1 - o) - 0.08co$ .

## 16.69 Virtual-lab sessions and mechanistic experiments

biosim\_vr pairs a scripted virtual bench (move, pipette, incubate, measure) with mechanistic experiments: rigid-pose Lennard-Jones docking by differential evolution, CRISPR guide enumeration, and a stochastic protocol simulator with 51 asserted diagnostics. Two defects were found and fixed. Bug 59: the scripted transfection session set the plate signal to 0.72 but measured the plate-reader object, which has no signal state, so the log read 0.0 while the conclusion claimed 0.72 RFU; the measurement now targets the plate. Bug 60: the docking session crashed outright, calling dock with swapped arguments and a nonexistent seed keyword; it now reports  $-2.94 \text{ kcal/mol}$ . The guide enumerator reproduces crispr\_opt's forward-strand set exactly on the real  $bla_{NDM-1}$  CDS (65 guides), the rotation is a proper rotation, the returned docked pose reproduces the returned score exactly with rigid bonds, and on a spherical ground-truth pocket the ligand settles at the LJ well ( $3.00 \text{ \AA}$ ).

**(F116) Virtual-lab kinetics and scoring.** Incubation gain  $g = (1 - e^{-t/240})e^{-((T-37)/12)^2}$ ; shifted LJ well  $u(r) = (3/r)^{12} - 2(3/r)^6$  with minimum  $-1$  at  $r = 3$ ; guide score  $\max(0, 1 - 2|GC - 0.5| - 0.15 \max(0, h - 3))/(1 + n_{\text{dup}})$ .

## 16.70 Digital cell twins and virtual drug trials

cell\_twin builds a multi-omic patient-cell state and screens drug responses with a viable/apoptotic/proliferating/damage ODE. Two defects were found and fixed. Bug 61: the reported maximum apoptosis was taken at the last-listed dose, so a descending dose list silently returned the minimum-dose value. Bug 62: the combination screen claimed a mechanistic combined-exposure rerun but merely added a constant  $+0.08$  synergy bonus to the Bliss expectation for every pair; combinations are now genuinely rerun with the Bliss-combined effect scalar  $1 - (1 - e_a)(1 - e_b)$  in the response ODEs, so the Bliss excess is solved, not assumed. The pathway activities  $s/(s + 100)$ , the effect scalar  $k \cdot d/(d + 1) \cdot (1 - r)$ , and the Bliss expectation all recompute exactly, and the single-drug ODE matches an independent RK4 integration to  $10^{-3}$  in terminal viability.

**(F117) Twin response model.** Pathway activity  $A = s/(s + 100)$ ; effect  $e = kd/(d + 1)(1 - r)$ ; ODE  $\dot{v} = g - d_{\text{kill}}, \dot{a} = d_{\text{kill}} - 0.02a, \dot{p} = -0.05ep + 0.02A_p(1 - p), \dot{u} = i - 0.03A_{\text{rep}}u - 0.02u$  with  $g = 0.04A_p v(1 - v/2), i = 0.08ev, d_{\text{kill}} = i(0.5 + A_a)(1 - 0.5A_s)$ ; Bliss combination  $e_{ab} = 1 - (1 - e_a)(1 - e_b)$ .

## 16.71 Module 91: cellpainter (Cell Painting morphological profiling)

**Finding (BUG 63).** Mechanism similarity was cosine over raw signature vectors whose entries sit near 1.0 (fold change vs vehicle), so the shared baseline dominated: distinct mechanisms scored about 0.999 (sugarcodes commit e6fb780) and the novel-mechanism hint could never trigger. After the fix, similarity is computed on vehicle-subtracted deltas: the three distinct mechanism pairs score 0.0, the same mechanism at two doses scores 1.0, and the novel-mechanism hint is reachable (benchmarks/sweep\_cellpainter.json, 6/6 checks passed). The image path on real BBBC001 stacks is deferred to cellpainter\_4d.

**(F118) Delta-cosine and signature model.** Signature feature  $s_j = 1 + (\delta_j - 1) \log_{10}(1 + d) \min(1, h/24)$  for dose  $d$  and time  $h$ ; similarity  $\cos(\mathbf{s}_a - \mathbf{1}, \mathbf{s}_b - \mathbf{1})$ ; plate normalisation  $z = (x - \bar{x}) / (1.4826 \text{ MAD})$ ; vehicle ( $d = 0$ ) gives  $\mathbf{s} = \mathbf{1}$  exactly. Recomputed signature value 1.12 matched the module; QC CV 0.0816 matched.

## 16.72 Module 92: cellpainter\_4d (temporal morphology and real nuclear images)

**Finding (BUG 64).** The morphology rate constant was set from the observation window ( $\tau = T_{\text{obs}}/5$ ) while cell death used fixed hourly rates, so the same apoptosis run gave 12 h nuclear areas of 71.49, 120.84 and 152.29  $\mu\text{m}^2$  for 24, 72 and 144 h windows. Kinetics now use a fixed program clock and the 12 h area is 120.84 for every window (benchmarks/sweep\_cellpainter\_4d.json).

**Real images.** New segment\_nuclei on BBBC001 (6 HT29 Hoechst fields, two human counters): mean count deviation 8.18% (inter-human 11%, CellProfiler 6.2%). Negative result: two earlier drafts scored 7.0% and 5.8% but leaked across the watershed background (synthetic 121-px disks measured 145/109/109, then 81 px); the shipped version is the geometry-correct one and nothing was tuned to the ground truth.

**(F119) Program-clock relaxation and segmentation geometry.**  $a(t) = a_0 t_a + a_0(1 - t_a)e^{-t/\tau}$ ,  $\tau = T_{\text{prog}}/(5s)$ ;  $n(t) = n_0 e^{-0.04t}$  (apoptosis); Otsu  $t^* = \arg \max_t [\mu_T \omega(t) - \mu(t)]^2 / [\omega(t)(1 - \omega(t))]$ ; seed radius  $r = \frac{1}{2} \sqrt{A/\pi}$ ; circularity  $4\pi A/P^2$  with  $P = \frac{\pi}{4} E$  for  $E$  exposed pixel edges; aspect ratio  $\sqrt{\lambda_1/\lambda_2}$  of the pixel covariance.

## 16.73 Module 93: crispr\_cargo (delivery vehicle selection and PK)

**Finding (BUG 65).** pk\_model added a zero-width-time trapezoid  $(c_0 + c_0)/2 \cdot dt$  at  $t = 0$ , overstating AUC by exactly  $c_0 dt$  at every step size: +5.8% for LNP over 96 h at  $dt = 1$  (608.317 vs analytic 574.824). The fixed grid matches the closed form  $C_0/k_e(1 - e^{-k_e t})$  to  $2.8 \times 10^{-4}$  relative at  $dt = 1$  and exactly at  $dt = 0.01$  (benchmarks/sweep\_crispr\_cargo.json).

**(F120) One-compartment PK and payload accounting.**  $C(t) = (D/V_d)e^{-k_e t}$ ,  $k_e = \ln 2/t_{1/2}$ ; AUC =  $C_0(1 - e^{-k_e T})/k_e$ ; CDS length =  $3N_{aa} + 3$  bp (UniProt Q99ZW2 SpCas9 1368 aa  $\rightarrow$  4107 bp, J7RUA5 SaCas9 1053 aa  $\rightarrow$  3162 bp, fetched live); AAV's 4.7 kb cargo limit leaves under 0.6 kb once the SpCas9 CDS is in. Compartment mass balance error  $< 10^{-6}$ .

## 16.74 Module 94: crispr\_muse (adaptive guide design)

**Finding (BUG 66).** enumerate\_configurations only searched 3' of the spacer for the PAM, but Cas12a (TTTV) and CasX (TTCN) PAMs are 5' of the spacer, so every valid Cas12a/CasX site was missed (0

configurations on a 5'-TTTA + 23-nt guide construct). A `pam_side` field fixes this; Cas9 enumeration is unchanged (benchmarks/sweep\_crispr\_muse.json).

**(F121) Posterior and policy quantities.** Beta posterior  $\alpha' = a + s$ ,  $\beta' = b + f$ ,  $\mu = \alpha' / (\alpha' + \beta')$ ,  $\sigma^2 = \alpha' \beta' / [(\alpha' + \beta')^2 (\alpha' + \beta' + 1)]$  (matches Beta(9,3): 0.75, 0.1201); clonal selection  $f_i^{t+1} = f_i^t w_i / \sum_j f_j^t w_j$  (b after 10 doublings: 102.4/103.3); policy entropy  $-\sum p \ln p$ , uniform =  $\ln 4$ ; repair prior  $[1.1 - 0.35m, 0.2 + 1.5m, 0.05 + 1.4d]$  with microhomology  $m$ , donor  $d$ , S-phase  $s$ .

## 16.75 Module 95: ecosystem (plugin, SDK and standards tooling)

**Finding (BUG 67).** `resolve_plugin_order` keyed dependencies by name, so two different manifests with the same name collapsed into one plugin (`plugin_count = 1`) and the second plugin's capabilities vanished. Duplicate names now raise; self-cycles, two-plugin cycles and missing dependencies are all rejected (benchmarks/sweep\_ecosystem.json).

**(F122) Plugin graph resolution.** Kahn topological order over the dependency digraph  $G = (V, E)$  with in-degree  $d^-(v) = |\text{deps}(v)|$ ; acceptance iff  $|\text{order}| = |V|$ ; dependency density  $|E| / (n(n-1))$ ; integration readiness  $(c/a + r/n)/2$  with  $c/a$  compliant assets and  $r/n$  parallel roots. Fifty deterministic diagnostics over the live registry (95 modules, 9 subnetworks).

## 16.76 Module 96: enterprise\_bio (tiered entitlements and governed operations)

**Finding.** No defect found under adversarial probes: a single-field tamper breaks the audit chain; cross-tenant vault decryption fails (per-tenant Fernet keys); academic blocks module 41/vault/robot while allowing module 40; illegal robot transitions and budget overspend are rejected (benchmarks/sweep\_enterprise\_bio.json, 6/6 checks, 8 new tests).

**(F123) Audit chain and usage forecast.** Event digest  $h_i = \text{SHA256}(\text{canon}(e_i))$  with  $e_i$  carrying `prev =  $h_{i-1}$`  ( $h_0 = \text{genesis}$ ); verification replays links and digests. Per-tenant vault key  $k_t = \text{HMAC-SHA256}(k_m, \text{sugarcodes-vault})$ . Forecast  $\hat{h} = h_{\text{used}} \cdot 30/d$ ; over-quota iff  $\hat{h} > q$ .

## 16.77 Module 97: epi\_edit (CRISPRa/i epigenome perturbation design)

**Findings (BUGs 68 and 69).** `expression_trajectory` reported `steady_state = target/decay`, ignoring its own feedback term: it reported 1.577 while the ODE actually converged to 1.385. The fixed point now solves  $\delta f y^2 + \delta y - T = 0$  and matches the 72 h trajectory endpoint. Separately, CRISPRa with the TSS too close to the sequence start silently fell back to designing guides  $-300..+100$  (inside the gene, the CRISPRi window) while reporting `guide_window [0,0]`; CRISPRa is now strictly upstream-only and reports zero guides honestly (benchmarks/sweep\_epi\_edit.json).

**(F124) Effector response and guide windows.** Recruitment  $r = o^n / (0.5^n + o^n)$  ( $r = 0.5$  at  $o = 0.5$ ); state  $s_k = \text{clip}(b_k + r\Delta_k)$ ; expression  $= 0.45a + 0.4ac + 0.15(1-m)$ . CRISPRi window  $[-50, +300]$  around TSS, CRISPRa  $[-400, -50]$  upstream - verified on FN396876 (4318 bp): 5 CRISPRi guides ( $0.395 \times$  repression), 5 CRISPRa guides ( $4.46 \times$  activation).

**(F125) Expression trajectory fixed point.**  $\dot{y} = T/(1 + fy) - \delta y$ ,  $\delta = \ln 2/t_{1/2}$ ; steady state  $y^* = (\sqrt{\delta^2 + 4\delta fT} - \delta)/(2\delta f)$  for  $f > 0$ , else  $T/\delta$ .

## 16.78 Module 98: fate\_predictor (cell-fate reprogramming design)

**Finding.** No code defect; validated against live PubMed: fibroblast→neuron ASCL1/BRN2/MYT1L (Vierbuchen 2010, PMID 20107439), fibroblast→cardiomyocyte GMT (Ieda 2010, PMID 20691899), B-cell→macrophage CEBPA (Xie 2004, PMID 15163413), fibroblast→endothelial ETV2 (Morita 2015, PMID 25540418). The exact-subset optimizer rediscovers the published factor sets. Negative: the hermetic ODE peaks at 5.65% target fraction versus the curated 35% literature rate for the same protocol - documented as a calibration gap, not refitted (benchmarks/sweep\_fate\_predictor.json).

**(F126) TF subset selection and state transitions.** Score =  $\sum_{g \in T} e_g^+ w_g + \sum_{g \in S} e_g^- w_g - 0.08k^{1.5}$  over all  $\binom{n}{k}$  subsets; ODE  $\dot{s} = -c$ ,  $\dot{\tau} = c - m - 0.01x\tau$ ,  $\dot{f} = m - 0.005f$ ,  $\dot{x} = b(s + \tau) - 0.02x$  with conversion  $c = 0.02 \sigma A s$  and maturation  $m = 0.035 \tau e^{-x}$ .

## 16.79 Module 99: gene\_explorer (central-dogma trace)

**Finding (BUG 70).** explore() translated from frame 0 of the raw input, so on a real RefSeq mRNA with a 5' UTR (NM\_000518.5 HBB, 628 nt) it reported a 20-aa UTR peptide (2260 Da) as the protein while its own longest\_orf knew the true 147-aa hemoglobin. The protein block now annotates the longest ORF (translation\_mode longest\_orf), falling back to frame translation only when no ORF exists (benchmarks/sweep\_gene\_explorer.json).

**(F127) ORF-anchored translation and variant cascade.** ORF translation of NM\_000518.5 matches UniProt P68871 exactly (147 aa); sickle-cell E6V (GAG→GTG) gives exactly one protein change (MVHLTPEE→MVHLTP); silent CCT→CTT gives zero. Cascade  $\Delta\Delta G = 0.5 + 2c + 0.8i + 0.4n_{aa}$ ; expression ratio  $\max(0.05, 1 - r - 0.1n_{aa})$ ; export probability  $p_{cap}(1 - e^{-L_{polyA}/50})$ .

## 16.80 Module 100: gene\_tx\_opt (gene-therapy program design)

**Finding.** No defect found: canonical programs verified (liver AAV8+TBG, CNS AAV9+Syn1, retina AAV2, muscle AAV9); an oversized transgene selects nothing; score and immune formulas recompute exactly; delivery mass declines monotonically over 720 h (benchmarks/sweep\_gene\_tx\_opt.json, 5/5 checks, 8 new tests). Documented negative: the 168 h default window ends before the expression peak (true peak 198 h).

**(F128) Vector-promoter scoring and delivery cascade.** Expression =  $\tau a s_p f_c (1 - p_i)$ ; score =  $0.55E + 0.2\text{spec} + 0.15f_c + 0.1(1 - p_i)$ ; immune risk  $p_i = 1 - (1 - p_{\text{nab}})(1 - p_{\text{route}})$ ; delivery  $\dot{P} = -(k_e + u)P$ ,  $\dot{T} = uP - 0.015T$ ,  $\dot{I} = 0.015T - 0.008I$ ,  $\dot{E} = 0.004I - 0.02E$  with uptake  $u = 0.025 a \tau P$ .

**(F129) Splice-site scoring on the HBB locus.** Donor log-odds  $s_d = \sum_i \log_2 p_i(b_i)/0.25$  normalized by the consensus maximum (Shapiro-Senapathy MAG|GTRAGT, invariant GT at  $p = 1$ ). On the real NG\_000007.3 HBB locus (GenBank CDS join 70595..70686, 70817..71039, 71890..72018; translates to the exact 147-aa

P68871 protein): IVS2 donor scores 0.823 at the exact annotated junction, IVS2 acceptor 0.805, IVS1 acceptor 0.683 within 2 nt. Negative: this RefSeqGene haplotype carries a non-consensus IVS1 donor GTTGGT scoring 0.476, correctly below the 0.8 default threshold. Loop confidence  $c = \min(1, 20000/\text{span})$ ; contact probability  $p(d, \text{conv})$  reproduces 0.9103/0.354/0.1619/0.0629 at 1 kb convergent, 1 kb non-convergent, 10 kb convergent, 10 kb non-convergent.

## 16.81 Module 102: infinite\_diagnosis (cross-domain case reasoning)

**Findings (BUG 71, BUG 72).** BUG 71: the raw case text went to the catalog ranker, so function words drove every angle; the query and alone ranks SynDroid (13.033), Liquid Biopsy (12.427), NeoHunter (11.215), Micro-Tx (10.457), while the content words tall stature lens dislocation match no module. The ranker now receives content words only. BUG 72: no module spec names a disease, symptom or gene, so cystic fibrosis and CFTR matched 0 of 95 catalog documents although RareNet AI holds curated records; case terms are now bridged to those records by whole-phrase match (Marfan via lens dislocation and tall stature, PKU via musty odor and seizures, CF via name or gene). `joint_case_view` validated on live ClinVar: VCV000007105 CFTR p.Phe508del (Pathogenic, practice guideline) is the lead hypothesis at support 2.5; VCV000015333 HBB p.Glu7Val (Pathogenic, 2-star) scores 1.5 but is correctly not a lead because HBB lies outside the CF symptom differential. Negatives: gnomAD skipped for HGVS-only input; the bridge covers only six curated diseases (benchmarks/sweep\_infinite\_diagnosis.json).

**(F130) Catalog ranking and curated bridge.** Score  $S(d, q) = \sum_{t \in q \setminus W} \text{idf}(t) \text{tf}(t, d) \cdot (1.5 \text{ if } t \in \text{name}(d))$  with  $\text{idf}(t) = \ln(1 + (N - n_t + 0.5)/(n_t + 0.5))$ ,  $N = 95$ , and  $W$  the function-word set; bridge  $B(q) = \{D : \text{name}(D) \cup \text{sym}(D) \cup \text{genes}(D) \cap \text{phrases}(q) \neq \emptyset\}$  adds RareNet AI when  $B(q) \neq \emptyset$ . Variant support =  $2.0w_*/0.8$  for a ClinVar pathogenic match ( $w_*$  the review-star weight), giving +2.5 for a practice guideline and +1.5 for 2-star.

## 16.82 Module 103: living\_computer (boolean gene-circuit compiler)

**Finding (BUG 73).** Compiled circuits were not digital at the module's documented input normalization (fully induced = 1.0 REU). A repress edge with Hill coefficient  $n = 4$  leaks  $1/(1 + 2^4) = 5.9\%$  of  $v_{\max}$  at a 1.0 input; with internal protein at  $5 \times$  transcription (decay 0.2), one leaked edge leaves  $\approx 0.33$  REU - comparable to  $K = 0.5$  - which re-activates downstream gates and compounds through cascades. Measured at 1.0/0.0 inputs, identical on both simulation engines: XOR(1, 1) = 7.452 (reads ON, should be OFF), XNOR(1, 1) = 0.148, AND-NOT(1, 1) = 1.208 (OFF above the 0.5 threshold), 3-input XOR has 3 wrong rows. The verify path masked it by testing only at HIGH = 3.0 (leak 0.08%). Fix:  $n = 8$  (leak  $1/(1 + 2^8) = 0.39\%$ ), one constant, timing untouched. After: 14 expressions  $\times$  all input rows  $\times$  both engines  $\times$  both input conventions = 0 wrong rows; margins at documented levels min-ON 7.55 vs max-OFF 0.079 (benchmarks/sweep\_living\_computer.json).

**(F131) Hill repression leak and cascade compounding.** A repress edge passes  $\ell(x) = 1/(1 + (x/K)^n)$  of  $v_{\max}$ ; at documented input  $x = 1$ ,  $K = 0.5$ :  $\ell = 1/17$  for  $n = 4$  vs  $1/257$  for  $n = 8$ . Internal protein steady state  $p = \text{tx}/\delta = 5\text{tx}$  ( $\delta = 0.2$ ), so a leaked edge deposits  $5\ell v_{\max} \approx 0.33$  REU; a downstream gate reads it as  $(p/K)^n/(1 + (p/K)^n) \approx 0.16$  each, and an OR of two such leaks re-opens the reporter at 7.45 REU. With  $n = 8$ :  $5\ell v_{\max} \approx 0.02$  REU, downstream activation  $(0.02/0.5)^8 \approx 6 \times 10^{-12}$  - the cascade stays shut.

### 16.83 Module 104: living\_tx (engineered-probiotic design)

**Finding.** No defect; the gut-community ODE is a single-inoculum challenge whose only stable equilibrium has the engineered strain extinct: growth vanishes at the shared carrying capacity while the washout loss  $0.08(1 - \text{engraftment})$  persists (per-capita rates  $-0.008$  to  $-0.056 \text{ h}^{-1}$  at the resident equilibrium, substrate  $0.0385$ ). All four chassis wash out by day 56 at doses  $10^5$ - $10^{12}$  CFU, matching the clinical need for repeated dosing of Nissle-class chassis (PMID 30102294). Two additive honesty fixes shipped: the verdict was window-blind (1-day simulations read “persists”/ready only because decay had not acted; 14 days reads washed\_out for the same design), so it now carries the assessment window, the washout timescale, and a short-window caution; model\_status now declares the single-inoculum semantics. Literature: PMID 10958782 (IL-10 *L. lactis*) and 30102294 (SYNB1618) match the top payload fits; PMID 25626737 (GLP-1 *L. gasseri*) contradicts the GLP-1 fit (0.65) - calibration gap documented, not refitted (benchmarks/sweep\_living\_tx.json).

**(F132) Single-inoculum extinction and washout timescale.** Engineered per-capita rate  $g(E) = \frac{1}{2}eS(1 - T/10) - 0.08(1 - e)$  with  $T$  the total biomass; at the resident-only equilibrium ( $R=10$ ,  $S^* = \frac{0.02}{0.5+0.02} = 0.0385$ ),  $g = -0.08(1 - e) < 0$  for every chassis ( $e < 1$ ), so  $E \rightarrow 0$  exponentially with timescale  $\tau = 1/[0.08(1 - e)] \text{ h}$  (41.7 h for *Lactobacillus*). The verdict caution fires when the assessment window  $w < 3\tau$ : a “persists” verdict inside  $3\tau$  only means decay has not finished. Combined containment risk is a probability union  $1 - \prod_i (1 - r_i)$  over payload, persistence, transfer and shedding components.

### 16.84 Module 105: micro\_tx (microbe-prebiotic therapeutic design)

**Findings (BUG 74, BUG 75).** BUG 74: SCFA\_YIELD credited butyrate to Bifidobacterium (0.10), Akkermansia (0.15) and *L. reuteri* (0.10) - none produce butyrate. Bifidobacterium makes acetate and lactate and reaches butyrate only by cross-feeding partners (PMID 38126785); colonic butyrate producers are specific clostridial clusters (PMIDs 26925050, 19807780), here *F. prausnitzii* (0.75) and Roseburia (0.70), left intact. Deltas: resistant-starch ecology butyrate  $0.5425 \rightarrow 0.5141$ ; Bifidobacterium flux butyrate  $0.29 \rightarrow 0$ . BUG 75: compartment migration used a periodic (np.roll) Laplacian, so the proximal compartment exchanged directly with the distal one as if the gut were a ring; replaced with a zero-padded chain Laplacian (identical mass conservation, column sums 0); final abundances shift  $\leq 2 \times 10^{-5}$  because compartments start identical - a structural fix with near-zero numeric effect. Verified exact: the LP flux equals its analytic optimum  $b = Y_i \sum_j d_j A_{ij}$  for all five strains; Michaelis-Menten fermentation conserves mass to  $3.6 \times 10^{-15}$ ; host-outcome formulas reproduce exactly (benchmarks/sweep\_micro\_tx.json).

**(F133) Constraint flux, fermentation and chain migration.** Biomass flux  $b^* = Y_i \sum_j d_j A_{ij}$  (uptake saturates its enzyme-access bounds; LP and analytic optima agree to  $10^{-6}$ ). Fermentation  $\dot{S} = -v_{\max} S / (K_m + S)$  with  $v_{\max} = 0.12 + 0.18 \bar{A}_j$ , products  $(0.55, 0.25, 0.20) \dot{S}$  so  $S + \sum \text{products}$  is conserved to solver tolerance. Migration is the chain Laplacian  $\mu(\uparrow x + \downarrow x - \deg x)$ ,  $\mu = 0.04$ ,  $\deg = (1, 2, 1)$ ; the periodic form it replaces coupled compartment 1 to compartment 3 directly. Ecology:  $\dot{x}_c = x_c \odot (\rho_c r + A x_c)$  with retention  $\rho = (0.92, 1, 1.08)$ , competition  $A_{ii} = -0.85$ ,  $A_{ij} = -0.18$ , cross-feeding  $A_{21} = 0.22$  (acetate  $\rightarrow$  butyrate producers).

## 16.85 Module 106: microaiverse (cultivation solver)

**Findings (BUG 76, BUG 77).** BUG 76: predicted\_success was unbounded -  $0.3 + 0.2$  per supplement  $+0.2$  for partners reached 1.5 with five gaps, and duplicate gaps stacked identical supplements to inflate it further (five copies of the same B12 gap also scored 1.5 with five duplicate supplements). Gaps are now deduplicated and the score capped at 0.95; one- and two-gap scores are unchanged (0.7, 0.9). BUG 77: optimize\_medium crashed inside linprog with a raw solver error for a target with no requirements; it now returns the zero-cost empty medium. Verified exact: the LP medium equals the analytic optimum  $x_i = \text{req}_i$ ; partner coverage =  $\sum \min(\text{sec}, \text{req}) / \sum \text{req}$ , competition and net score recomputed exactly. Anchors: PMID 12004133 (diffusion-chamber isolation of “uncultivable” microbes), PMID 35536023 (Bacteroides metabolic synergy) (benchmarks/sweep\_microaiverse.json).

**(F134) Medium LP and cross-feeding score.** The medium problem  $\min_x c^\top x$  s.t.  $x_i \geq \text{req}_i$ ,  $x \geq 0$  has the closed-form optimum  $x^* = \text{req}$ , cost  $c^\top \text{req}$  - the LP reproduces it exactly, and the empty-requirements case is the zero vector at cost 0. Partner score =  $\text{cov} - 0.4 \text{comp}$  with  $\text{cov} = \sum_m \min(s_m, r_m) / \sum_m r_m$  over secreted versus required metabolites and  $\text{comp} = \sum_m \min(u_m^i, u_m^p) / \sum_m u_m^i$  over shared uptake. Success heuristic  $\min(0.95, 0.3 + 0.2 n_{\text{supp}} + 0.2 \lceil \text{partners} \rceil)$  - bounded after BUG 76.

## 16.86 Module 107: microbiome\_rx (community-metabolite intervention design)

**Finding (BUG 78).** optimize\_intervention’s input penalty (0.02 per gram) was  $\sim 10^3$  times larger than the squared metabolite errors this model produces ( $\sim 10^{-5}$  at its natural  $\sim 10^{-2}$  output scale), so the penalty always won: every target received fiber=sugar=0 - even reachable ones (butyrate 0.005  $\rightarrow$  diet  $\{0,0\}$ , achieved 0.00201; butyrate 0.5  $\rightarrow \{0,0\}$  with converged=True and no reported gap). The penalty is now a  $10^{-6}$ /g tie-breaker and the result reports target versus achieved per metabolite. After: butyrate 0.005  $\rightarrow$  fiber 1.011 g, achieved 0.00474 (error  $2.6 \times 10^{-4}$ ); butyrate 0.5  $\rightarrow$  fiber 3 g best-effort, achieved 0.00731, error 0.493 visible. Metabolic sim responds to diet (butyrate 0.00212  $\rightarrow$  0.01133 for fiber 0.2  $\rightarrow$  2.5); diagnostics exactly 50 for 4/5/6-species communities (benchmarks/sweep\_microbiome\_rx.json).

**(F135) Penalty scaling in intervention optimization.** Objective  $J(z) = \sum_m (M_m(z) - v_m)^2 + \lambda \sum z$ ; the coupled ODE yields  $M_m = O(10^{-2})$ , so errors are  $O(10^{-5})$ - $10^{-1}$  while  $\lambda \sum z$  with  $\lambda = 0.02$  is  $O(10^{-2})$  per gram - the penalty swamps the signal and  $\arg \min J = \{0,0\}$  for all  $v$ . A tie-breaker needs  $\lambda \ll \min_z \|\Delta M(z)\|^2 / \|z\|$ , here  $\lambda = 10^{-6}$ : reachable targets are hit (error  $2.6 \times 10^{-4}$ ), unreachable ones return the best-effort boundary  $z = 3$  with the gap reported. Coupled dynamics:  $\dot{x}_i = x_i(g_i \text{mean}_{m \in C_i} \frac{M_m}{1+M_m} + \sum_j a_{ij} x_j - 0.08 \sum x - \text{abx}_i)$ ,  $\dot{M}_m = 0.02(d_m - M_m) + \sum_i (\frac{0.12}{|P_i|} [m \in P_i] - \frac{0.15}{|C_i|} [m \in C_i]) [\dot{x}_i]^+$ .

**(F136) Path-count ranking versus embedding noise.** The repurposing resilience index  $\text{tri} = .45 \min(1, .4 n_{\text{paths}}) + .25 \min(n_{\text{targets}}/3, 1) + .3 d_{\text{target}}$  reproduces the scan exactly: aspirin  $\rightarrow$  arthritis gives  $.45(.8) + .25(2/3) + .3 = 0.8267$  ( $n_{\text{paths}} = 2$ , targets COX1/COX2). The graph-convolution score  $\text{tri}_{\text{graph}} = .4(\cos+1)/2 + .35 \text{path\_support} + .25 d_{\text{target}}$  with untrained random embeddings  $\cos \sim U(-1, 1)$  has  $\mathbb{E}[\text{tri}_{\text{graph}} | n_{\text{paths}} = 0] = .4 \cdot .5 + .25d$  - a drug with zero mechanistic paths can outrank one with a path purely by embedding luck, which is BUG 79: filtering  $n_{\text{paths}} = 0$  removes 7/16 diseases whose entire candidate lists were such draws (T2D top: thalidomide 0.338, 0 paths).

## 17 Adding evolutionary conservation to the splice-region model

The best splice-region model so far is a logistic regression on PWM and MaxEntScan features. Neither feature sees conservation. We fetched phyloP100way at each variant position from the UCSC Genome Browser REST API for a stratified subsample of 1,501 of the 9,235 modelled variants (379 pathogenic, 1,054 genes, 0 missing). The subsample keeps the class-by-site proportions of the full set; the API rate limit (one request per second) is why we did not fetch all 9,235. Median phyloP is 3.214 at pathogenic and 0.191 at benign positions. On its own, phyloP reaches AUROC 0.8377 (0.9108 at donors, 0.6966 at acceptors). Added to logit+MaxEnt under gene-grouped 5-fold cross-validation repeated over 20 seeds, AUROC rises from 0.963 to 0.9678. A paired bootstrap on the averaged out-of-fold predictions gives a 95% interval for the gain of [+0.0008, +0.0091], which excludes zero. The gain is concentrated at donor-region positions

| site     | $n$  | phyloP alone | logit+MaxEnt | + phyloP |
|----------|------|--------------|--------------|----------|
| donor    | 876  | 0.9108       | 0.9557       | 0.9634   |
| acceptor | 625  | 0.6966       | 0.9702       | 0.9723   |
| all      | 1501 | 0.8377       | 0.9630       | 0.9678   |

Table 20: AUROC with and without phyloP (discovery/splice\_region\_vus/phylop\_lift.json). Per-site values use the seed-averaged predictions.

(+3 to +6), where the intronic consensus is short and conservation carries information the PWM misses. At acceptors (the polypyrimidine tract, −3 to −14), MaxEntScan already scores well and conservation adds little.

**Caveat.** ClinVar submitters may cite conservation as supporting evidence (ACMG criteria PP3/BP4). Some of the gain may therefore be label circularity rather than new biology. We report the gain as modest and conditional on that caveat.

**(F47) Model with conservation.**  $\text{logit } P(\text{path} \mid x) = \beta_0 + \beta^\top \mathbf{z}(x) + \gamma \phi(x) + \delta \phi(x) \mathbb{I}[\text{donor}]$ , with  $\mathbf{z}$  the standardised PWM/MaxEnt/position features and  $\phi$  the phyloP score; the interaction lets conservation matter differently at donors.

**(F48) Paired bootstrap.** For resamples  $b = 1..B$  of variant indices  $I_b$ ,  $\Delta_b = \text{AUC}(y_{I_b}, \hat{p}_{I_b}^+) - \text{AUC}(y_{I_b}, \hat{p}_{I_b}^0)$ ; the interval is the 2.5th and 97.5th percentile of  $\{\Delta_b\}$ . Pairing on the same resample removes the shared sampling noise of the two models.

## 18 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

## 19 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is met (manifests/tools.tsv). The manifests in the appendix list exactly what was used.

## 20 Module inventory and validation coverage

sugarcode-ai contains 95 application modules and 32 bio/ libraries. 83 of these 127 have been checked against an independent reference on real data in this work (51 modules, 32 libraries). The rest are not yet validated. That does not mean they are wrong, only that no claim about them is made here. Validation status is a curated list tied to committed result files (manifests/module\_inventory.tsv, built by sweep/module\_inventory.py), not a keyword match.

| kind   | name           | py lines | status    |
|--------|----------------|----------|-----------|
| bio    | align          | 164      | validated |
| bio    | bed            | 180      | validated |
| bio    | chembl         | 184      | validated |
| bio    | codon          | 170      | validated |
| bio    | de             | 218      | validated |
| bio    | entrez         | 196      | validated |
| bio    | fasta          | 55       | validated |
| bio    | fastq          | 183      | validated |
| bio    | genbank        | 103      | validated |
| bio    | gff            | 237      | validated |
| bio    | gnomad         | 122      | validated |
| bio    | gstats         | 194      | validated |
| bio    | kmer           | 261      | validated |
| bio    | motif          | 198      | validated |
| bio    | newick         | 248      | validated |
| bio    | orf            | 189      | validated |
| bio    | pdb            | 346      | validated |
| bio    | phylo          | 232      | validated |
| bio    | pileup         | 148      | validated |
| bio    | primer         | 278      | validated |
| bio    | proteinprops   | 261      | validated |
| bio    | pwm            | 63       | validated |
| bio    | reactome       | 63       | validated |
| bio    | restriction    | 150      | validated |
| bio    | rnaseq         | 201      | validated |
| bio    | sam            | 221      | validated |
| bio    | sequence       | 174      | validated |
| bio    | splice         | 682      | validated |
| bio    | stockholm      | 260      | validated |
| bio    | structures     | 267      | validated |
| bio    | uniprot        | 106      | validated |
| bio    | vcf            | 263      | validated |
| module | acmg_bayesian  | 469      | validated |
| module | alpha_fold_ui  | 442      | validated |
| module | bio_material   | 116      | validated |
| module | bio_switch     | 511      | validated |
| module | biofactory_1_a | 105      | validated |
| module | biogpt_lit     | 258      | validated |
| module | car_t_designer | 104      | validated |
| module | cell_free_opt  | 109      | validated |
| module | cellfatenet    | 207      | validated |
| module | cf_d_offtarget | 157      | validated |

| kind   | name                  | py lines | status           |
|--------|-----------------------|----------|------------------|
| module | chem_descriptors      | 923      | <b>validated</b> |
| module | chem_similarity       | 234      | <b>validated</b> |
| module | chemgpt_engine        | 305      | <b>validated</b> |
| module | codon_opt             | 289      | <b>validated</b> |
| module | crispr_opt            | 648      | <b>validated</b> |
| module | crisprater            | 97       | <b>validated</b> |
| module | crisprscan_score      | 178      | <b>validated</b> |
| module | dark_genome           | 207      | <b>validated</b> |
| module | deepsplice            | 751      | <b>validated</b> |
| module | dna_to_code           | 132      | <b>validated</b> |
| module | docking_studio        | 535      | <b>validated</b> |
| module | dti_bench             | 201      | <b>validated</b> |
| module | evidence_mining       | 418      | <b>validated</b> |
| module | evofold_4d            | 239      | <b>validated</b> |
| module | gene_analysis         | 278      | <b>validated</b> |
| module | liquid_biopsy         | 411      | <b>validated</b> |
| module | metabodesigner        | 190      | <b>validated</b> |
| module | microbiome_exp        | 120      | <b>validated</b> |
| module | mit_offtarget         | 154      | <b>validated</b> |
| module | molecule_eval         | 149      | <b>validated</b> |
| module | mutdock               | 337      | <b>validated</b> |
| module | neohunter             | 358      | <b>validated</b> |
| module | neuroplan_ai          | 387      | <b>validated</b> |
| module | omega_stats           | 161      | <b>validated</b> |
| module | oncocircuit           | 124      | <b>validated</b> |
| module | openclinvar           | 514      | <b>validated</b> |
| module | pgx_guidelines        | 287      | <b>validated</b> |
| module | phage_designer        | 82       | <b>validated</b> |
| module | prime_design          | 263      | <b>validated</b> |
| module | profile_hmm           | 747      | <b>validated</b> |
| module | promoter_lib          | 143      | <b>validated</b> |
| module | qsar_bench            | 389      | <b>validated</b> |
| module | rarenet_ai            | 366      | <b>validated</b> |
| module | riboswitch            | 142      | <b>validated</b> |
| module | rna_decoder           | 144      | <b>validated</b> |
| module | rna_nussinov          | 387      | <b>validated</b> |
| module | stability_ai          | 199      | <b>validated</b> |
| module | str_scope             | 340      | <b>validated</b> |
| module | structural_biophysics | 528      | <b>validated</b> |
| module | syn_stab_ai           | 77       | <b>validated</b> |
| module | vector_opt            | 100      | <b>validated</b> |
| module | bio_copilot           | 245      | not yet          |
| module | bioimage_ai           | 146      | not yet          |
| module | bioplayground         | 98       | not yet          |
| module | bioprint_pro          | 143      | not yet          |
| module | biosimvr              | 200      | not yet          |
| module | cell_twin             | 170      | not yet          |
| module | cellpainter           | 289      | not yet          |
| module | cellpainter_4d        | 164      | not yet          |
| module | crispr_cargo          | 156      | not yet          |
| module | crispr_muse           | 278      | not yet          |
| module | ecosystem             | 175      | not yet          |
| module | enterprise_bio        | 260      | not yet          |
| module | epi_edit              | 178      | not yet          |
| module | fate_predictor        | 131      | not yet          |
| module | gene_explorer         | 140      | not yet          |
| module | gene_tx_opt           | 149      | not yet          |
| module | genomegpt             | 279      | not yet          |
| module | infinite_diagnosis    | 130      | not yet          |
| module | living_computer       | 857      | not yet          |
| module | living_tx             | 184      | not yet          |
| module | micro_tx              | 218      | not yet          |
| module | microaverse           | 138      | not yet          |
| module | microbiome_rx         | 172      | not yet          |
| module | neodti_engine         | 257      | not yet          |
| module | neuro_hub             | 44       | not yet          |
| module | neuro_hub_dashboard   | 25       | not yet          |

| kind   | name            | py lines | status  |
|--------|-----------------|----------|---------|
| module | neuro_pipeline  | 326      | not yet |
| module | nexus_support   | 299      | not yet |
| module | organoid_ai     | 216      | not yet |
| module | organoid_screen | 275      | not yet |
| module | pdx_insight     | 138      | not yet |
| module | phage_tx        | 161      | not yet |
| module | phageforge      | 87       | not yet |
| module | proteinPainter  | 161      | not yet |
| module | report_studio   | 213      | not yet |
| module | robotic_flow    | 128      | not yet |
| module | synBioStudio    | 43       | not yet |
| module | synbio_studio   | 113      | not yet |
| module | synbio_wizard   | 224      | not yet |
| module | syndroid        | 231      | not yet |
| module | synlife_evo     | 278      | not yet |
| module | synthetic_life  | 197      | not yet |
| module | tissue_eng      | 193      | not yet |
| module | virtual_cell    | 504      | not yet |

## A Tools Table

| Tool                                 | Version               | Status/used in                           |
|--------------------------------------|-----------------------|------------------------------------------|
| pyhmmer                              | 0.12.3                | installed                                |
| ViennaRNA                            | 2.7.2                 | installed                                |
| RDKit                                | 2026.3.6              | installed                                |
| SpliceAI                             | 1.3.1                 | installed                                |
| TensorFlow                           | 2.15.1                | installed                                |
| Keras                                | 2.15.0                | installed                                |
| scikit-learn                         | 1.7.2                 | installed                                |
| SciPy                                | 1.15.3                | installed                                |
| NumPy                                | 1.26.4                | installed                                |
| Biopython                            | 1.88                  | installed                                |
| twobitreader                         | 4.0.2                 | installed                                |
| AutoDock Vina (python)               | 1.2.7                 | installed                                |
| Open Babel                           | 3.1.1.23              | installed                                |
| pytest                               | 9.1.1                 | installed                                |
| pandas                               | 2.3.3                 | installed                                |
| MHCflurry                            | 2.2.1                 | sweep/neohunter_mhcflurry                |
| CRISPOR reference scorers (vendored) | web service/vendored  | sweep/_crispor_*_ref.py                  |
| EBI Clustal Omega REST               | web service/vendored  | data/README.md                           |
| CPIC API                             | web service/vendored  | sweep/pgx_cpic_bench.py                  |
| NCBI E-utilities                     | web service/vendored  | src/sc_validate                          |
| LOVD3 API                            | web service/vendored  | discovery/splice_region_vus              |
| MaxEntScan (maxentpy)                | 0.0.4                 | discovery/splice_region_vus              |
| codon-bias                           | 0.5.0                 | sweep/codon_tai_enc.py                   |
| statsmodels                          | 0.15.0                | benchmarks/sweep_codon_c                 |
| PyTorch                              | 2.14.0+cpu            | discovery/splice_region_vus (splice CNN) |
| Matplotlib                           | 3.10.9                | paper/make_figs.py (all figs)            |
| FreeSASA                             | 2.x (python)          | sweep/sasa_vs_freesasa.py                |
| primer3-py                           | 2.3.1                 | marks/sweep_sasa_freesasa                |
| DendroPy                             | 5.1.0                 | sweep/primer_vs_primer3.p                |
| orfipy                               | 0.0.4                 | marks/sweep_primer_prime                 |
| pysam (htslib)                       | 0.24.1                | bio.primer.hairpin_dg                    |
| PyDESeq2                             | 0.5.2                 | sweep/phylo_vs_dendropy.p                |
| parasail                             | 2.6.1                 | marks/sweep_phylo_dendro                 |
| g:Profiler g:GOST API                | e114_eg62_p19_374p083 | sweep/orf_vs_orfipy.py                   |
| STRING API                           | 12.0                  | marks/sweep_orf_orfipy.js                |
| Reactome AnalysisService             | live 2026-09-24       | sweep/vcf_vs_pysam.py                    |
| networkx                             | local 3.4.2           | marks/sweep_vcf_pysam.js                 |
| gseapy                               | local 1.3.1           | of ClinVar VCF)                          |

## B Dataset Manifest

| #  | Dataset                                                                        | Accession/version                                               | Used in                                                   |
|----|--------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------|
| 1  | ClinVar variant_summary                                                        | variant_summary.txt.gz (GRCh38)                                 | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2  | UCSC hg38 reference genome                                                     | hg38.2bit                                                       | discovery/splice_region_vus/extract.py                    |
| 3  | LOVD3 shared variant records                                                   | top-20 VUS query                                                | discovery/splice_region_vus/lovd_top20.json               |
| 4  | SKEMPI 2.0                                                                     | skempi_v2.csv (PMID 30598530)                                   | data/skempi_v2.csv                                        |
| 5  | BiGG e.coli_core                                                               | e.coli_core (PMID 21988831)                                     | data/e.coli_core.json                                     |
| 6  | NCBI RefSeqGene BRCA1                                                          | NG_005905.2:60000-70000                                         | data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv           |
| 7  | CRISPOR reference CFD/MIT matrices                                             | crisporWebsite (PMID 30341956)                                  | data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py     |
| 8  | CRISPOR Rule Set 1 reference                                                   | crisporEffScores.py                                             | data/doench_rs1_oracle.tsv                                |
| 9  | UniProtKB reviewed globins                                                     | 25 sequences                                                    | data/globins25.fa                                         |
| 10 | UniProtKB reviewed non-globins                                                 | 25 sequences                                                    | data/nonglobins25.fa                                      |
| 11 | Pfam Globin HMM                                                                | PF00042.29                                                      | data/pfam_PF00042_globin.hmm                              |
| 12 | Rfam tRNA seed                                                                 | RF00005                                                         | data/rfam/RF00005.sto                                     |
| 13 | Rfam 5S rRNA seed                                                              | RF00001                                                         | data/rfam/RF00001.sto                                     |
| 14 | Rfam RNase P RNA seed                                                          | RF00010                                                         | data/rfam/RF00010.sto                                     |
| 15 | ChEMBL approved drugs                                                          | 1000-compound slice                                             | data/chembl_approved_1000.tsv                             |
| 16 | Kazusa CUTG E. coli K12                                                        | taxid 83333                                                     | data/README.md (sanity only)                              |
| 17 | NCBI RefSeq E. coli K-12 MG1655 CDS                                            | GCF_000005845.2                                                 | sweep/codon_cai_bench.py                                  |
| 18 | PaxDb E. coli integrated abundance                                             | 511145-WHOLE_ORGANISM-integrated                                | sweep/codon_cai_bench.py                                  |
| 19 | CPIC CYP2C19 diplotype table                                                   | 666 diplotypes                                                  | sweep/pgx_cpic_bench.py                                   |
| 20 | CPIC CYP2D6 diplotype table                                                    | 16836 diplotypes                                                | sweep/pgx_cpic_bench.py                                   |
| 21 | CPIC CYP2D6 allele activity table                                              | 208 alleles                                                     | sugarcode-ai pgx_guidelines/data                          |
| 22 | Edinburgh Genome Foundry codon tables                                          | e.coli_316407                                                   | sweep/codon_cai_bench.py (via sugarcode-ai)               |
| 23 | IEDB MHC-I binding data 2013 (Kim et al. 2014)                                 | bdata.20130222.mhci.txt                                         | sweep/neohunter_iedb_bench.py                             |
| 24 | LP-PDBBind (Li et al. leak-proof split)                                        | LP.PDBBind.csv                                                  | sweep/docking_bench.py                                    |
| 25 | NCBI RefSeq transcripts (TP53, BRCA1, BRCA2, BRAF, EGFR)                       | NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5 | sweep/primer_vs_primer3.py                                |
| 26 | ClinVar GRCh38 VCF (tabix regions TP53, BRCA1, BRCA2, BRAF, EGFR)              | clinvar.vcf.gz (46,603 records)                                 | sweep/vcf_vs_pysam.py                                     |
| 27 | GEO GSE52778 airway smooth muscle RNA-seq (Himes et al. 2014), NCBI raw counts | GSE52778_raw_counts_GRCh38.p13NCBI raw counts                   | sweep/ncbi_raw_counts.py                                  |
|    | Platinum flat file (experimental ligand-binding ddG)                           | 1008 mutation records                                           | data/platinum/platinum_flat_file.csv                      |

## C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

| # | Kind  | Study | Accession                       | Used in                                                   |
|---|-------|-------|---------------------------------|-----------------------------------------------------------|
| 1 | study | 1     | variant_summary.txt.gz (GRCh38) | data/clinvar_sample_500.tsv; discovery/splice_region_vus/ |
| 2 | study | 2     | hg38.2bit                       | discovery/splice_region_vus/extract.py                    |
| 3 | study | 3     | top-20 VUS query                | discovery/splice_region_vus/lovd_top20.json               |
| 4 | study | 4     | skempi_v2.csv (PMID 30598530)   | data/skempi_v2.csv                                        |
| 5 | study | 5     | e.coli_core (PMID 21988831)     | data/e.coli_core.json                                     |
| 6 | study | 6     | NG_005905.2:60000-70000         | data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv           |

| #  | Kind      | Study | Accession                                | Used in    |
|----|-----------|-------|------------------------------------------|------------|
| 7  | study     | 7     | crisporWebsite (PMID 30341956)           | data/crisp |
| 8  | study     | 8     | crisporEffScores.py                      | sweep/.d   |
| 9  | study     | 9     | 25 sequences                             | data/doi   |
| 10 | study     | 10    | 25 sequences                             | data/glob  |
| 11 | study     | 11    | PF00042.29                               | data/non   |
| 12 | study     | 12    | RF00005                                  | data/pfar  |
| 13 | study     | 13    | RF00001                                  | data/rfar  |
| 14 | study     | 14    | RF00010                                  | data/rfar  |
| 15 | study     | 15    | 1000-compound slice                      | data/che   |
| 16 | study     | 16    | taxid 83333                              | data/RE    |
| 17 | study     | 17    | GCF_000005845.2                          | sweep/c    |
| 18 | study     | 18    | 511145-<br>WHOLE_ORGANISM-<br>integrated | sweep/c    |
| 19 | study     | 19    | 666 diplotypes                           | sweep/p    |
| 20 | study     | 20    | 16836 diplotypes                         | sweep/p    |
| 21 | study     | 21    | 208 alleles                              | sugarco    |
| 22 | study     | 22    | e_coli_316407                            | sweep/c    |
| 23 | study     | 23    | bdata.20130222.mhci.txt                  | sugarco    |
| 24 | study     | 24    | LP_PDBBind.csv                           | sweep/n    |
| 25 | pdb_entry |       | 1CNW                                     | sweep/d    |
| 26 | pdb_entry |       | 1EJN                                     | sweep/d    |
| 27 | pdb_entry |       | 1F5L                                     | sweep/d    |
| 28 | pdb_entry |       | 1G54                                     | sweep/d    |
| 29 | pdb_entry |       | 1GI1                                     | sweep/d    |
| 30 | pdb_entry |       | 1K1J                                     | sweep/d    |
| 31 | pdb_entry |       | 1O3L                                     | sweep/d    |
| 32 | pdb_entry |       | 1P1O                                     | sweep/d    |
| 33 | pdb_entry |       | 1Y3X                                     | sweep/d    |
| 34 | pdb_entry |       | 1ZGI                                     | sweep/d    |
| 35 | pdb_entry |       | 2ERZ                                     | sweep/d    |
| 36 | pdb_entry |       | 2J4I                                     | sweep/d    |
| 37 | pdb_entry |       | 2P4J                                     | sweep/d    |
| 38 | pdb_entry |       | 2X7T                                     | sweep/d    |
| 39 | pdb_entry |       | 2XAB                                     | sweep/d    |
| 40 | pdb_entry |       | 2XJG                                     | sweep/d    |
| 41 | pdb_entry |       | 2ZDN                                     | sweep/d    |
| 42 | pdb_entry |       | 3B3C                                     | sweep/d    |
| 43 | pdb_entry |       | 3C84                                     | sweep/d    |
| 44 | pdb_entry |       | 3D8W                                     | sweep/d    |
| 45 | pdb_entry |       | 3HKQ                                     | sweep/d    |
| 46 | pdb_entry |       | 3HLL                                     | sweep/d    |
| 47 | pdb_entry |       | 3IBI                                     | sweep/d    |
| 48 | pdb_entry |       | 3M35                                     | sweep/d    |
| 49 | pdb_entry |       | 3PO6                                     | sweep/d    |
| 50 | pdb_entry |       | 3R17                                     | sweep/d    |
| 51 | pdb_entry |       | 3R4N                                     | sweep/d    |
| 52 | pdb_entry |       | 4B2I                                     | sweep/d    |
| 53 | pdb_entry |       | 4DB7                                     | sweep/d    |
| 54 | pdb_entry |       | 4F5Y                                     | sweep/d    |
| 55 | pdb_entry |       | 4FL1                                     | sweep/d    |
| 56 | pdb_entry |       | 4J93                                     | sweep/d    |
| 57 | pdb_entry |       | 4JFM                                     | sweep/d    |
| 58 | pdb_entry |       | 4KNJ                                     | sweep/d    |
| 59 | pdb_entry |       | 4LXZ                                     | sweep/d    |
| 60 | pdb_entry |       | 4MO8                                     | sweep/d    |
| 61 | pdb_entry |       | 4NH7                                     | sweep/d    |
| 62 | pdb_entry |       | 4O04                                     | sweep/d    |
| 63 | pdb_entry |       | 4OAG                                     | sweep/d    |
| 64 | pdb_entry |       | 4R76                                     | sweep/d    |
| 65 | pdb_entry |       | 4RFC                                     | sweep/d    |
| 66 | pdb_entry |       | 4TIM                                     | sweep/d    |
| 67 | pdb_entry |       | 4XIT                                     | sweep/d    |
| 68 | pdb_entry |       | 4YTC                                     | sweep/d    |
| 69 | pdb_entry |       | 4YXI                                     | sweep/d    |

| #   | Kind      | Study | Accession                                      | Used in |
|-----|-----------|-------|------------------------------------------------|---------|
| 70  | pdb_entry |       | 4ZME                                           | sweep/d |
| 71  | pdb_entry |       | 5AML                                           | sweep/d |
| 72  | pdb_entry |       | 5CAP                                           | sweep/d |
| 73  | pdb_entry |       | 5CAU                                           | sweep/d |
| 74  | pdb_entry |       | 5D3L                                           | sweep/d |
| 75  | pdb_entry |       | 5EWK                                           | sweep/d |
| 76  | pdb_entry |       | 5EWY                                           | sweep/d |
| 77  | pdb_entry |       | 5F60                                           | sweep/d |
| 78  | pdb_entry |       | 5F62                                           | sweep/d |
| 79  | pdb_entry |       | 5G5V                                           | sweep/d |
| 80  | pdb_entry |       | 5HTZ                                           | sweep/d |
| 81  | pdb_entry |       | 5IX0                                           | sweep/d |
| 82  | pdb_entry |       | 5L8C                                           | sweep/d |
| 83  | pdb_entry |       | 5MJN                                           | sweep/d |
| 84  | pdb_entry |       | 5MPN                                           | sweep/d |
| 85  | pdb_entry |       | 5N24                                           | sweep/d |
| 86  | pdb_entry |       | 5O87                                           | sweep/d |
| 87  | pdb_entry |       | 5SZ4                                           | sweep/d |
| 88  | pdb_entry |       | 5SZ5                                           | sweep/d |
| 89  | pdb_entry |       | 5Y8Y                                           | sweep/d |
| 90  | pdb_entry |       | 6EQ5                                           | sweep/d |
| 91  | pdb_entry |       | 6G38                                           | sweep/d |
| 92  | pdb_entry |       | 6G6T                                           | sweep/d |
| 93  | pdb_entry |       | 6HQY                                           | sweep/d |
| 94  | pdb_entry |       | 6HRQ                                           | sweep/d |
| 95  | pdb_entry |       | 6OE0                                           | sweep/d |
| 96  | pdb_entry |       | 6RFN                                           | sweep/d |
| 97  | pdb_entry |       | 6U6W                                           | sweep/d |
| 98  | species   |       | 10090-ASTROCYTE-<br>integrated.txt             | sweep/c |
| 99  | species   |       | GCF_000001635.27                               | sweep/c |
| 100 | species   |       | 10116-<br>CARDIAC_VENTRICLE-<br>integrated.txt | sweep/c |
| 101 | species   |       | GCF_036323735.1                                | sweep/c |
| 102 | species   |       | 121845-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 103 | species   |       | GCF_000475195.1                                | sweep/c |
| 104 | species   |       | 1295-<br>WHOLE_ORGANISM-<br>integrated.txt     | sweep/c |
| 105 | species   |       | GCF_025558325.1                                | sweep/c |
| 106 | species   |       | 1314-<br>WHOLE_ORGANISM-<br>integrated.txt     | sweep/c |
| 107 | species   |       | GCF_900475035.1                                | sweep/c |
| 108 | species   |       | 185431-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 109 | species   |       | GCF_000002445.2                                | sweep/c |
| 110 | species   |       | 192222-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 111 | species   |       | GCF_000009085.1                                | sweep/c |
| 112 | species   |       | 196627-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 113 | species   |       | GCF_000011325.1                                | sweep/c |
| 114 | species   |       | 198214-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 115 | species   |       | GCF_000006925.2                                | sweep/c |
| 116 | species   |       | 208964-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 117 | species   |       | GCF_000006765.1                                | sweep/c |

| #   | Kind    | Study | Accession                                    | Used in |
|-----|---------|-------|----------------------------------------------|---------|
| 118 | species |       | 224308-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 119 | species |       | GCF_000009045.1                              | sweep/c |
| 120 | species |       | 224911-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 121 | species |       | GCF_001642675.1                              | sweep/c |
| 122 | species |       | 243159-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 123 | species |       | GCF_000021485.1                              | sweep/c |
| 124 | species |       | 243230-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 125 | species |       | GCF_020546685.1                              | sweep/c |
| 126 | species |       | 246200-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 127 | species |       | GCF_000011965.2                              | sweep/c |
| 128 | species |       | 2903-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 129 | species |       | GCF_000372725.1                              | sweep/c |
| 130 | species |       | 343509-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 131 | species |       | GCF_000010085.1                              | sweep/c |
| 132 | species |       | 36329-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/c |
| 133 | species |       | GCF_000002765.6                              | sweep/c |
| 134 | species |       | 403677-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 135 | species |       | GCF_000142945.1                              | sweep/c |
| 136 | species |       | 4081-PERICARP-<br>integrated.txt             | sweep/c |
| 137 | species |       | GCF_036512215.1                              | sweep/c |
| 138 | species |       | 4097-LEAF-integrated.txt                     | sweep/c |
| 139 | species |       | GCF_000715075.1                              | sweep/c |
| 140 | species |       | 418985-<br>WHOLE_ORGANISM-<br>integrated.txt | sweep/c |
| 141 | species |       | GCA_002081605.1                              | sweep/c |
| 142 | species |       | 44689-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/c |
| 143 | species |       | GCF_000004695.1                              | sweep/c |
| 144 | species |       | 4932-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 145 | species |       | GCF_000146045.2                              | sweep/c |
| 146 | species |       | 54291-<br>WHOLE_ORGANISM-<br>integrated.txt  | sweep/c |
| 147 | species |       | GCF_030505655.1                              | sweep/c |
| 148 | species |       | 6239-<br>WHOLE_ORGANISM-<br>integrated.txt   | sweep/c |
| 149 | species |       | GCF_000002985.6                              | sweep/c |
| 150 | species |       | 7227-SPERM-integrated.txt                    | sweep/c |
| 151 | species |       | GCF_000001215.4                              | sweep/c |
| 152 | species |       | 7460-ABDOMEN-<br>integrated.txt              | sweep/c |
| 153 | species |       | GCF_003254395.2                              | sweep/c |
| 154 | species |       | 8030-LIVER-integrated.txt                    | sweep/c |

| #   | Kind              | Study                | Accession                                               | Used in  |
|-----|-------------------|----------------------|---------------------------------------------------------|----------|
| 155 | species           |                      | GCF_965601325.2                                         | sweep/c  |
| 156 | species           |                      | 83332-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/c  |
| 157 | species           |                      | GCF_000195955.2                                         | sweep/c  |
| 158 | species           |                      | 9031-HEART-integrated.txt                               | sweep/c  |
| 159 | species           |                      | GCF_016699485.2                                         | sweep/c  |
| 160 | species           |                      | 93061-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/c  |
| 161 | species           |                      | GCF_000013425.1                                         | sweep/c  |
| 162 | species           |                      | 9544-BILE_DUCT-<br>integrated.txt                       | sweep/c  |
| 163 | species           |                      | GCF_049350105.2                                         | sweep/c  |
| 164 | species           |                      | 9606-ADRENAL_GLAND-<br>integrated.txt                   | sweep/c  |
| 165 | species           |                      | GCF_000001405.40                                        | sweep/c  |
| 166 | species           |                      | 9615-PLASMA-<br>integrated.txt                          | sweep/c  |
| 167 | species           |                      | GCF_011100685.1                                         | sweep/c  |
| 168 | species           |                      | 9823-LIVER-integrated.txt                               | sweep/c  |
| 169 | species           |                      | GCF_054392235.1                                         | sweep/c  |
| 170 | species           |                      | 9913-<br>LENS_OF_CAMERA_TYPE_EYE-<br>integrated.txt     | sweep/c  |
| 171 | species           |                      | GCF_002263795.3                                         | sweep/c  |
| 172 | species           |                      | 99287-<br>WHOLE_ORGANISM-<br>integrated.txt             | sweep/c  |
| 173 | species           |                      | GCF_000006945.2                                         | sweep/c  |
| 174 | species           |                      | 9986-<br>PERIPHERAL_REGION_OF_RETINA-<br>integrated.txt | sweep/c  |
| 175 | species           |                      | GCF_964237555.1                                         | sweep/c  |
| 176 | gtrnadb           |                      | Esch.coli_K_12.MG1655                                   | sweep/c  |
| 177 | gtrnadb           |                      | Baci.subt_subtilis_168                                  | sweep/c  |
| 178 | gtrnadb           |                      | Scere3                                                  | sweep/c  |
| 179 | refseq_transcript | 25                   | NM_000546.6                                             | sweep/p  |
| 180 | refseq_transcript | 25                   | NM_007294.4                                             | sweep/p  |
| 181 | refseq_transcript | 25                   | NM_000059.4                                             | sweep/p  |
| 182 | refseq_transcript | 25                   | NM_004333.6                                             | sweep/p  |
| 183 | refseq_transcript | 25                   | NM_005228.5                                             | sweep/p  |
| 184 | study             | 26                   | clinvar.vcf.gz                                          | sweep/v  |
| 185 | study             | 27                   | GSE52778                                                | sweep/d  |
| 186 | rfam_family       | RF00004              | RF00004                                                 | sweep/st |
| 187 | rfam_family       | RF00015              | RF00015                                                 | sweep/st |
| 188 | rfam_family       | RF00017              | RF00017                                                 | sweep/st |
| 189 | rfam_family       | RF00023              | RF00023                                                 | sweep/st |
| 190 | rfam_family       | RF00050              | RF00050                                                 | sweep/st |
| 191 | rfam_family       | RF00059              | RF00059                                                 | sweep/st |
| 192 | rfam_family       | RF00162              | RF00162                                                 | sweep/st |
| 193 | rfam_family       | RF00167              | RF00167                                                 | sweep/st |
| 194 | rfam_family       | RF00168              | RF00168                                                 | sweep/st |
| 195 | rfam_family       | RF00174              | RF00174                                                 | sweep/st |
| 196 | rfam_family       | RF00234              | RF00234                                                 | sweep/st |
| 197 | rfam_family       | RF00504              | RF00504                                                 | sweep/st |
| 198 | refseq_genome     | J01749.1             | J01749.1                                                | sweep/p  |
| 199 | refseq_genome     | NC_001416.1          | NC_001416.1                                             | sweep/p  |
| 200 | opentree_node     | ott1062253           | ott1062253                                              | sweep/n  |
| 201 | opentree_node     | ott392223            | ott392223                                               | sweep/n  |
| 202 | opentree_node     | ott44565             | ott44565                                                | sweep/n  |
| 203 | opentree_node     | ott691846            | ott691846                                               | sweep/n  |
| 204 | opentree_node     | ott81461             | ott81461                                                | sweep/n  |
| 205 | opentree_node     | ott913935            | ott913935                                               | sweep/n  |
| 206 | study             | NA12878_chr20_lowcov | NA12878.chrom20.ILLUMINA                                | sweep/f  |
| 207 | study             | SRR622461            | SRR622461                                               | sweep/f  |
| 208 | jaspar_matrix     | MA0003.4             | MA0003.4                                                | sweep/m  |

| #   | Kind          | Study          | Accession                 | Used in |
|-----|---------------|----------------|---------------------------|---------|
| 209 | jaspar_matrix | MA0079.3       | MA0079.3                  | sweep/m |
| 210 | jaspar_matrix | MA0080.4       | MA0080.4                  | sweep/m |
| 211 | jaspar_matrix | MA0099.3       | MA0099.3                  | sweep/m |
| 212 | jaspar_matrix | MA0105.4       | MA0105.4                  | sweep/m |
| 213 | jaspar_matrix | MA0106.3       | MA0106.3                  | sweep/m |
| 214 | jaspar_matrix | MA0108.2       | MA0108.2                  | sweep/m |
| 215 | jaspar_matrix | MA0112.3       | MA0112.3                  | sweep/m |
| 216 | jaspar_matrix | MA0113.3       | MA0113.3                  | sweep/m |
| 217 | jaspar_matrix | MA0139.1       | MA0139.1                  | sweep/m |
| 218 | nucore_record | NC.000913.3    | NC.000913.3               | sweep/g |
| 219 | nucore_record | NC.001422.1    | NC.001422.1               | sweep/g |
| 220 | nucore_record | NC.001477.1    | NC.001477.1               | sweep/g |
| 221 | nucore_record | NC.001498.1    | NC.001498.1               | sweep/g |
| 222 | nucore_record | NC.001802.1    | NC.001802.1               | sweep/g |
| 223 | nucore_record | NC.001806.2    | NC.001806.2               | sweep/g |
| 224 | nucore_record | NC.002549.1    | NC.002549.1               | sweep/g |
| 225 | nucore_record | NC.003977.2    | NC.003977.2               | sweep/g |
| 226 | nucore_record | NC.012920.1    | NC.012920.1               | sweep/g |
| 227 | nucore_record | NC.045512.2    | NC.045512.2               | sweep/g |
| 228 | nucore_record | NG.005905.2    | NG.005905.2               | sweep/g |
| 229 | nucore_record | NM.000038.6    | NM.000038.6               | sweep/g |
| 230 | nucore_record | NM.000249.4    | NM.000249.4               | sweep/g |
| 231 | nucore_record | NM.000251.3    | NM.000251.3               | sweep/g |
| 232 | nucore_record | NM.000314.8    | NM.000314.8               | sweep/g |
| 233 | nucore_record | NM.001126112.3 | NM.001126112.3            | sweep/g |
| 234 | nucore_record | NM.001904.4    | NM.001904.4               | sweep/g |
| 235 | nucore_record | NM.002524.5    | NM.002524.5               | sweep/g |
| 236 | nucore_record | NM.004985.5    | NM.004985.5               | sweep/g |
| 237 | study         | 29             | hg38.cpgIslandExt.chr20   | sweep/b |
| 238 | study         | 30             | hg38.rmsk.chr20:0-5000000 | sweep/b |
| 239 | nucore_record | MG772933.1     | MG772933.1                | sweep/k |
| 240 | nucore_record | MN996532.2     | MN996532.2                | sweep/k |
| 241 | nucore_record | MT040333.1     | MT040333.1                | sweep/k |
| 242 | nucore_record | NC.004718.3    | NC.004718.3               | sweep/k |
| 243 | nucore_record | NC.019843.3    | NC.019843.3               | sweep/k |
| 244 | dbnp_variant  | rs1042713      | rs1042713                 | sweep/g |
| 245 | dbnp_variant  | rs10427255     | rs10427255                | sweep/g |
| 246 | dbnp_variant  | rs1229984      | rs1229984                 | sweep/g |
| 247 | dbnp_variant  | rs12913832     | rs12913832                | sweep/g |
| 248 | dbnp_variant  | rs1426654      | rs1426654                 | sweep/g |
| 249 | dbnp_variant  | rs16891982     | rs16891982                | sweep/g |
| 250 | dbnp_variant  | rs17822931     | rs17822931                | sweep/g |
| 251 | dbnp_variant  | rs1799945      | rs1799945                 | sweep/g |
| 252 | dbnp_variant  | rs1800562      | rs1800562                 | sweep/g |
| 253 | dbnp_variant  | rs1801133      | rs1801133                 | sweep/g |
| 254 | dbnp_variant  | rs1815739      | rs1815739                 | sweep/g |
| 255 | dbnp_variant  | rs2231142      | rs2231142                 | sweep/g |
| 256 | dbnp_variant  | rs334          | rs334                     | sweep/g |
| 257 | dbnp_variant  | rs3827760      | rs3827760                 | sweep/g |
| 258 | dbnp_variant  | rs429358       | rs429358                  | sweep/g |
| 259 | dbnp_variant  | rs4680         | rs4680                    | sweep/g |
| 260 | dbnp_variant  | rs4988235      | rs4988235                 | sweep/g |
| 261 | dbnp_variant  | rs53576        | rs53576                   | sweep/g |
| 262 | dbnp_variant  | rs6025         | rs6025                    | sweep/g |
| 263 | dbnp_variant  | rs671          | rs671                     | sweep/g |
| 264 | dbnp_variant  | rs7412         | rs7412                    | sweep/g |
| 265 | dbnp_variant  | rs762551       | rs762551                  | sweep/g |
| 266 | hgnc_record   | HGNC:ABCC11    | HGNC:14639                | sweep/u |
| 267 | hgnc_record   | HGNC:ABL1      | HGNC:76                   | sweep/u |
| 268 | hgnc_record   | HGNC:ACTN3     | HGNC:165                  | sweep/u |
| 269 | hgnc_record   | HGNC:ADH1B     | HGNC:250                  | sweep/u |
| 270 | hgnc_record   | HGNC:ALDH2     | HGNC:404                  | sweep/u |
| 271 | hgnc_record   | HGNC:APC       | HGNC:583                  | sweep/u |
| 272 | hgnc_record   | HGNC:APOE      | HGNC:613                  | sweep/u |
| 273 | hgnc_record   | HGNC:ASXL1     | HGNC:18318                | sweep/u |
| 274 | hgnc_record   | HGNC:ATM       | HGNC:795                  | sweep/u |

| #   | Kind               | Study              | Accession                              | Used in                   |
|-----|--------------------|--------------------|----------------------------------------|---------------------------|
| 275 | hgnc_record        | HGNC:BRAF          | HGNC:1097                              | sweep/u                   |
| 276 | hgnc_record        | HGNC:BRCA1         | HGNC:1100                              | sweep/u                   |
| 277 | hgnc_record        | HGNC:BRCA2         | HGNC:1101                              | sweep/u                   |
| 278 | hgnc_record        | HGNC:CDKN2A        | HGNC:1787                              | sweep/u                   |
| 279 | hgnc_record        | HGNC:CFTR          | HGNC:1884                              | sweep/u                   |
| 280 | hgnc_record        | HGNC:CHEK2         | HGNC:16627                             | sweep/u                   |
| 281 | hgnc_record        | HGNC:COMT          | HGNC:2228                              | sweep/u                   |
| 282 | hgnc_record        | HGNC:DMD           | HGNC:2928                              | sweep/u                   |
| 283 | hgnc_record        | HGNC:DNMT3A        | HGNC:2978                              | sweep/u                   |
| 284 | hgnc_record        | HGNC:EDAR          | HGNC:2895                              | sweep/u                   |
| 285 | hgnc_record        | HGNC:EGFR          | HGNC:3236                              | sweep/u                   |
| 286 | hgnc_record        | HGNC:ERBB2         | HGNC:3430                              | sweep/u                   |
| 287 | hgnc_record        | HGNC:F5            | HGNC:3542                              | sweep/u                   |
| 288 | hgnc_record        | HGNC:FLT3          | HGNC:3765                              | sweep/u                   |
| 289 | hgnc_record        | HGNC:FMR1          | HGNC:3775                              | sweep/u                   |
| 290 | hgnc_record        | HGNC:GBA1          | HGNC:4177                              | sweep/u                   |
| 291 | hgnc_record        | HGNC:HBB           | HGNC:4827                              | sweep/u                   |
| 292 | hgnc_record        | HGNC:HERC2         | HGNC:4868                              | sweep/u                   |
| 293 | hgnc_record        | HGNC:HFE           | HGNC:4886                              | sweep/u                   |
| 294 | hgnc_record        | HGNC:HTT           | HGNC:4851                              | sweep/u                   |
| 295 | hgnc_record        | HGNC:IDH1          | HGNC:5382                              | sweep/u                   |
| 296 | hgnc_record        | HGNC:IDH2          | HGNC:5383                              | sweep/u                   |
| 297 | hgnc_record        | HGNC:JAK2          | HGNC:6192                              | sweep/u                   |
| 298 | hgnc_record        | HGNC:KCNQ1         | HGNC:6294                              | sweep/u                   |
| 299 | hgnc_record        | HGNC:KRAS          | HGNC:6407                              | sweep/u                   |
| 300 | hgnc_record        | HGNC:LCT           | HGNC:6530                              | sweep/u                   |
| 301 | hgnc_record        | HGNC:LMNA          | HGNC:6636                              | sweep/u                   |
| 302 | hgnc_record        | HGNC:MLH1          | HGNC:7127                              | sweep/u                   |
| 303 | hgnc_record        | HGNC:MSH2          | HGNC:7325                              | sweep/u                   |
| 304 | hgnc_record        | HGNC:MTHFR         | HGNC:7436                              | sweep/u                   |
| 305 | hgnc_record        | HGNC:MYC           | HGNC:7553                              | sweep/u                   |
| 306 | hgnc_record        | HGNC:MYH7          | HGNC:7577                              | sweep/u                   |
| 307 | hgnc_record        | HGNC:NF1           | HGNC:7765                              | sweep/u                   |
| 308 | hgnc_record        | HGNC:NF2           | HGNC:7773                              | sweep/u                   |
| 309 | hgnc_record        | HGNC:NPM1          | HGNC:7910                              | sweep/u                   |
| 310 | hgnc_record        | HGNC:NRAS          | HGNC:7989                              | sweep/u                   |
| 311 | hgnc_record        | HGNC:OCA2          | HGNC:8101                              | sweep/u                   |
| 312 | hgnc_record        | HGNC:PALB2         | HGNC:26144                             | sweep/u                   |
| 313 | hgnc_record        | HGNC:PIK3CA        | HGNC:8975                              | sweep/u                   |
| 314 | hgnc_record        | HGNC:PTEN          | HGNC:9588                              | sweep/u                   |
| 315 | hgnc_record        | HGNC:RB1           | HGNC:9884                              | sweep/u                   |
| 316 | hgnc_record        | HGNC:RUNX1         | HGNC:10471                             | sweep/u                   |
| 317 | hgnc_record        | HGNC:SCN5A         | HGNC:10593                             | sweep/u                   |
| 318 | hgnc_record        | HGNC:SLC24A5       | HGNC:20611                             | sweep/u                   |
| 319 | hgnc_record        | HGNC:SMAD4         | HGNC:6770                              | sweep/u                   |
| 320 | hgnc_record        | HGNC:SOD1          | HGNC:11179                             | sweep/u                   |
| 321 | hgnc_record        | HGNC:STK11         | HGNC:11389                             | sweep/u                   |
| 322 | hgnc_record        | HGNC:TET2          | HGNC:25941                             | sweep/u                   |
| 323 | hgnc_record        | HGNC:TP53          | HGNC:11998                             | sweep/u                   |
| 324 | hgnc_record        | HGNC:TTN           | HGNC:12403                             | sweep/u                   |
| 325 | hgnc_record        | HGNC:VHL           | HGNC:12687                             | sweep/u                   |
| 326 | study              | 31                 | UniProt2Reactome.txt (current release) | sweep/r                   |
| 327 | study              | 32                 | gnomad_r4                              | gnomad.constraint sweep/g |
| 328 | study              | 33                 | sramp_mature_test.csv (133,826 labels) | sweep/r                   |
| 329 | study              | 34                 | GSE63753                               | sweep/r                   |
| 330 | ensembl_transcript | ENSMUST00000056677 | ENSMUST00000056677                     | sweep/r                   |
| 331 | ensembl_transcript | ENSMUST00000106089 | ENSMUST00000106089                     | sweep/r                   |
| 332 | ensembl_transcript | ENSMUST00000125127 | ENSMUST00000125127                     | sweep/r                   |
| 333 | ensembl_transcript | ENST00000234590    | ENST00000234590                        | sweep/r                   |
| 334 | ensembl_transcript | ENST00000481115    | ENST00000481115                        | sweep/r                   |
| 335 | ensembl_transcript | ENST00000525069    | ENST00000525069                        | sweep/r                   |
| 336 | ensembl_transcript | ENST00000602390    | ENST00000602390                        | sweep/r                   |
| 337 | ensembl_transcript | ENSMUST00000145952 | ENSMUST00000145952                     | sweep/r                   |
| 338 | ensembl_transcript | ENST00000227520    | ENST00000227520                        | sweep/r                   |

| #   | Kind               | Study              | Accession              | Used in |
|-----|--------------------|--------------------|------------------------|---------|
| 339 | ensembl_transcript | ENST00000255641    | ENST00000255641        | sweep/r |
| 340 | ensembl_transcript | ENSMUST00000015889 | ENSMUST00000015889     | sweep/r |
| 341 | ensembl_transcript | ENSMUST00000038791 | ENSMUST00000038791     | sweep/r |
| 342 | ensembl_transcript | ENSMUST00000022709 | ENSMUST00000022709     | sweep/r |
| 343 | ensembl_transcript | ENST00000223368    | ENST00000223368        | sweep/r |
| 344 | ensembl_transcript | ENST00000284551    | ENST00000284551        | sweep/r |
| 345 | ensembl_transcript | ENST00000305885    | ENST00000305885        | sweep/r |
| 346 | ensembl_transcript | ENST00000357186    | ENST00000357186        | sweep/r |
| 347 | ensembl_transcript | ENST00000418388    | ENST00000418388        | sweep/r |
| 348 | ensembl_transcript | ENST00000622639    | ENST00000622639        | sweep/r |
| 349 | ensembl_transcript | ENSMUST00000179669 | ENSMUST00000179669     | sweep/r |
| 350 | ensembl_transcript | ENST00000356872    | ENST00000356872        | sweep/r |
| 351 | ensembl_transcript | ENST00000359867    | ENST00000359867        | sweep/r |
| 352 | ensembl_transcript | ENST00000361952    | ENST00000361952        | sweep/r |
| 353 | ensembl_transcript | ENST00000404824    | ENST00000404824        | sweep/r |
| 354 | ensembl_transcript | ENST00000428417    | ENST00000428417        | sweep/r |
| 355 | ensembl_transcript | ENSMUST00000035295 | ENSMUST00000035295     | sweep/r |
| 356 | ensembl_transcript | ENSMUST00000058667 | ENSMUST00000058667     | sweep/r |
| 357 | ensembl_transcript | ENST00000200453    | ENST00000200453        | sweep/r |
| 358 | ensembl_transcript | ENST00000316836    | ENST00000316836        | sweep/r |
| 359 | ensembl_transcript | ENST00000575785    | ENST00000575785        | sweep/r |
| 360 | ensembl_transcript | ENSMUST00000018716 | ENSMUST00000018716     | sweep/r |
| 361 | ensembl_transcript | ENSMUST00000061425 | ENSMUST00000061425     | sweep/r |
| 362 | ensembl_transcript | ENSMUST00000081747 | ENSMUST00000081747     | sweep/r |
| 363 | ensembl_transcript | ENSMUST00000084743 | ENSMUST00000084743     | sweep/r |
| 364 | ensembl_transcript | ENST00000618250    | ENST00000618250        | sweep/r |
| 365 | ensembl_transcript | ENSMUST00000022272 | ENSMUST00000022272     | sweep/r |
| 366 | ensembl_transcript | ENSMUST00000037235 | ENSMUST00000037235     | sweep/r |
| 367 | ensembl_transcript | ENSMUST00000040967 | ENSMUST00000040967     | sweep/r |
| 368 | ensembl_transcript | ENSMUST00000042203 | ENSMUST00000042203     | sweep/r |
| 369 | ensembl_transcript | ENSMUST00000042227 | ENSMUST00000042227     | sweep/r |
| 370 | pmid               | 37278811           | PMID 37278811          | sweep/e |
| 371 | pmid               | 40163448           | PMID 40163448          | sweep/e |
| 372 | pmid               | 32154747           | PMID 32154747          | sweep/e |
| 373 | pmid               | 31626570           | PMID 31626570          | sweep/e |
| 374 | pmid               | 27394157           | PMID 27394157          | sweep/e |
| 375 | pmid               | 30334692           | PMID 30334692          | sweep/e |
| 376 | pmid               | 40418577           | PMID 40418577          | sweep/e |
| 377 | pmid               | 39235967           | PMID 39235967          | sweep/e |
| 378 | pmid               | 31999662           | PMID 31999662          | sweep/e |
| 379 | pmid               | 27252987           | PMID 27252987          | sweep/e |
| 380 | nct                | NCT04235140        | NCT04235140            | sweep/e |
| 381 | nct                | NCT03029455        | NCT03029455            | sweep/e |
| 382 | nct                | NCT01614457        | NCT01614457            | sweep/e |
| 383 | nct                | NCT01937325        | NCT01937325            | sweep/e |
| 384 | nct                | NCT03525574        | NCT03525574            | sweep/e |
| 385 | nct                | NCT02797132        | NCT02797132            | sweep/e |
| 386 | nct                | NCT05699148        | NCT05699148            | sweep/e |
| 387 | nct                | NCT03085485        | NCT03085485            | sweep/e |
| 388 | nct                | NCT02965326        | NCT02965326            | sweep/e |
| 389 | nct                | NCT07363304        | NCT07363304            | sweep/e |
| 390 | study              | 35                 | genes_to_phenotype.txt | sweep/r |
| 391 | omim               | OMIM:219700        | OMIM:219700            | sweep/r |
| 392 | omim               | OMIM:143100        | OMIM:143100            | sweep/r |
| 393 | omim               | OMIM:154700        | OMIM:154700            | sweep/r |
| 394 | omim               | OMIM:261600        | OMIM:261600            | sweep/r |
| 395 | omim               | OMIM:230800        | OMIM:230800            | sweep/r |
| 396 | omim               | OMIM:310200        | OMIM:310200            | sweep/r |
| 397 | study              | 36                 | pathology.tsv          | sweep/o |
| 398 | ensembl_gene       | ENSG00000164362    | ENSG00000164362        | sweep/o |
| 399 | ensembl_gene       | ENSG00000089685    | ENSG00000089685        | sweep/o |
| 400 | ensembl_gene       | ENSG00000081051    | ENSG00000081051        | sweep/o |
| 401 | ensembl_gene       | ENSG00000142515    | ENSG00000142515        | sweep/o |
| 402 | ensembl_gene       | ENSG00000141736    | ENSG00000141736        | sweep/o |
| 403 | chembl_target      | CHEMBL203          | CHEMBL203              | sweep/d |
| 404 | chembl_target      | CHEMBL279          | CHEMBL279              | sweep/d |
| 405 | chembl_target      | CHEMBL1862         | CHEMBL1862             | sweep/d |

| #   | Kind          | Study       | Accession   | Used in |
|-----|---------------|-------------|-------------|---------|
| 406 | chembl_target | CHEMBL2971  | CHEMBL2971  | sweep/d |
| 407 | chembl_target | CHEMBL4282  | CHEMBL4282  | sweep/d |
| 408 | chembl_target | CHEMBL2842  | CHEMBL2842  | sweep/d |
| 409 | uniprot       | P00533      | P00533      | sweep/d |
| 410 | uniprot       | P35968      | P35968      | sweep/d |
| 411 | uniprot       | P00519      | P00519      | sweep/d |
| 412 | uniprot       | O60674      | O60674      | sweep/d |
| 413 | uniprot       | P31749      | P31749      | sweep/d |
| 414 | uniprot       | P42345      | P42345      | sweep/d |
| 415 | pdb           | 1BWB        | 1BWB        | sweep/m |
| 416 | pdb           | 1JPZ        | 1JPZ        | sweep/m |
| 417 | pdb           | 1MUI        | 1MUI        | sweep/m |
| 418 | pdb           | 1TSL        | 1TSL        | sweep/m |
| 419 | pdb           | 1VKJ        | 1VKJ        | sweep/m |
| 420 | pdb           | 2AOD        | 2AOD        | sweep/m |
| 421 | pdb           | 2IEN        | 2IEN        | sweep/m |
| 422 | pdb           | 2IKI        | 2IKI        | sweep/m |
| 423 | pdb           | 2Q88        | 2Q88        | sweep/m |
| 424 | pdb           | 2R5Q        | 2R5Q        | sweep/m |
| 425 | pdb           | 2Z4O        | 2Z4O        | sweep/m |
| 426 | pdb           | 3FQL        | 3FQL        | sweep/m |
| 427 | pdb           | 3IJW        | 3IJW        | sweep/m |
| 428 | pdb           | 3KYF        | 3KYF        | sweep/m |
| 429 | pdb           | 3NTY        | 3NTY        | sweep/m |
| 430 | pdb           | 3NU3        | 3NU3        | sweep/m |
| 431 | pdb           | 3TKW        | 3TKW        | sweep/m |
| 432 | pdb           | 3UM8        | 3UM8        | sweep/m |
| 433 | pdb_ligand    | 017         | 017         | sweep/m |
| 434 | pdb_ligand    | 065         | 065         | sweep/m |
| 435 | pdb_ligand    | 140         | 140         | sweep/m |
| 436 | pdb_ligand    | 146         | 146         | sweep/m |
| 437 | pdb_ligand    | 1CY         | 1CY         | sweep/m |
| 438 | pdb_ligand    | 1UN         | 1UN         | sweep/m |
| 439 | pdb_ligand    | 2NC         | 2NC         | sweep/m |
| 440 | pdb_ligand    | 388         | 388         | sweep/m |
| 441 | pdb_ligand    | 478         | 478         | sweep/m |
| 442 | pdb_ligand    | 4CS         | 4CS         | sweep/m |
| 443 | pdb_ligand    | 5GP         | 5GP         | sweep/m |
| 444 | pdb_ligand    | 5P3         | 5P3         | sweep/m |
| 445 | pdb_ligand    | 79Z         | 79Z         | sweep/m |
| 446 | pdb_ligand    | A15         | A15         | sweep/m |
| 447 | pdb_ligand    | A3P         | A3P         | sweep/m |
| 448 | pdb_ligand    | AB1         | AB1         | sweep/m |
| 449 | pdb_ligand    | ACO         | ACO         | sweep/m |
| 450 | kegg_compound | C00010      | cpd:C00010  | sweep/m |
| 451 | kegg_compound | C00011      | cpd:C00011  | sweep/m |
| 452 | kegg_compound | C00022      | cpd:C00022  | sweep/m |
| 453 | kegg_compound | C00024      | cpd:C00024  | sweep/m |
| 454 | kegg_compound | C00031      | cpd:C00031  | sweep/m |
| 455 | kegg_compound | C00042      | cpd:C00042  | sweep/m |
| 456 | kegg_compound | C00048      | cpd:C00048  | sweep/m |
| 457 | kegg_compound | C00074      | cpd:C00074  | sweep/m |
| 458 | kegg_compound | C00085      | cpd:C00085  | sweep/m |
| 459 | kegg_compound | C00092      | cpd:C00092  | sweep/m |
| 460 | kegg_compound | C00111      | cpd:C00111  | sweep/m |
| 461 | kegg_compound | C00118      | cpd:C00118  | sweep/m |
| 462 | kegg_compound | C00149      | cpd:C00149  | sweep/m |
| 463 | kegg_compound | C00158      | cpd:C00158  | sweep/m |
| 464 | kegg_compound | C00186      | cpd:C00186  | sweep/m |
| 465 | kegg_compound | C00197      | cpd:C00197  | sweep/m |
| 466 | kegg_compound | C00236      | cpd:C00236  | sweep/m |
| 467 | kegg_compound | C00354      | cpd:C00354  | sweep/m |
| 468 | kegg_compound | C00469      | cpd:C00469  | sweep/m |
| 469 | kegg_compound | C01089      | cpd:C01089  | sweep/m |
| 470 | kegg_enzyme   | EC 1.1.1.1  | ec:1.1.1.1  | sweep/m |
| 471 | kegg_enzyme   | EC 1.1.1.27 | ec:1.1.1.27 | sweep/m |
| 472 | kegg_enzyme   | EC 1.2.1.12 | ec:1.2.1.12 | sweep/m |

| #   | Kind        | Study                               | Accession                                    | Used in |
|-----|-------------|-------------------------------------|----------------------------------------------|---------|
| 473 | kegg_enzyme | EC 1.2.4.1                          | ec:1.2.4.1                                   | sweep/m |
| 474 | kegg_enzyme | EC 2.3.1.9                          | ec:2.3.1.9                                   | sweep/m |
| 475 | kegg_enzyme | EC 2.3.3.1                          | ec:2.3.3.1                                   | sweep/m |
| 476 | kegg_enzyme | EC 2.3.3.9                          | ec:2.3.3.9                                   | sweep/m |
| 477 | kegg_enzyme | EC 2.7.1.1                          | ec:2.7.1.1                                   | sweep/m |
| 478 | kegg_enzyme | EC 2.7.1.11                         | ec:2.7.1.11                                  | sweep/m |
| 479 | kegg_enzyme | EC 2.7.1.40                         | ec:2.7.1.40                                  | sweep/m |
| 480 | kegg_enzyme | EC 2.7.2.3                          | ec:2.7.2.3                                   | sweep/m |
| 481 | kegg_enzyme | EC 4.1.2.13                         | ec:4.1.2.13                                  | sweep/m |
| 482 | kegg_enzyme | EC 4.1.3.1                          | ec:4.1.3.1                                   | sweep/m |
| 483 | kegg_enzyme | EC 4.2.1.11                         | ec:4.2.1.11                                  | sweep/m |
| 484 | kegg_enzyme | EC 5.3.1.1                          | ec:5.3.1.1                                   | sweep/m |
| 485 | kegg_enzyme | EC 5.3.1.9                          | ec:5.3.1.9                                   | sweep/m |
| 486 | pubchem_cid | 54675776                            | CID 54675776                                 | sweep/b |
| 487 | pubchem_cid | 6134                                | CID 6134                                     | sweep/b |
| 488 | pubchem_cid | 439195                              | CID 439195                                   | sweep/b |
| 489 | pubchem_cid | 2153                                | CID 2153                                     | sweep/b |
| 490 | pubchem_cid | 241                                 | CID 241                                      | sweep/b |
| 491 | pubmed      | 7510417                             | PMID 7510417                                 | sweep/b |
| 492 | pubmed      | 1812784                             | PMID 1812784                                 | sweep/b |
| 493 | jaspar      | MA0099.3                            | MA0099.3                                     | sweep/d |
| 494 | jaspar      | MA0107.1                            | MA0107.1                                     | sweep/d |
| 495 | jaspar      | MA0106.3                            | MA0106.3                                     | sweep/d |
| 496 | jaspar      | MA0108.2                            | MA0108.2                                     | sweep/d |
| 497 | jaspar      | MA0093.3                            | MA0093.3                                     | sweep/d |
| 498 | jaspar      | MA0079.4                            | MA0079.4                                     | sweep/d |
| 499 | jaspar      | MA0060.2                            | MA0060.2                                     | sweep/d |
| 500 | jaspar      | MA0114.4                            | MA0114.4                                     | sweep/d |
| 501 | ucsc_hg38   | chr3:36990000-37000000              | hg38 chr3:36,990,000-37,000,000              | sweep/d |
| 502 | ucsc_hg38   | chr9:21990000-22000000              | hg38 chr9:21,990,000-22,000,000              | sweep/d |
| 503 | ucsc_hg38   | chr9:21976000-21989000              | hg38 chr9:21,976,000-21,989,000              | sweep/d |
| 504 | ucsc_cpg    | cpgIslandExt chr3:36900000-37100000 | hg38 cpgIslandExt chr3:36,900,000-37,100,000 | sweep/d |
| 505 | ucsc_cpg    | cpgIslandExt chr9:21940000-22000000 | hg38 cpgIslandExt chr9:21,940,000-22,000,000 | sweep/d |
| 506 | pubmed      | 8242752                             | PMID 8242752                                 | sweep/d |
| 507 | hgnc        | OCT4                                | OCT4                                         | sweep/c |
| 508 | hgnc        | SOX2                                | SOX2                                         | sweep/c |
| 509 | hgnc        | NANOG                               | NANOG                                        | sweep/c |
| 510 | hgnc        | GATA4                               | GATA4                                        | sweep/c |
| 511 | hgnc        | MEF2C                               | MEF2C                                        | sweep/c |
| 512 | hgnc        | TBX5                                | TBX5                                         | sweep/c |
| 513 | hgnc        | ASCL1                               | ASCL1                                        | sweep/c |
| 514 | hgnc        | NEUROD1                             | NEUROD1                                      | sweep/c |
| 515 | hgnc        | MYT1L                               | MYT1L                                        | sweep/c |
| 516 | hgnc        | MYOD1                               | MYOD1                                        | sweep/c |
| 517 | hgnc        | TNNT2                               | TNNT2                                        | sweep/c |
| 518 | hgnc        | MYH6                                | MYH6                                         | sweep/c |
| 519 | hgnc        | NKX2-5                              | NKX2-5                                       | sweep/c |
| 520 | hgnc        | TUBB3                               | TUBB3                                        | sweep/c |
| 521 | hgnc        | MAP2                                | MAP2                                         | sweep/c |
| 522 | hgnc        | SCN1A                               | SCN1A                                        | sweep/c |
| 523 | hgnc        | MYOG                                | MYOG                                         | sweep/c |
| 524 | hgnc        | MYH1                                | MYH1                                         | sweep/c |
| 525 | hgnc        | COL1A1                              | COL1A1                                       | sweep/c |
| 526 | hgnc        | VIM                                 | VIM                                          | sweep/c |
| 527 | hgnc_alias  | OCT4-¿POU5F1                        | POU5F1                                       | sweep/c |
| 528 | pubmed      | 16904174                            | PMID 16904174                                | sweep/c |
| 529 | pubmed      | 20691899                            | PMID 20691899                                | sweep/c |
| 530 | pubmed      | 24243019                            | PMID 24243019                                | sweep/c |
| 531 | pubmed      | 42775788                            | PMID 42775788                                | sweep/b |
| 532 | pubmed      | 42759458                            | PMID 42759458                                | sweep/b |

| #   | Kind       | Study                                                                         | Accession                  | Used in  |
|-----|------------|-------------------------------------------------------------------------------|----------------------------|----------|
| 533 | pubmed     | 42746974                                                                      | PMID 42746974              | sweep/b  |
| 534 | pubmed     | 42731586                                                                      | PMID 42731586              | sweep/b  |
| 535 | pubmed     | 42727368                                                                      | PMID 42727368              | sweep/b  |
| 536 | pubmed     | 42701997                                                                      | PMID 42701997              | sweep/b  |
| 537 | pubmed     | 42693013                                                                      | PMID 42693013              | sweep/b  |
| 538 | pubmed     | 42641908                                                                      | PMID 42641908              | sweep/b  |
| 539 | icite      | pmids_42775788_42759458_42746974_42731586_42727368_42701997_42693013_42641908 | citation_count map         | sweep/b  |
| 540 | refseq     | NM_000518.5                                                                   | RefSeq FASTA               | sweep/g  |
| 541 | refseq     | NM_000546.6                                                                   | RefSeq FASTA               | sweep/g  |
| 542 | refseq     | NM_000492.4                                                                   | RefSeq FASTA               | sweep/g  |
| 543 | ncbi_gene  | 7157                                                                          | esummary                   | sweep/g  |
| 544 | ncbi_gene  | 672                                                                           | esummary                   | sweep/g  |
| 545 | ncbi_gene  | 1080                                                                          | esummary                   | sweep/g  |
| 546 | ncbi_gene  | 3043                                                                          | esummary                   | sweep/g  |
| 547 | ncbi_gene  | 1956                                                                          | esummary                   | sweep/g  |
| 548 | uniprot    | P04637                                                                        | UniProt REST               | sweep/g  |
| 549 | uniprot    | P38398                                                                        | UniProt REST               | sweep/g  |
| 550 | uniprot    | P13569                                                                        | UniProt REST               | sweep/g  |
| 551 | uniprot    | P68871                                                                        | UniProt REST               | sweep/g  |
| 552 | cbioportal | coadread_tcga_pan_can_atlas_2018                                              | cBioPortal mutations/fetch | sweep/li |
| 553 | cbioportal | luad_tcga_pan_can_atlas_2018                                                  | cBioPortal mutations/fetch | sweep/li |
| 554 | cbioportal | brca_tcga_pan_can_atlas_2018                                                  | cBioPortal mutations/fetch | sweep/li |
| 555 | cbioportal | paad_tcga_pan_can_atlas_2018                                                  | cBioPortal mutations/fetch | sweep/li |
| 556 | cbioportal | prad_tcga_pan_can_atlas_2018                                                  | cBioPortal mutations/fetch | sweep/li |
| 557 | cbioportal | skcm_tcga_pan_can_atlas_2018                                                  | cBioPortal mutations/fetch | sweep/li |
| 558 | pubmed     | 25079552                                                                      | PMID 25079552              | sweep/li |
| 559 | pubmed     | 28810144                                                                      | PMID 28810144              | sweep/li |
| 560 | pubmed     | 27428049                                                                      | PMID 27428049              | sweep/li |
| 561 | pubmed     | 26771485                                                                      | PMID 26771485              | sweep/li |
| 562 | uniprot    | P03135                                                                        | UniProt REST               | sweep/v  |
| 563 | uniprot    | Q6JC40                                                                        | UniProt REST               | sweep/v  |
| 564 | pubmed     | 9445046                                                                       | PMID 9445046               | sweep/v  |
| 565 | pubmed     | 16378984                                                                      | PMID 16378984              | sweep/v  |
| 566 | rcsb       | 1UBQ                                                                          | RCSB PDB                   | sweep/a  |
| 567 | pubchem    | 243                                                                           | PubChem PUG-REST           | sweep/c  |
| 568 | pubchem    | 996                                                                           | PubChem PUG-REST           | sweep/c  |
| 569 | pubchem    | 1001                                                                          | PubChem PUG-REST           | sweep/c  |
| 570 | pubchem    | 6115                                                                          | PubChem PUG-REST           | sweep/c  |
| 571 | pubchem    | 10008                                                                         | PubChem PUG-REST           | sweep/c  |
| 572 | pubchem    | 10009                                                                         | PubChem PUG-REST           | sweep/c  |
| 573 | pubchem    | 13195                                                                         | PubChem PUG-REST           | sweep/c  |
| 574 | chembl     | CHEMBL2107497                                                                 | ChEMBL similarity API      | sweep/c  |

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